Foreword

Notice

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COMPAQ PORTABLE II PERSONAL COMPUTER
MAINTENANCE AND SERVICE GUIDE

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Preface
This publication is intended as a reference for technicians trained to repair the COMPAQ PORTABLE II Personal Computer.

Summary of Text

This guide contains the following eight chapters:

Chapter 1, "Operating and Performance Specifications," provides physical, environmental, and performance specifications for the COMPAQ PORTABLE II Personal Computer.

Chapter 2, "Power On Self Test," describes the internal system diagnostic programs that execute automatically when the system is turned on. It also includes a quick reference for identifying and correcting problems that could occur during the Power On Self Test procedure.

Chapters 3 and 4: Please see the SUPPORT SOFTWARE MAINTENANCE AND SERVICE GUIDE for information on the SETUP and ADVANCED DIAGNOSTICS programs.

Chapter 5, "Error Messages and Codes," lists the Power On Self Test (POST) and DIAGNOSTICS Error Codes and provides the course of action necessary to resolve the problem.

Chapter 6, "Illustrated Parts Catalog," provides an illustrated reference for COMPAQ PORTABLE II Personal Computer spare parts.

Chapter 7, "Removal and Replacement Procedures," describes how to remove and replace field replaceable subassemblies.

Chapter 8, "Jumper Settings, Switch Settings, and Adjustments," provides detailed information for setting jumpers and switches and performing monitor adjustments.

Tools and Supplies

The following tools and supplies are required to service the COMPAQ PORTABLE II Personal Computer:

- Torx screwdrivers, sizes T-15 and T-10
- Slotted blade screwdriver, 1/4 inch
- Needlenosed pliers
- 25 pin printer loopback plug
- 25 pin serial loopback plug
- 9 pin serial loopback plug
- DIAGNOSTICS diskette
- Formatted diskette

Optional Tools

- Memory chip insertion tool
- Memory chip removal tool
- Diskette drive signal extension cable (PN 100546-001)
- Diskette drive power extension cable (PN 100545-001)
- Socket/Torx tool kit (PN 101089-001)
  - Torx screwdrivers, sizes T-15 and T-10
  - Adjustable socket tool with hex sockets and one Torx socket (T-10)

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CAUTION

Only trained technicians should attempt to repair this equipment. All troubleshooting and repair procedures are detailed to allow subassembly/module level repair only.

Because of the complexity of the individual boards and subassemblies, no one should attempt to make repairs at the component level or make modifications to any printed wiring board. Improper repairs may create a safety hazard. Indications of component replacement or printed wiring board modifications may void any warranty or exchange allowances.

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Additional Reference Information

The following support documentation is available for the COMPAQ PORTABLE II

Personal Computer.

- Operations Guide
- MS-DOS VERSION 3 REFERENCE GUIDE
- Technical Reference Guide
- HOW TO DO BUSINESS WITH COMPAQ SERVICE
- COMPAQ SERVICE QUICK REFERENCE GUIDE
- COMPAQ Service Advisories and Bulletins
- COMPAQ QuickFind Support Reference Library

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WARNING

The computer is designed to be electrically grounded. To ensure proper operation of the computer, plug the AC power cord only into a properly grounded AC outlet.

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Compaq Computer Corporation requires all peripheral devices be connected to this computer via shielded cables with metal RFI/EMI connector hoods.

WIRE TYPE: Multipaired, overall shielded; Belden #98XX; Alpha #54XX; or equivalent.

CONNECTOR HOOD: RFI/EMI metal shield; AMP #7451 7X-X; or equivalent.

It is important that the chassis ground of any peripheral device be connected to the computer chassis. An Alpha #1221 flat braid strap is sufficient. This strap is not necessary if the shielded cable connects the two chassis.

Chapter 1. Operating and Performance Specifications

Chapter 1.0 Introduction

This chapter contains operating and performance specifications for the COMPAQ PORTABLE II Personal Computer.

Chapter 1.1 System Unit

Table 1-1. Electrical and Mechanical Specifications

Dimensions:		
Height	7.5 in. (19.0 cm)	
Depth	13.9 in. (35.3 cm)	
Width	17.7 in. (45.0 cm)	
Weight	Model 1 23.6 lb (10.7 kg)	
	Model 2 25.6 lb (11.6 kg)	
	Model 3 26.2 lb (11.9 kg)	
	Model 4 26.3 lb (12.2 kg)	
Power Requirements:		
Voltage	120 VAC	220 - 240 VAC
Frequency	60 Hz	50 Hz
Current	3A	2.5A
AC Power Cable:		
Length	79 in. (2 m)	
Gauge	18 AWG	
Environmental Requirements:		
Temperature:		
Operating	10oC to 40oC (50oF to 104oF)	
Nonoperating	10oC to 60oC (50oF to 140oF)	
Shipping	-30oC to 60oC (-22oF to 140oF)	
Humidity:		
Operating	20% to 80% (noncondensing)	
Nonoperating	5% to 90% (noncondensing	
Shock:		
Operating	5 g, 11 ms, half sine	
Nonoperating	35 g, 11 ms, half sine	
Vibration:		
Operating	0.5 g, 5 - 550 Hz	
Nonoperating	1.0 g, 5 - 550 Hz	
Heat Dissipation:		
Delivered	410 BTU/hr at 120W	
Total Watts	615 BTU/hr at 180W	
Maximum Unpressurized Altitude:		
Operating	10,000 ft (3048 m)	
Nonoperating	30,000 ft (9144 m)	

Chapter 1.2 Keyboard

Table 1-2. Keyboard Specifications

Dimensions:	
Height	1.3 in. (3.30 cm)
Depth	7.0 in. (17.78 cm)
Width	17.1 in. (43.43 cm)
Number of Keys	84
Keyboard:	
Cable Length	60.00 in. (152.4 cm)
Coil Diameter	0.75 in. (1.9 cm)
Coil Length	11.00 in. (27.94 cm)

Chapter 1.3 Video Display Unit

Table 1-3. Video Display Unit Specifications

Display	9 in. diagonal 90 degree deflection High persistence green phosphor Etched surface to reduce glare
Display Size	80 character by 25 line screen 40 character by 25 line screen
Graphic Resolution	640 x 200 dot resolution 320 x 200 dot resolution
Text Resolution	720 x 350 dot resolution

Chapter 1.4 10 Megabyte Fixed Disk Drive

Table 1-4. 10 Megabyte Fixed Disk Drive Specifications

Dimensions:	
Height	1.7 in. (4.32 cm)
Depth	5.80 in. (14.73 cm)
Width	3.75 in. (9.54 cm)
Weight	2.2 lb (1.0 kg)
Media:	1 Platter
Number of Data Heads	4
Number of Cylinders	306
Average Access Time	105 ms

Data Transfer Rate	5.0 Mb/s
Type	1

Chapter 1.5 20 Megabyte Fixed Disk Drive

Table 1-5. 20 Megabyte Fixed Disk Drive Specifications

Dimensions:	
Height	1.7 in. (4.32 cm)
Depth	4.1 in. (10.41 cm)
Width	5.75 in. (14.61 cm)
Weight	1.8 lb (0.8 kg)
Media:	2 Platter
Number of Data Heads	4
Number of Cylinders	615
Average Access Time	105 ms
Data Transfer Rate	5.0 Mb/s
Type	2

Chapter 1.6 360 KByte Diskette Drive

Table 1-6. 360 Kbyte Diskette Drive Specifications

Dimensions:	
Height	1.1 in. (2.85 cm)
Depth	7.9 in. (20.3 cm)
Width	5.9 in. (15.0 cm)
Weight	1.98 lb (0.9 kg)
Media:	
Tracks per inch	48
Number of Tracks	40 (when formatted with MS-DOS)
Data Transfer Rate	250 Kb/s

Chapter 1.7 1.2 Megabyte Diskette Drive

Table 1-7. 1.2 Megabyte Diskette Drive Specifications

Dimensions:	
Height	1.1 in. (2.8 cm)
Depth	8.0 in. (20.3 cm)
Width	5.8 in. (14.6 cm)
Weight	1.9 lb (0.9 kg)
Media:	

Tracks per Inch	96 (1.2 megabyte format) 48 (360 Kbyte format)
Number of Tracks	80 (1.2 megabyte format) 40 (360 Kbyte format)
Data Transfer Rate	500 Kb/s (1.2 megabyte format) 300 Kb/s (360 Kbyte format)

Chapter 1.8 1.44 Megabyte Diskette Drive Specifications

Table 1-8. 1.44 Megabyte Diskette Drive Specifications

Dimensions:	
Height	0.90 in. (2.3 cm)
Depth	6.47 in. (16.42 cm)
Width	5.95 in. (11.1 cm)
Weight	1.98 lb (0.9 kg)
Media:	
Tracks per Inch	135
Number of Tracks	80
Data Transfer Rate	500 Kb/s (high density) 250 Kb/s (low density)

Chapter 1.9 Power Supply

Table 1-9. Power Supply Specifications

Input Requirements:	
Line Voltage	102 to 132 VAC. single phase 180 to 264 VAC. single phase 3.0A fuse, 125 VAC for 120 VAC operations 2.5A fuse, 150 VAC for 220/240 VAC operation
Line Frequency	47 - 62 Hz
Cooling	Forced air provided by internal 12 VDC fan

Volts Direct Current Output

Nominal Voltage (VDC)	Current Amperes Minimum	Nominal Continuous Current (Amperes) Maximum *	Maximum Peak Current (Amperes)	Regulation Tolerance Percentage
+5.0	3.0	13.9	13.9	ñ2
-5.0	0.0	0.5	0.5	ñ5
+12.2	1.0	4.0	7.0	ñ2
-12.0	0.0	0.5	0.5	ñ5

* These values are maximum values based on nominal operating conditions for temperature, line voltage, frequency and altitude. Also, a minimum load on l2 VDC must be maintained for proper power supply operation.

Chapter 2. Power On Self Test/Problem Isolation

2.1 Power On Self Test

The Power On Self Test (POST) is a series of diagnostic tests performed automatically when the system is turned on. These tests verify that the computer system is operating properly. If a subassembly fails POST, an error code appears on the screen.

To start POST, follow these steps.

1. Turn off the computer.

2. Disconnect the signal cables of any peripheral devices (printers, modems, and so on) from the computer.

3. Remove the outer access casing to expose the system LED located on the system board (see Chapter 7 for this procedure).

4. Install a loopback plug connector at each location from which a peripheral signal cable was removed.

   NOTE: To test your printer during the printer test or to log errors to a printer, reconnect the printer now.

5. Verify that the AC power cable is connected to a usable, grounded, three pronged AC power outlet.

6. Insert the DIAGNOSTICS diskette into drive A.

7. Turn on the computer. The system automatically runs POST. See Section 2.2, "System Problem Isolation," for information on handling system problems.

Chapter 2.2 System Problem Isolation

This section contains problem isolation procedures for identifying and correcting system related problems.

To isolate system problems, perform the following steps:

1. Verify that the AC power switch is on.

2. Listen for the sound of the fan. Did you hear the fan?

   YES: Proceed to step 3.

   NO: Turn off the computer and check the condition of the AC power fuse.

   Is the fuse good?

   YES: Return the AC power fuse to its fuse holder and turn on the system. Proceed to step 3.

NO: Replace the AC power fuse (refer to Chapter 7 in this guide for AC power fuse removal and replacement information). Repeat steps 1 and 2.

3. Did you hear an audible beep or see an error code on the screen?

YES: See Chapter 5, "Error Messages and Codes" for information on how to identify and correct either audible or error code related problems.

NO: Proceed to step 4.

4. Check the system LED on the system board. Is it lit?

YES: Proceed to step 5. If you did not hear the fan, replace it now.

NO: Replace the power supply subassembly chassis and repeat the problem isolation procedures.

If the problem continues, replace the system board and repeat the problem isolation procedures.

If the problem has not been corrected, test each controller or option board by turning off the system, removing a board and turning the power back on.

If the LED is still not on, test each video and storage device by turning off the system, disconnecting the device power cable and then turning on the system.

When the system LED lights, power off the system and replace the defective subassembly.

NOTE: For removal and replacement procedures, refer to Chapter 7.

5. Check the screen for a blinking cursor. Do you see a cursor on the screen?

YES: Proceed to step 6.

NO: Listen for an audible tone (beep) from the system speaker.

If you hear one or two short beeps, POST is complete. The system now boots the DIAGNOSTICS program. If the screen remains blank, adjust the brightness knob.

If the display screen remains blank after adjusting the brightness knob, the video display unit subsystem is not functioning properly.

o If a color graphics or composite video monitor is available, connect either device to the correct video outputs on the video display controller board. Using the CTRL + ALT + < command, change the display mode to the graphics mode.

If the first screen of the DIAGNOSTICS program appears on the external monitor, replace the internal video display unit.

If the screen on the external monitor is blank, replace the video display

controller board.

- If a color graphics or composite video monitor is not available, replace the video display controller board.

- If a color graphics or composite video monitor is not available and replacing the video display controller board does not solve the problem, replace the internal video display unit.

If you do not hear a beep, replace the system board. If you hear any other audible sounds, refer to Chapter 5 for more information on POST related error messages.

6. Did the cursor disappear from the screen, replaced by the number of Kbytes (shown in increments) on the display screen?

   YES: This indicates the memory self test portion of POST. Proceed to step 7.

   NO: Replace the system board. Restart the system problem isolation procedures.

7. Did the LEDs on the mass storage device flash, followed by one or two short beeps?

   YES: This indicates that POST and the SETUP utility have completed successfully. The DIAGNOSTICS program should then boot and the DIAGNOSTICS menu should appear on the display screen.

   Refer to Chapter 4 for additional information on the COMPAQ DIAGNOSTICS program.

   NO: If the cursor reappears and remains on the screen, replace the system board.

   If one audible tone is heard and an error message appears, the SETUP function was not completed on the system. Press the F1 key and proceed to Chapter 3 in this guide for information on the SETUP function.

Chapter 3. Setup

Please consult the SUPPORT SOFTWARE MAINTENANCE AND SERVICE GUIDE for current information on SETUP.

Chapter 4. Diagnostic Procedures

Please consult the SUPPORT SOFTWARE MAINTENANCE AND SERVICE GUIDE for current information on DIAGNOSTICS.

Chapter 5. Error Messages and Codes

5.0 Introduction

This section contains Power On Self Test (POST) messages, DIAGNOSTIC error codes, and memory error codes. The messages and codes are given in tables that list the message or error code, a description of the error, the probable cause of the error, and the action required to resolve the error condition.

Chapter 5.1 Power On Self Test Messages

An error message results if a problem is encountered during the Power On Self Test. This utility runs automatically when the system is turned on. Table 5-1 lists the message, audible (beep) message, probable cause, and the recommended action.

Table 5-1. Power On Self Test Error Messages

Message	Beeps	Probable Cause	Recommended Action
101 - I/O ROM Error	1 Long, 1 Short	Option ROM	1. Inspect ROM placement. 2. Verify correct ROM location.
101 - ROM Error	1 Long, 1 Short	System ROM	3. Replace the ROM.
102 - System Board	None	System board	The following recommended action applies to error codes 102 through 115:
110 - System Failure	None	Timer failure	
111 - System Failure	None	Memory refresh failure	Replace the system board.
112 - System Failure	None	CMOS failure	
113 - System Failure	None	DMA failure	
114 - System Failure	None	Timer interrupt failure	
115 - System Failure	None	Protected mode failure	

Message	Beeps	Probable Cause	Recommended Action
162 - System Options Error	2 Short	SETUP utility incorrect	Run SETUP
163 - Time and Date Not Set	2 Short	Invalid time or date	Run SETUP
164 - Memory Size Error	2 Short	System configuration incorrect	Check the memory configuration and run SETUP

XX000B YYZZ 201 - Memory Error	None	RAM failure	Refer to Section 5.3, "Memory Related Error Codes."
301 - Keyboard Error	None	Keyboard failure	Replace the keyboard.
303 - Keyboard Controller Error	None	Keyboard controller failure	Replace the system board.
304 - Keyboard or System Unit Error	None	Keyboard or system board failure	1. Replace the keyboard. 2. Replace the system board.
401 - Printer Interface Error	None	System board failure	Replace the system board.
402 - Monochrome Display Controller Failure	1 Long, 2 Short	System board or display controller board	Replace the monochrome display controller board or system board.
501 - Display Controller Failure	1 Long, 2 Short	Video display or controller board	Replace the video board.
Message	**Beeps**	**Probable Cause**	**Recommended Action**
601 - Diskette Controller Error	None	Diskette controller board	1. Replace multipurpose controller board. 2. Replace multipurpose fixed disk drive controller board.
602 - Diskette Boot Record Error	None	Diskette in Drive A not bootable	Replace diskette.
1780 - Disk 0 Failure	None	Fixed disk drive/ format error	Run DIAGNOSTICS.
1782 - Disk Controller failure	None	Fixed disk drive/ controller error	Run DIAGNOSTICS.
1790 - Disk 0 Error	None	Fixed disk drive error or wrong drive type	Run SETUP and DIAGNOSTICS.
XX000Y ZZ Parity Check 2	None	Expansion RAM	Run DIAGNOSTICS.
Audible	1 Short	Power on successful	None
Audible	2 Short	Power on successful	None
(RESUME = "F1" KEY)	None	As indicated	Press F1 key to continue

Chapter 5.2 Diagnostics Error Codes

DIAGNOSTICS error codes occur if the system recognizes a problem while running the DIAGNOSTICS program. These error codes help identify possible defective subassemblies.

Table 5-2 through 5-12 list possible error codes, a description of the error condition, and the action required to correct the condition.

In each case, the Recommended Action column lists steps necessary to correct the problem. After completing each step, run the DIAGNOSTICS program to verify that the error condition has been corrected. If the error code reappears, perform the next step, then run the DIAGNOSTICS program. Continue until the DIAGNOSTICS program no longer detects an error condition.

The error codes are in the form of AYY - XX or AAYY - XX, A or AA denotes the test or action that failed. XX denotes a specific problem. For example, error code 603 - 07 indicated that the diskette drive write/read/compare test failed because of a fatal error while writing.

For assistance in the removal and replacement of a particular subassembly, see Chapter 7, " Removal and Replacement Procedures."

Table 5-2. Processor Test Error Codes

Error Codes	Error Description	Recommended Action
101 - 01	CPU test failed	Replace the system board and retest for error codes 101 - 01 through 101 - 94
102 - 01	Numeric coprocessor initial status word incorrect	The following steps apply to error codes 102 - 01 through 102 - 15: 1. Replace the numeric coprocessor and retest. 2. Replace the system board and retest.
102 - 02	Numeric coprocessor initial control word incorrect	
102 - 03	Numeric coprocessor tag word not all ones	
102 - 04	Numeric coprocessor tag word not all zeros	
102 - 05	Numeric coprocessor exchange command failed	
102 - 06	Numeric coprocessor masked exception incorrectly handled	
102 - 07	Numeric coprocessor unmasked exception incorrectly handled	
102 - 08	Numeric coprocessor wrong mask bit set in status register	
102 - 09	Numeric coprocessor unable to store real number	

Codes	Error Description	Recommended Action
102 - 10	Numeric coprocessor real number calculation test failed	The following steps apply to error codes 102 - 01 through 102 - 15:
102 - 11	Numeric coprocessor speed test failed	1. Replace the numeric coprocessor and retest.
102 - 12	Numeric coprocessor pattern test failed	2. Replace the system board and retest.
102 - 15	Numeric coprocessor inoperative or socket unoccupied	
103 - 01	DMA page registers test failed	Replace the system board and retest for error codes 103 - 01 through 113 - 01.
103 - 02	DMA byte controller test failed	
103 - 03	DMA word controller test failed	
104 - 01	Interrupt controller master test failed	
104 - 02	Interrupt controller slave test failed	
104 - 03	Interrupt controller software RTC inoperative	

Error Codes	Error Description	Recommended Action
105 - 01	Port 61 bit 6 not at zero	Replace the system board and retest for error codes 103 - 01 through 113 - 01.
105 - 02	Port 61 bit 5 not at zero	
105 - 03	Port 61 bit 3 not at zero	
105 - 04	Port 61 bit 1 not at zero	
105 - 05	Port 61 bit 0 not at zero	
105 - 06	Port 61 bit 5 not at one	
105 - 07	Port 61 bit 3 not at one	
105 - 08	Port 61 bit 1 not at one	
105 - 09	Port 61 bit 0 not at one	
105 - 10	Port 61 I/O test failed	
105 - 11	Port 61 bit 7 not at zero	
105 - 12	Port 61 bit 2 not at zero	
106 - 01	Keyboard controller self test failed	

107 - 01 CMOS RAM test failed

108 - 02 CMOS interrupt test failed

Error Codes	Error Description	Recommended Action
108 - 03	CMOS interrupt test, CMOS not properly initialized	Replace the system board and retest for error codes 103 - 01 through 113 - 01.
109 - 01	CMOS clock load data test failed.	
109 - 02	CMOS clock rollover test failed	
109 - 03	CMOS clock test, CMOS not properly initialized	
110 - 01	Programmable timer load data test failed	
110 - 02	Programmable timer dynamic test failed	
111 - 01	Refresh detect test failed	
112 - 01	Speed test slow mode out of range	
112 - 02	Speed test mixed mode out of range	
112 - 03	Speed test fast mode out of range	
112 - 04	Speed test unable to enter slow mode	
112 - 05	Speed test unable to enter mixed mode	
112 - 06	Speed test unable to enter fast mode	

Error Codes	Error Description	Recommended Action
112 - 07	Speed test system error	Replace the system board and retest for error codes 103 - 01 through 113 - 01.
112 - 08	Speed test unable to enter AUTO mode	
112 - 09	Speed test unable to enter HIGH mode	
112 - 10	Speed test HIGH mode out of range	
112 - 11	Speed test AUTO mode out of range	
113 - 01	Protected mode test failed	

114 - 01	Speaker test failed	1. Verify speaker connection. 2. Replace the speaker and retest. 3. Replace the system board and retest.

Table 5-3. Memory Test Error Codes

Error Codes	Error Description	Recommended Action
201 - 01	Memory machine ID test failed	The following steps apply to error codes 201 - 01 through 202 - 03: 1. Replace VDU controller board and retest. 2. Replace the system ROM and retest. 3. Replace the system board and retest.
201 - 02	Memory machine type test failed	
201 - 03	Memory machine ID and type tests failed	
202 - 01	Memory system ROM checksum failed	
202 - 02	Failed RAM/ROM map test	
202 - 03	Failed RAM/ROM protect test	
203 - 01	Memory write/read test failed	The following steps apply to error codes 203 -01 through 211 - 03: 1. Replace the defective memory chip (or option memory if installed, error code 208) if an error code is accompanied by an XX000 BYYZZ code and retest. 2. Replace the system memory board or the system board.
203 - 02	Error during saving program memory write/read test	
203 - 03	Error during restore of program memory in write/read test	
204 - 01	Memory address test failed	
204 - 02	Error during saving program memory in address test	
204 - 03	Error during restore of program memory in address test	
204 - 04	A20 address test failed	
205 - 01	Walking I/O test failed	
205 - 02	Error during saving program memory in walking I/O test	
205 - 03	Error during restore of program memory in walking I/O test	
205 - 04	Insufficient memory to perform test	

Error Codes	Error Description	Recommended Action
208 - 01	Memory refresh test failed	The following steps apply

		to error codes 203 - 01 through 211 - 03:
208 - 02	Error during saving program memory in refresh test	1. Replace the defective memory chip (or option memory if installed, error code 208) if an error code is accompanied by an XX000 BYYZZ code and retest. 2. Replace the system memory board or the system board.
208 - 03	Error during restore program memory in refresh test	
210 - xx	Incrementing pattern test failed (expanded memory test)	
210 - 01	Data read from block did not match data written	
210 - 02	Scan of test block prior to testing	
210 - 03	Error while restoring block under test	
210 - 05	Inverted pattern compare, but RAM was correct	
210 - 06	Error while restoring block under test	
210 - 07	Main memory update failure	
211 - xx	Random data pattern test	
211 - 01	Memory random pattern test failed	
211 - 02	Error during saving program memory and random pattern in write/read test	
211 - 03	Error during restore of program memory and random pattern in write/read test	

Table 5-4. Keyboard Test Error Code

Error Codes	Error Description	Recommended Action
301 - 01	Keyboard short test, 8042 self test failed	The following steps apply to error codes 301 - 01 through 304 - 06: 1. Check the keyboard cable connector. If disconnected, turn off computer and connect keyboard. 2. Check internal keyboard connections to system board. 3. Replace the keyboard and retest. 4. Replace the system board and retest.
301 - 02	Keyboard short test. interface test failed	
301 - 03	Keyboard short test. echo test failed	
301 - 04	Keyboard short test failed	
302 - 01	Keyboard long test failed	
303 - 01	Keyboard LED test, 8042 self test failed	

Error Codes	Error Description	Recommended Action
303 - 02	Keyboard LED test, reset test failed	
303 - 03	Keyboard LED test, reset test failed	
303 - 04	Keyboard LED test, LED command test failed	
303 - 05	Keyboard LED test, LED command test failed	
303 - 06	Keyboard LED test, LED command test failed	
303 - 07	Keyboard LED test, LED command test failed	
303 - 08	Keyboard LED test, command byte restore test failed	The following steps apply to error codes 301 - 01 through 304 - 06:
303 - 09	Keyboard LED test, LEDs failed to light	1. Check the keyboard cable connector. If disconnected, turn off computer and connect keyboard.
304 - 01	Keyboard typematic test failed	2. Check internal keyboard connections to system board.
304 - 02	Unable to enter mode 3	3. Replace the keyboard and retest.
304 - 03	Incorrect scan code from keyboard	4. Replace the system board and retest.
304 - 04	No keyboard make code observed	
304 - 05	Unable to disable typematic feature	
304 - 06	Unable to return to normal mode	

Table 5-5. Parallel Printer Interface Test Error Codes

Error Codes	Error Description	Recommended Action
401 - 01	Printer failed or not connected	The following steps apply to error codes 401 - 01 through 498 - 00:
402 - 01	Printer data register failed	
402 - 02	Printer control register failed	1. If a printer is connected, be sure it is turned on and in the online mode.
402 - 03	Printer data and control register failed	2. Check the switch settings on the multipurpose fixed disk drive controller board or the multipurpose controller board (see Chapter 8, "Jumper
402 - 04	Printer loopback failed	
402 - 05	Printer loopback and data failed	
402 - 06	Printer loopback and control registers failed	

		Positions and Switch Settings and Adjustments").
402 - 07	Printer loopback, data, and control registers failed	3. Replace the printer and/or the printer cable and retest.
402 - 08	Printer interrupt test failed	4. Replace the multipurpose fixed disk drive controller board or the multipurpose controller board and retest.
402 - 09	Printer interrupt and data register failed	5. Replace the system board and retest.
402 - 10	Printer interrupt and control register failed	
402 - 11	Printer interrupt, data, and control register failed	
402 - 12	Printer interrupt and loopback failed	
402 - 13	Printer interrupt, loopback and data register failed	
402 - 14	Printer interrupt, loopback and control register failed	
402 - 15	Printer interrupt, loopback, data, and control register failed	
402 - 16	Printer unexpected interrupt received	
403 - 01	Printer pattern test failed	
498 - 00	Printer failed or not connected	

Table 5-6. Video Display Unit Test Error Codes

Error Codes	Error Description	Recommended Action
501 - 01	Video controller test failed	The following steps apply to error codes 501 - 01 through 516 - 01:
502 - 01	Video memory test failed	
503 - 01	Video attribute test failed	1. Replace the video display controller board and retest.
504 - 01	Video character set test failed	2. Replace the monitor and retest.
505 - 01	Video 80 x 25 mode 9 x 14 character cell test failed	3. Check the VDU fuse on the system board and retest.
506 - 01	Video 80 x 25 mode 8 x 8 character cell test failed	4. Replace the system board and retest.
507 - 01	Video 40 x 25 mode test failed	
508 - 01	Video 320 x 200 mode color set 0 test failed	

509 - 01	Video 320 x 200 mode color set 1 test failed	
510 - 01	Video 640 x 200 mode test failed	
511 - 01	Video screen memory page test failed	
512 - 01	Video gray scale test failed	
514 - 01	Video white screen test failed	
516 - 01	Video noise pattern test failed	
517 - 01	Lightpen text mode test failed - no response	The following steps apply to error codes 517 - 01 through 517 - 04:
517 - 02	Lightpen text mode test failed - invalid response	1. Replace the lightpen and retest.
517 - 03	Lightpen medium resolution mode test failed - no response	2. Replace the display controller board and retest.
517 - 04	Lightpen medium resolution modes test failed - invalid response	3. Replace the system board and retest.

Table 5-7. Diskette Drive Test Error Codes

Error Codes	Error Description	Recommended Action
600 - xx	Diskette ID drive types test failed	The following steps apply to error codes 600 - xx through 614 - xx:
601 - xx	Diskette format failed	
602 - xx	Diskette read test failed	NOTE: If the COMPAQ Enhanced Color Graphics Board is installed, check the interrupt setting.
603 - xx	Diskette write/read/compare test failed	
604 - xx	Diskette random seek test failed	1. Replace the diskette and retest.
605 - xx	Diskette ID media failed	2. Check the switch and jumper settings on the multipurpose fixed disk drive controller board or the multipurpose controller board.
606 - xx	Diskette speed test failed	
607 - xx	Diskette wrap test failed	
608 - xx	Diskette write protect test failed	3. Replace the diskette power and signal cables and retest.
609 - xx	Diskette reset controller failed	4. Replace the multipurpose fixed disk drive controller board or multipurpose controller board and retest.
610 - xx	Diskette change line test failed	
614 - xx	Diskette drive file write test	
614 - 01	Exceeded maximum soft error limit	5. Replace the diskette

Error Codes	Error Description	Recommended Action
		drive and retest.
614 - 02	Exceeded maximum hard error limit	6. Replace the system board and retest.
614 - 03	Previously exceeded maximum soft error limit	
614 - 04	Previously exceeded maximum hard error limit	

Error Codes	Error Description	Recommended Action
614 - 05	Failed to reset controller	The following steps apply to error codes 600 - xx through 614 - xx:
614 - 06	Fatal error while reading	
614 - 07	Fatal error while writing	NOTE: If the COMPAQ Enhanced Color Graphics
614 - 08	Failed compare of write/read buffers	Board is installed, check the interrupt setting.
614 - 09	Failed to format a track	1. Replace the diskette and retest.
614 - 10	Failed sector wrap test	2. Check the switch and jumper settings on the multipurpose fixed disk drive controller board or the multipurpose controller board.
614 - 20	Failed to get drive type	3. Replace the diskette power and signal cables and retest.
614 - 21	Failed to get change line status	4. Replace the multipurpose fixed disk drive controller board or multipurpose controller board and retest.
614 - 22	Failed to clear change line status	5. Replace the diskette drive and retest.
614 - 23	Failed to set drive type in ID media	6. Replace the system board and retest.
614 - 24	Failed to read diskette media	
614 - 25	Failed to verify diskette media	
614 - 26	Failed to read media in speed test	
614 - 27	Failed speed limits	
614 - 28	Failed write protect test	
614 - 29	Failed to read media in file write test	
614 - 40	Cylinder 0 error	

Error Codes	Error Description	Recommended Action
694 - 00	Pin 34 is not cut on 360 KB diskette drive	See Service Bulletin 25, Modification of 360 Kbyte Diskette Drives.
697 - 00	Diskette type error	1. Replace the diskette and retest.
698 - 00	Diskette drive speed not within limits	2. Check the switch and jumper settings on the multipurpose fixed disk drive controller board

		or the multipurpose controller board. 3. Replace the diskette power and signal cables and retest. 4. Replace the multipurpose fixed disk drive controller board or multipurpose controller board and retest. 5. Replace the diskette drive and retest 6. Replace the system board and retest.
699 - 00	Diskette drive/media ID error	Rerun the SETUP program.

Table 5-8. Serial Interface Test Error Codes

Error Codes	Error Description	Recommended Action
1101 - 01	Serial port test, UART DLAB bit failure	The following steps apply to error codes 101 - 01 through 1101 - 14; 1. Check jumper and switch settings on the serial/parallel interface board, multipurpose fixed disk drive controller board, or the multipurpose controller board and retest. 2. Replace the serial/parallel interface board, multipurpose fixed disk drive controller board or multipurpose controller board and retest.
1101 - 02	Serial port test, line input or UART fault	
1101 - 03	Serial port test, address line fault	
1101 - 04	Serial port test, data line fault	
1101 - 05	Serial port test, UART control signal failure	
1101 - 06	Serial port test, UART THRE bit failure	
1101 - 07	Serial port test, UART DATA READY bit failure	
1101 - 08	Serial port test, UART TX/RX buffer failure	
1101 - 09	Serial port test, INTERRUPT circuit failure	
1101 - 10	Serial port test, COM1 set to invalid interrupt	
1101 - 11	Serial port test, COM2 set to invalid interrupt	
1101 - 12	Serial port test, DRIVER/RECEIVER control signal failure	
1101 - 13	Serial port test, UART control signal interrupt failure	

1101 - 14	Serial port test, DRIVER/RECEIVER data failure	
1102 - 58	Serial Port COM1 set to INTR 3	
1102 - 60	Serial Port COM2 set to INTR 4	
1102 - 64	Serial Port data compare error	
1102 - 72	Serial Port no TX RDY	
1102 - 88	Serial Port no RX RDY	

Table 5-9. Modem Communications Test Error Codes

Error Codes	Error Description	Recommended Action
1201 - xx	Modem Internal Loopback test	The following apply to error codes 120l - xx through 1210 - xx: 1. Refer to modem documentation for correct setup procedures. 2. Check the modem line. 3. Replace the modem and retest.
1201 - 01	UART DLAB bit failure	
1201 - 02	Line input or UART failure	
1201 - 03	Address line fault	
1201 - 04	Data line fault	
1201 - 05	UART control signal failure	
1201 - 07	UART DATA READY bit failure	
1201 - 08	UART TX/RX buffer failure	
1201 - 09	INTERRUPT circuit failure	
1201 - 10	COM1 set to invalid interrupt	
1201 - 11	COM2 set to invalid interrupt	
1201 - 12	DRIVER/RECEIVER control signal failure	
1201 - 13	UART control signal interrupt failure	
1201 - 14	DRIVER/RECEIVER data failure	

Error Codes	Error Description	Recommended Action
1201 - 15	Modem detection failure	The following apply to error codes 120l - xx

Error Codes	Error Description	Recommended Action
1201 - 16	Modem ROM, checksum failure	through 1210 - xx:
1201 - 17	Tone detection failure	1. Refer to modem documentation for correct setup procedures.
1202 - xx	Modem Internal Loopback test	2. Check the modem line.
1202 - 01	Timed out waiting for SYNC (local loopback mode)	3. Replace the modem and retest.
1202 - 02	Timed out waiting for modem response (local loopback mode)	
1202 - 03	Exceeded data block retry limit (local loopback mode)	
1202 - 11	Timed out waiting for SYNC (analog loopback originate mode)	
1202 - 12	Timed out waiting for modem response (analog loopback originate mode)	
1202 - 13	Exceeded data block retry limit (analog loopback originate mode)	
1202 - 21	Timed out waiting for SYNC (analog loopback answer mode)	
1202 - 22	Timed out waiting for modem response (analog loopback answer mode)	
1202 - 23	Exceeded data block retry limit (analog loopback answer mode)	

Error Codes	Error Description	Recommended Action
1203 - xx	Modem External Termination test	The following apply to error codes 1201 - xx
1203 - 01	Modem external TIP/RING failure	through 1210 - xx:
1203 - 02	Modem external DATA TIP/RING failure	1. Refer to modem documentation for correct setup procedures.
1203 - 03	Modem line termination failure	2. Check the modem line.
1204 - xx	Modem auto originate test	3. Replace the modem and retest.
1205 - xx	Modem auto answer test	
1206 - xx	Dial multifrequency tone test	
1210 - xx	Modem direct connect test	
1210 - 01	Modem timed out waiting for SYNC	
1210 - 02	Modem timed out waiting for response	
1210 - 03	Modem exceeded data block retry limit	
1210 - 04	RCV exceeded carrier lost limit	

Error Codes	Error Description	Recommended Action
1210 - 05	XMIT exceeded carrier lost limit	
1210 - 06	Timed out waiting for dial tone	
1210 - 07	Dial number string too long	
1210 - 08	Modem timed out waiting for remote response	The following apply to error codes 1201 - xx through 1210 - xx: 1. Refer to modem documentation for correct setup procedures. 2. Check the modem line. 3. Replace the modem and retest.
1210 - 09	Modem exceeded maximum redial limit	
1210 - 10	Line quality prevented remote connection	
1210 - 11	Modem timed out waiting for remote connection	
1210 - 12	Modem timed out waiting for modem response (analog loopback originate mode)	
1210 - 13	Modem exceeded data block retry limit (analog loopback originate mode)	
1210 - 14	DRIVER/RECEIVER data failure	
1210 - 15	Modem detection failure	
1210 - 16	Modem ROM, checksum failed	
1210 - 17	Tone detection failure	
1210 - 21	Modem timed out waiting SYNC (analog loopback answer mode)	
1210 - 23	Modem exceeded data block for retry limit (analog loopback answer mode)	

Table 5-10. Fixed Disk Drive Test Error Codes

Error Codes	Error Description	Recommended Action
1700 - xx	Fixed disk ID drive types test failed	The following steps apply to error codes 1700 - xx through 1716 - xx. 1. Check switch and jumper settings on the fixed disk drive controller board, multipurpose fixed disk drive
1701 - xx	Fixed disk drive format test failed	
1702 - xx	Fixed disk drive read test failed	
1703 - xx	Fixed disk drive write/read/compare test failed	

1704 - xx	Fixed disk drive random seek test failed	controller board, or multipurpose controller board and retest. 2. Replace the fixed disk drive signal and power cables and retest. 3. Replace the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller controller board and retest. 4. Replace the fixed disk drive and retest. 5. Replace the system board and retest.
1705 - xx	Fixed disk drive controller test failed	
1706 - xx	Fixed disk drive ready test failed	
1707 - xx	Fixed disk drive recalibrate test failed	
1708 - xx	Fixed disk drive format bad track test failed	
1709 - xx	Fixed disk drive reset controller test failed	
1710 - xx	Fixed disk drive park head test failed	
1714 - xx	Fixed disk drive file write test failed	
1715 - xx	Fixed disk drive head select test failed	
1716 - xx	Fixed disk drive conditional format test failed	

Error Codes	Error Description	Recommended Action
17yy - xx	Invalid fixed disk drive type failed	The following steps apply to error codes 1700 - xx through 1716 - xx. 1. Check switch and jumper settings on the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller board and retest. 2. Replace the fixed disk drive signal and power cables and retest. 3. Replace the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller controller board and retest. 4. Replace the fixed disk drive and retest. 5. Replace the system board and retest.
17yy - 01	Exceeded maximum soft error limit	
17yy - 02	Exceeded maximum hard error limit	
17yy - 03	Previously exceeded maximum soft error limit	
17yy - 04	Previously exceeded maximum hard error limit	
17yy - 05	Failed to reset controller	
17yy - 06	Fatal error while reading	
17yy - 07	Fatal error while writing	
17yy - 08	Failed compare of write/read/compare	
17yy - 09	Failed to format a track	
17yy - 10	Failed sector wrap test	
17yy - 19	Controller failed to deallocate bad sector	
17yy - 40	Failed cylinder 0	

17yy - 41 Drive not ready

17yy - 42 Recalibration failed

Error Codes	Error Description	Recommended Action
17yy - 43	Failed to format bad track	The following steps apply to error codes 1700 - xx through 1716 - xx: 1. Check switch and jumper settings on the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller board and retest. 2. Replace the fixed disk drive signal and power cables and retest. 3. Replace the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller board and retest. 4. Replace the fixed disk drive and retest. 5. Replace the system board and retest.
17yy - 44	Failed fixed disk drive controller board diagnostics	
17yy - 45	Failed to get drive parameters from ROM	
17yy - 46	Invalid drive parameters found in ROM	
17yy - 47	Failed to park heads	
17yy - 48	Failed to move disk table to RAM	
17yy - 49	Failed to read media in file write test	
17yy - 50	Failed file I/O write test	
17yy - 51	Failed file I/O read test	
17yy - 52	Failed file I/O compare test	

17yy - 53 Failed drive/head register test

17yy - 54 Failed digital input register test

Error Codes	Error Description	Recommended Action
17yy - 55	Failed cylinder 1 test	The following steps apply to error codes 1700 - xx through 1716 - xx: 1. Check switch and jumper settings on the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller board and retest. 2. Replace the fixed disk drive signal and power
17yy - 56	Fixed disk drive controller board RAM diagnostics failed	
17yy - 57	Fixed disk drive controller board to drive test failed	
17yy - 58	Failed to write sector buffer	
17yy - 59	Failed to read sector buffer	
17yy - 60	Failed to compare sector buffer	

17yy - 63	Failed soft error rate	cables and retest. 3. Replace the fixed disk drive controller board, multipurpose fixed disk drive controller board, or multipurpose controller board and retest. 4. Replace the fixed disk drive and retest. 5. Replace the system board and retest.
17yy - 65	Exceeded maximum bad sector per track	
17yy - 66	Failed initial drive parameter	
17yy - 67	Failed to write long	
17yy - 68	Failed to read long	
17yy - 69	Failed to read drive size from controller	
17yy - 70	Failed translate mode	
17yy - 71	Failed nontranslated mode	
17yy - 72	Bad track limit exceeded	
17yy - 73	Previously exceeded bad track limit	

Table 5-11. Enhanced Color Graphics Test Error Codes

Error Codes	Error Description	Recommended Action
2402 - 01	Video memory test failed	The following steps apply to error codes 2402 - 01 through 2416 - 01: 1. Verify the switch settings. 2. Replace the COMPAQ Enhanced Color Graphics Board ROM and retest. 3. Replace the COMPAQ Enhanced Color Graphics Board and retest. 4. Replace the system board and retest.
2403 - 01	Video attribute test failed	
2404 - 01	Video character set test failed	
2405 - 01	Video 80 x 25 mode 9 x 14 character cell test failed	
2406 - 01	Video 80 x 25 mode 8 x 8 character cell test failed	
2407 - 01	Video 40 x 25 mode test failed	
2408 - 01	Video 320 x 200 mode color set 0 test failed	
2409 - 01	Video 320 x 200 mode color set 1 test failed	
2410 - 01	Video 640 x 200 mode test failed	
2411 - 01	Video screen memory page test failed	
2412 - 01	Video gray scale test failed	
2414 - 01	Video white screen test failed	
2416 - 01	Video noise pattern test failed	

Error Codes	Error Description	Recommended Action
2417 - 01	Lightpen text mode failed - no response	The following steps apply to error codes 2417 - 01 through 2417 - 04:
2417 - 02	Lightpen text mode test failed - invalid response	1. Replace the lightpen and retest.
2417 - 03	Lightpen medium resolution mode test failed - no response	2. Replace the COMPAQ Enhanced Color Graphics Board ROM and retest.
2417 - 04	Lightpen medium resolution mode test failed - invalid response	3. Replace the COMPAQ Enhanced Color Graphics Board and retest.
		4. Replace the system board and retest.
2418 - 01	ECG memory test failed	The following steps apply to error codes 2418 - 01 through 2425 - 01:
2418 - 02	ECG shadow RAM test failed	
2419 - 01	ECG ROM checksum test failed	1. Verify switch settings.
2420 - 01	ECG attribute test failed	2. Replace the COMPAQ Enhanced Color Graphics Board ROM.
2421 - 01	ECG 640 x 200 graphics mode test failed	3. Replace the COMPAQ Enhanced Color Graphics Board and retest.
2422 - 01	ECG 640 x 350 16 Color Set test failed	4. Replace the system board and retest.
2423 - 01	ECG 640 x 350 64 Color Set test failed	
2424 - 01	ECG Monochrome Text Mode test failed	
2425 - 01	ECG Monochrome Graphics Mode test failed	

Table 5-12. Pointing Device Interface Test Error Codes

Error Codes	Error Description	Recommended Action
8601 - xx	Pointing device test	1. Verify switch settings on Multipurpose Controller Board, Multipurpose Fixed Disk Drive Controller Board, or Serial/Parallel Interface Board.
8601 - 02	Left button is inoperative	
8601 - 03	Left button is stuck closed	
8601 - 04	Right button is inoperative	
8601 - 05	Right button is stuck closed	2. Replace the pointing device with a working pointing device and retest.
8601 - 06	Left block not selected	

		3. Replace the Multipurpose Controller Board, Multipurpose Fixed Disk Drive Controller Board, or Serial/Parallel Interface Board and retest.
8601 - 07	Right block not selected	
8601 - 08	Timeout occurred	
8601 - 10	Pointing device inoperative	
		4. Replace the system board and retest.

Chapter 5.3 Memory Related Error Codes

Memory error codes result when the system detects a memory fault during the Power On Self Test or diagnostic test. The Test programs attempt to isolate the memory fault to a specific chip, then generate a memory error code that identifies the defective memory chip.

In some cases, replacing the memory chip will not solve the problem because:

- The system may not be able to accurately determine which chip or chips are at fault.
- The problem may be caused by to a failure in the memory support circuitry, not the memory.

The memory error code points to a specific memory address. The physical location of the memory address depends on the type of system, the number and type of memory boards installed, and the type of memory device used (64K x 1 Bit RAMs or 256K x 1 Bit RAMs).

Tables 5-14 through 5-17 identify which memory device to replace when the system shows a memory error. Table 5-13 indicates the table to reference to locate the defective chip.

Table 5-13. Memory Configurations

System Board RAM Type	Error Code XX Value	Memory Device Configuration	Tables
64K	00 to 02 (0 - 256 Kbyte)	System Board	5-14, 5-15
	10 to 26 (1 - 2.5 megabyte)	System Memory Board (512/1536 Kbyte)	5-17
256K	00 to 08 (0 - 640 Kbyte)	System Board	5-16

NOTE: XX codes 0A to 0F represent the memory space for the video memory and system ROMs. Errors in the video memory or system ROM space are not reported in the XX000B YYZZ error message format.

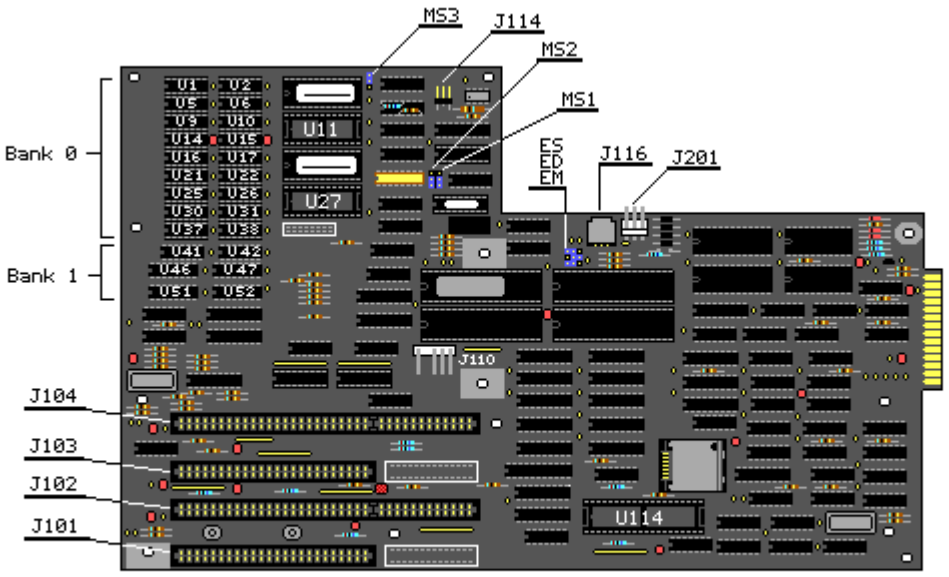


Figure 5-1. Memory Chip Locations on the COMPAQ PORTABLE II Personal Computer System Board Version 1 (Assy No. 000318)

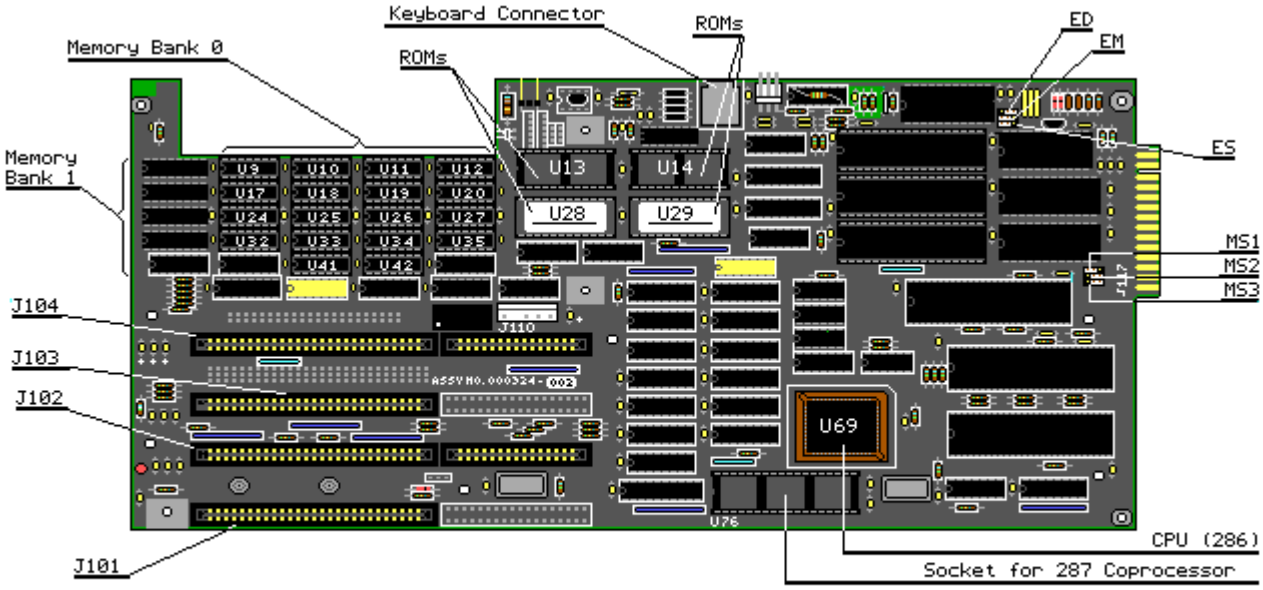


Figure 5-2. Memory Chip Locations on the COMPAQ PORTABLE II Personal Computer System Board Version 2 (Assy No. 000324)

Defective Memory Chip Isolation Map for System Boards using 64K RAMs and 256K RAMs

Memory error codes are in the format XX000B YYZZ.

XX is the memory segment of detected error.

000 is always equal to 000 (not used)

B Byte bank location (Low: B = 0, High: B = 1)

YY Bit location of the low byte (B = 0)

ZZ Bit location of the high byte (B =1)

For example, for the Version 1 system board, memory error message "02000 10010" specifies chip U47.

Use Tables 5-14 through 5-17 to locate the defective chip and determine the required action.

NOTE: If XX or YY is not documented in the following tables, this may mean that more than one area contributed to the error condition. If this is the case, the system adds the values of the failed data bits and displays the sum in the XX or YY location.

Figures 5-1 and 5-2 show the memory chip locations for the COMPAQ PORTABLE II Personal Computer System Boards - Version 1 and 2. For information on removing and replacing memory chips contained on the system board, refer to Chapter 7 in this guide.

Table 5-14. Version 1 System Board using 64K RAMs (0 - 256 Kbytes)

XX = 00 (Bank 0)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U37	U38	Replace memory chip
01	U30	U31	Replace memory chip
02	U25	U26	Replace memory chip
04	U21	U22	Replace memory chip
08	U16	U17	Replace memory chip
10	U14	U15	Replace memory chip
20	U9	U10	Replace memory chip
40	U5	U6	Replace memory chip
80	U1	U2	Replace memory chip

XX = 02 (Bank 1)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U41	U42	Replace system board
01	U51	U52	Replace system board
02	U51	U52	Replace system board
04	U51	U52	Replace system board
08	U51	U52	Replace system board

10	U46	U47	Replace system board
20	U46	U47	Replace system board
40	U46	U47	Replace system board
80	U46	U47	Replace system board

Table 5-15. Version 2 System Board using 64K RAMs (0 - 256 Kbytes)

XX = 00 (Bank 0)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U32	U35	Replace memory chip
01	U24	U27	Replace memory chip
02	U17	U20	Replace memory chip
04	U9	U12	Replace memory chip
08	U41	U42	Replace memory chip
10	U33	U34	Replace memory chip
20	U25	U26	Replace memory chip
40	U18	U19	Replace memory chip
80	U10	U11	Replace memory chip

XX = 02 (Bank 1)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U40	U43	Replace system board
01	U31	U16	Replace system board
02	U31	U16	Replace system board
04	U31	U16	Replace system board
08	U31	U16	Replace system board
10	U23	U8	Replace system board
20	U23	U8	Replace system board
40	U23	U8	Replace system board
80	U23	U8	Replace system board

Table 5-16. Version 1 System Board using 256K RAMs (0 - 640 Kbytes)

XX = 00, 02, 04, 06 (Bank 0)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U37	U38	Replace memory chip
01	U30	U31	Replace memory chip
02	U25	U26	Replace memory chip
04	U21	U22	Replace memory chip
08	U16	U17	Replace memory chip
10	U14	U15	Replace memory chip
20	U9	U10	Replace memory chip
40	U5	U6	Replace memory chip
80	U1	U2	Replace memory chip

XX = 08 (Bank 1)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U41	U42	Replace system board
01	U51	U52	Replace system board
02	U51	U52	Replace system board
04	U51	U52	Replace system board
08	U51	U52	Replace system board
10	U46	U47	Replace system board
20	U46	U47	Replace system board
40	U46	U47	Replace system board
80	U46	U47	Replace system board

Table 5-17. Version 2 System Board using 256K RAMs (0 - 640 Kbytes)

XX = 00, 02, 04, 06 (Bank 0)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U32	U35	Replace memory chip
01	U24	U27	Replace memory chip
02	U17	U20	Replace memory chip

04	U9	U12	Replace memory chip
08	U41	U42	Replace memory chip
10	U33	U34	Replace memory chip
20	U25	U26	Replace memory chip
40	U18	U19	Replace memory chip
80	U10	U11	Replace memory chip

XX = 08 (Bank 1)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U40	U43	Replace system board
01	U31	U16	Replace system board
02	U31	U16	Replace system board
04	U31	U16	Replace system board
08	U31	U16	Replace system board
10	U23	U8	Replace system board
20	U23	U8	Replace system board
40	U23	U8	Replace system board
80	U23	U8	Replace system board

Defective Memory Chip Isolation Map for the 512/1536 Kbyte System Memory Board Mapped into the 1 to 2.5 Megabyte Memory Space (256K x 1 Bit RAMs)

NOTE: This isolation map is valid for a system memory board that uses 256K x 1 bit RAMs.

Memory error codes are in the format XX000B YYZZ.

XX equals the 128 Kbyte memory segment where an error is detected.

000 is always equal to 000 (not used).

B Byte bank location (Low: B = 0, High: B = 1)

YY Bit location of the low byte (B = 0)

ZZ Bit location of the high byte (B = 1)

Use the following table to locate the defective memory chip and the required action. Figure 5-3 shows the memory chip locations for the COMPAQ PORTABLE II Personal Computer System Memory Board (512/1536 Kbyte). For information on removing and replacing memory chips contained on this board,

refer to Chapter 7.

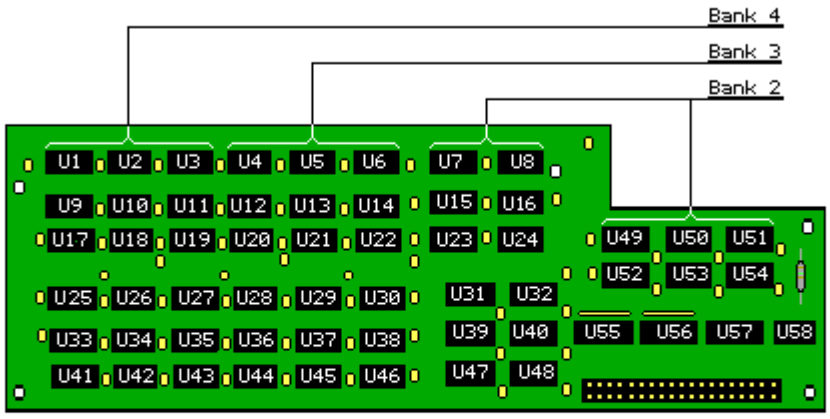


Figure 5-3. Memory Locations on the 512/1536 Kbyte Memory Board (Assy No. 000330)

Table 5-18. Memory Locations/Required Action for 512/1536 Kbyte System Memory Board Mapped to the 1 to 2.5 Megabyte Memory Space (256K x 1 Bit RAMs)

XX = 10, 12, 14, 16 (Bank 2)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U47	U23	Replace system memory board
01	U31	U7	Replace system memory board
02	U32	U8	Replace system memory board
04	U54	U51	Replace system memory board
08	U39	U15	Replace system memory board
10	U40	U16	Replace system memory board
20	U53	U50	Replace system memory board
40	U48	U24	Replace system memory board
80	U52	U49	Replace system memory board

XX = 18, 1A, 1C, 1E, (Bank 3)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U44	U20	Replace memory chip

01	U28	U4	Replace memory chip
02	U29	U5	Replace memory chip
04	U30	U6	Replace memory chip
08	U36	U12	Replace memory chip
10	U37	U13	Replace memory chip
20	U38	U14	Replace memory chip
40	U45	U21	Replace memory chip
80	U46	U22	Replace memory chip

XX = 20, 22, 24, 26 (Bank 4)

Data Bit YY or ZZ	B = 0 Low	B = 1 High	Required Action
00	U41	U17	Replace memory chip
01	U25	U1	Replace memory chip
02	U26	U2	Replace memory chip
04	U27	U3	Replace memory chip
08	U33	U9	Replace memory chip
10	U34	U10	Replace memory chip
20	U35	U11	Replace memory chip
40	U42	U18	Replace memory chip
80	U43	U19	Replace memory chip

Chapter 6. Illustrated Parts Catalog

6.0 Introduction

This section provides a spare parts reference for the COMPAQ PORTABLE II Personal Computer.

Each table in this section contains the following information:

- Item - reference number keyed to an illustration.
- Part Number - to be used when ordering parts from Compaq Computer Corporation.
- Description - name of the particular part or parts kit.

NOTE: When several parts have the same part number, the parts are available only as part of a kit.

Chapter 6.1 Compaq Portable II Personal Computer

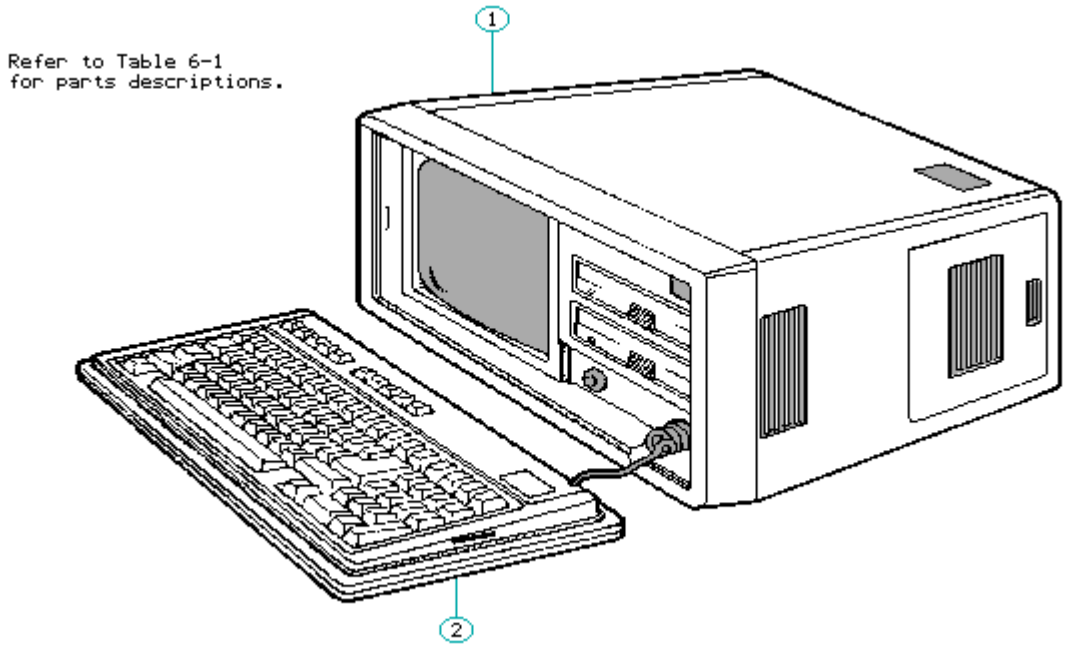


Figure 6-1. COMPAQ PORTABLE II Personal Computer

Table 6-1. COMPAQ PORTABLE II Personal Computer

Item	Part Number	Description
1	107899-001 **	Chassis Cover
2	104179-001 **	Keyboard Assembly

	107678-001	Spanish Enhanced Keyboard Assembly

** Part number no longer available.

Chapter 6.2 Chassis

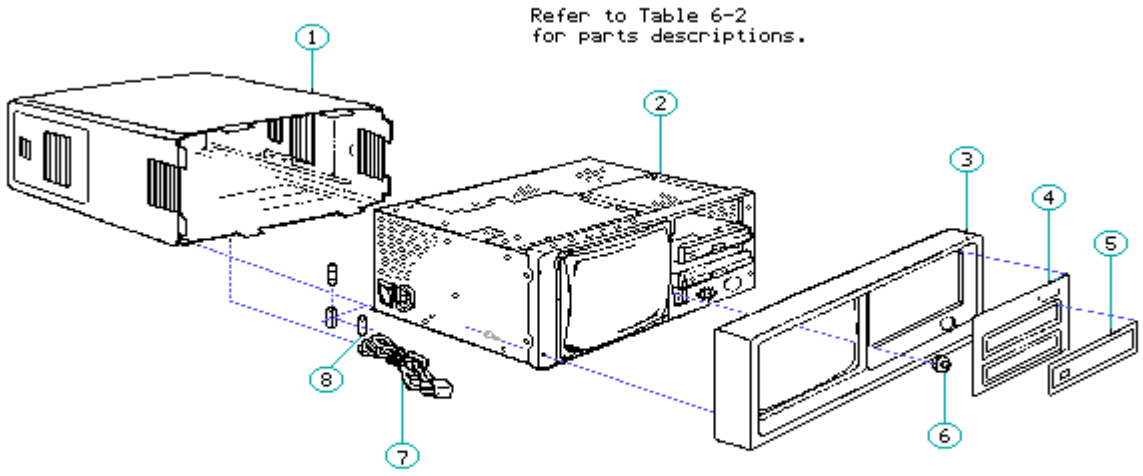


Figure 6-2. COMPAQ PORTABLE II Personal Computer Chassis

Table 6-2. COMPAQ PORTABLE II Personal Computer Chassis

Item	Part Number	Description
1	107899-001 **	Chassis Cover
2	104081-001 **	Chassis Assembly
3	104198-001 **	Front Bezel
4	104059-001 **	Drive Bezel
5	104050-001	Blank Drive Bezel (third height)
6	*	Brightness Knob Assembly
7	101155-001	AC Power Cord
8	102930-001	3-A AC Line Fuse (Quantity = 40)
	100228-002 **	3-A AC Line Fuse (Quantity = 1)

* Contained in the Miscellaneous Hardware Kit (Spare PN 104270-001 **). See Table 6-7.

** Part number no longer available.

Chapter 6.3 Common Chassis

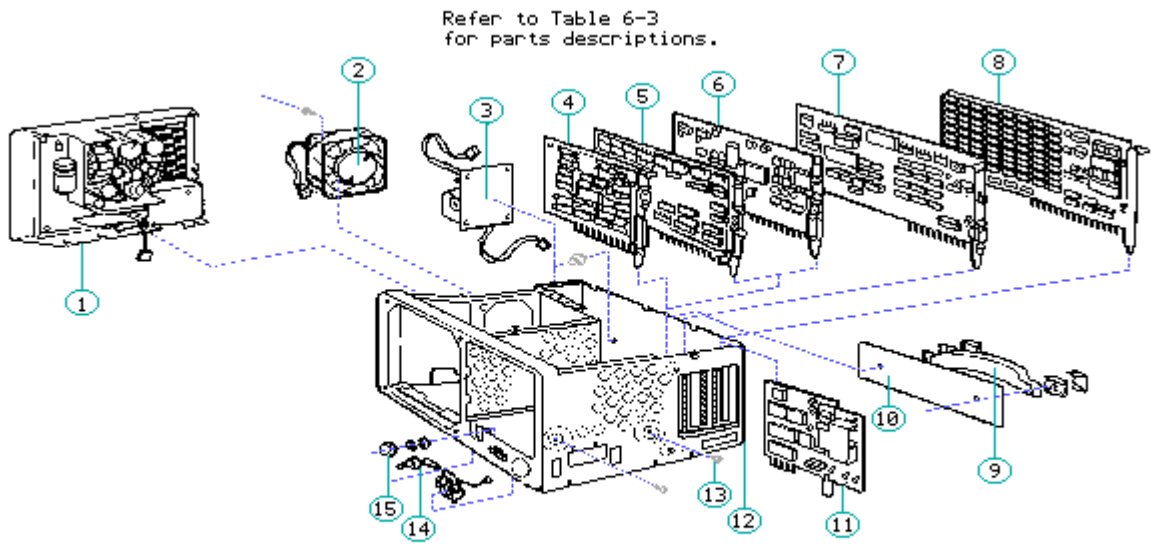


Figure 6-3. COMPAQ PORTABLE II Personal Computer Common Chassis

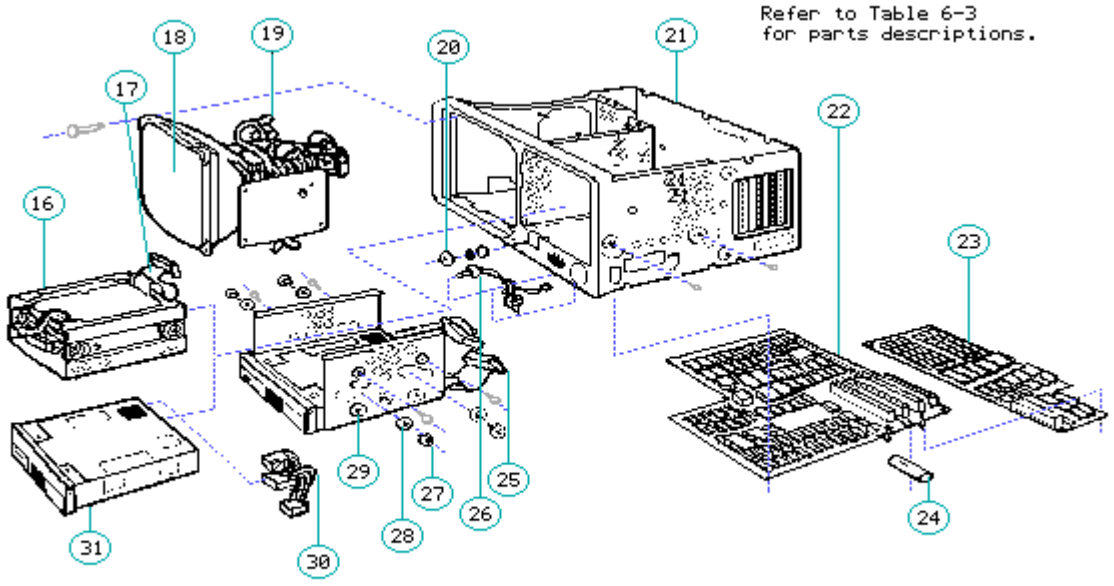


Figure 6-4. COMPAQ PORTABLE II Personal Computer Common Chassis

Table 6-3. COMPAQ PORTABLE II Personal Computer Common Chassis

Item	Part Number	Description

Item	Part Number	Description
1	104180-001 **	Power Supply Chassis
2	104178-001 **	Fan Assembly
3	104273-001 **	Automatic Power Switching (APS) Board
4	106886-001	Serial/Parallel Interface Board
5	104174-001	Multipurpose Fixed Disk Drive Controller Board
	102705-001 **	Multipurpose Controller Board
6	109196-001 **	Enhanced Color Graphics Board
	106686-001	Enhanced Color Graphics ROM
7	101340-001 **	Video Display Controller Board
8	105033-001 **	512/2048 Kbyte Memory Expansion Board
9	100025-001 **	Handle
10	104067-001 **	Handle Underlay
11	112693-001 (replaced by 197458-001)	2400 Baud Internal Modem
12	104081-001 **	Chassis Assembly
13	*	Torx 6-32 x 3/8 Screw
14	104185-001 **	Speaker/Brightness Potentiometer Assembly
15	*	Brightness Knob

* Contained in the Miscellaneous Hardware Kit (Spare PN 104270-001 **). See Table 6-7.

** Part number no longer available.

Item	Part Number	Description
16	104183-001 **	10 Megabyte Fixed Disk Drive
	104405-001 ***	20 Megabyte Fixed Disk Drive
17	104093-001 □ **	Fixed Disk Drive Signal Cable
18	104182-001 **	Monitor Assembly
19	104007-001 □ **	Video Display Signal Cable
20	*	Brightness Knob
21	104081-001 **	Chassis Assembly
22	104175-001 **	System Board (Assy No. 000318)
	104444-001 **	System Board (Assy No. 000324)

23	104176-001 **		512/1536 Kbyte System Memory Board
24	104186-001 **		Battery
25	104008-002 □ **		Diskette Drive Signal Cable
26	104185-001 **		Speaker/Brightness Potentiometer Assembly
27	*		Drive Shoulder Bolt
28	*		Vibration Isolators
29	104048-001 **		Storage Device Enclosure

* Contained in the Miscellaneous Hardware Kit (Spare PN 104270-001 **). See Table 6-7.

□ Contained in the Cable Kit, Spare PN 104191-001 **. See Table 6-6.

** Part number no longer available.

*** Part number 104405-001 was replaced by 142365-001.

Item	Part Number		Description
30	104092-002 □ **		Mass Storage Power Cable (Fixed Disk Drive)
	104006-001 □ **		Mass Storage Power Cable (Diskette Drive)
	101741-001		Mass Storage Power Cable (Diskette Drive C, 6 Inch)
31	104184-001 **		360 Kbyte (flip door) Diskette Drive %
	107360-001		360 Kbyte (push button) Diskette Drive
	104479-001 **		1.2 Megabyte (flip door) Diskette Drive
	107359-001		1.2 Megabyte (push button) Diskette Drive
	109595-001		1.44 Megabyte Diskette Drive ****
	106558-001		Weitek 287 Coprocessor (8 MHz)
	105151-001 **		150 ns DRAM Chips (256K x 1)
	105152-001 **		150 ns DRAM Chips (64K x 1)
	105035-001		System ROM

□ Contained in the Cable Kit, Spare PN 104191-001 **. See Table 6-6.

% When replacing a flip door style diskette drive with a push button style diskette drive, order a replacement drive bezel (PN 104059-001 **) for a better fit.

**** Requires system ROM N3 or later for use with MS-DOS and ROM P.1 or later for use with MS OS/2 Version 1.1.
Requires MS-DOS Version 3.31 Revision A if installed as Drive B, and MS-DOS Version 3.31 Revision B if installed as Drive A.

** Part number no longer available.

Chapter 6.4 Mechanical Chassis

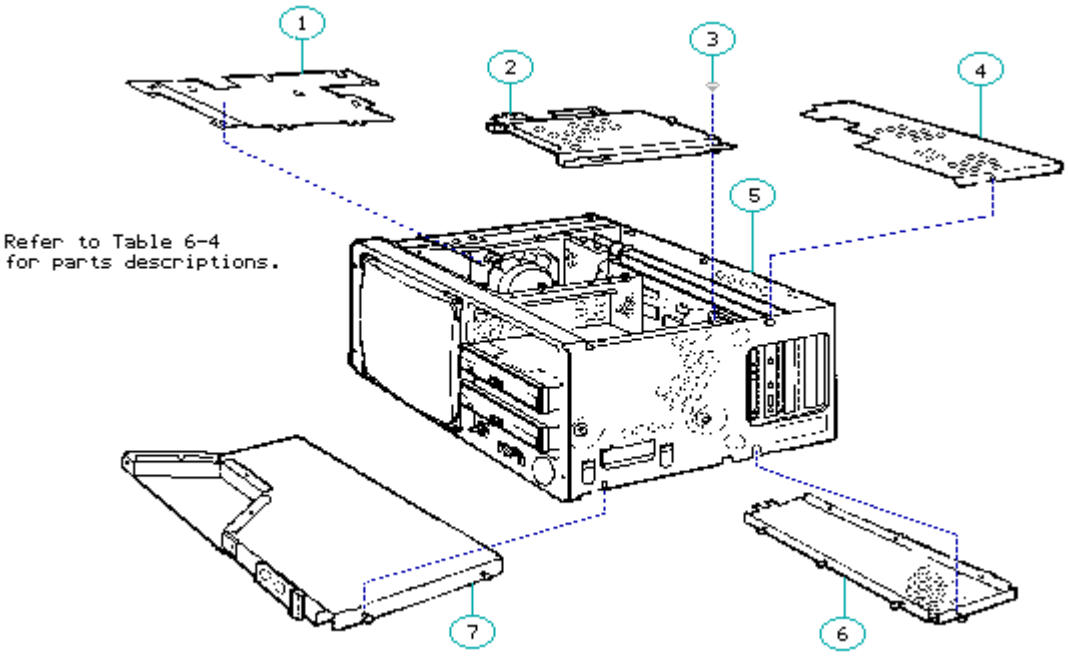


Figure 6-5. COMPAQ PORTABLE II Personal Computer Mechanical Chassis

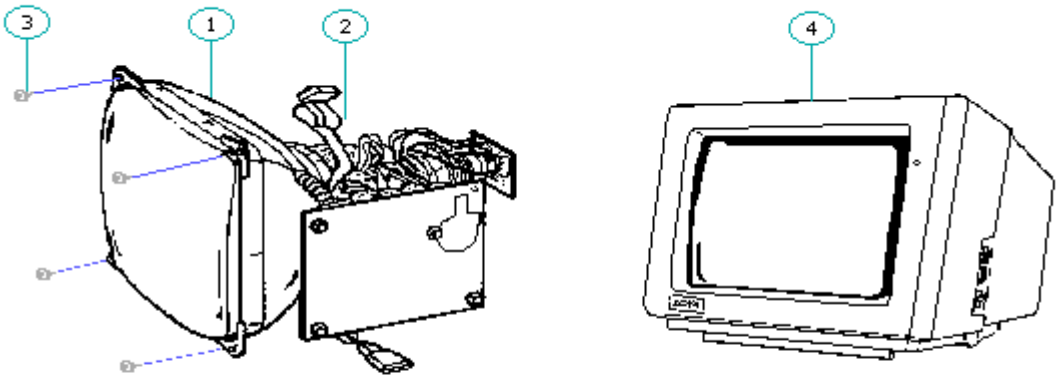
Table 6-4. COMPAQ PORTABLE II Personal Computer Mechanical Chassis

Item	Part Number	Description
1	104052-001 **	High Voltage Device Access Plate
2	104079-001 **	Mass Storage Device Access Plate
3	*	Torx 4-40 x 1/4 Screw
4	130770-001 **	Board Slots Access Plate
5	104081-001 **	Chassis Assembly
6	104098-001 **	Memory Expansion Access Plate
7	104099-001 **	Cable Tray Access Plate

* Contained in Miscellaneous Hardware Kit, Spare PN 104270-001 **.

** Part number no longer available.

Chapter 6.5 Monitor



Refer to Table 6-5
for parts descriptions.

Figure 6-6. COMPAQ PORTABLE II Personal Computer Monitor

Table 6-5. COMPAQ PORTABLE II Personal Computer Monitor

Item	Part Number	Description
1	104182-001 (No longer available)	Monitor Assembly
2	104007-001 * (No longer available)	Video Display Signal Cable
3	**	Mounting Screws (4)
4	106568-001	Color Monitor (external)

* Included in the Cable Kit. See Table 6.6.
** Included in the Miscellaneous Hardware Kit, Spare PN 104270-001 (No longer available). Refer to Table 6-7.

Chapter 6.6 Cable Kit

Table 6-6. COMPAQ PORTABLE II Personal Computer Cable Kit (PN 104191-001 (No longer available))

Description	Quantity
1. Diskette Drive Power Cable	1
2. Video Display Signal Cable	1

3. Fixed Disk Drive Controller Signal Cable	1
4. Speaker/Potentiometer Cable	1
5. Diskette Drive Signal Cable	1
6. Fixed Disk Drive Power Cable	1

Chapter 6.7 Miscellaneous Hardware Kit

Table 6-7. COMPAQ PORTABLE II Personal Computer Miscellaneous Hardware Kit (PN 104270-001 (No longer available))

Description	Quantity
Monitor/Drive Enclosure Screw	25
Screw Covers	25
Drive Shoulder Bolt	10
Drive Vibration Isolators	5
Ty-Wrap, 3/4 in. wide, 3 7/8 in. long, #18	100
PCB Board Guide	5
Fixed Disk Drive LED Cable	2
Brightness Knob	5
PCB Board Retainers	2
Push Mount Fasteners	5
Fan Well Nuts	5
High Voltage Monitor Grommet	2
System Board Connector Grommet	5
Keyboard Rubber Foot	10
Option Board Locking Standoffs	5
Monitor Sweep Board Standoffs	5
T-15 8/32 x 5/16 Screw	10
6-32 x 9/16 Screw	10
Power Grommet	2

Chapter 7. Removal and Replacement Procedures

7.1 Preparation

Before beginning the procedures in this section, complete the following steps:

1. Turn off the computer.

2. Close the keyboard and secure it by sliding the latches to the LOCK position (Figure 7-1) and folding the keyboard feet in place.

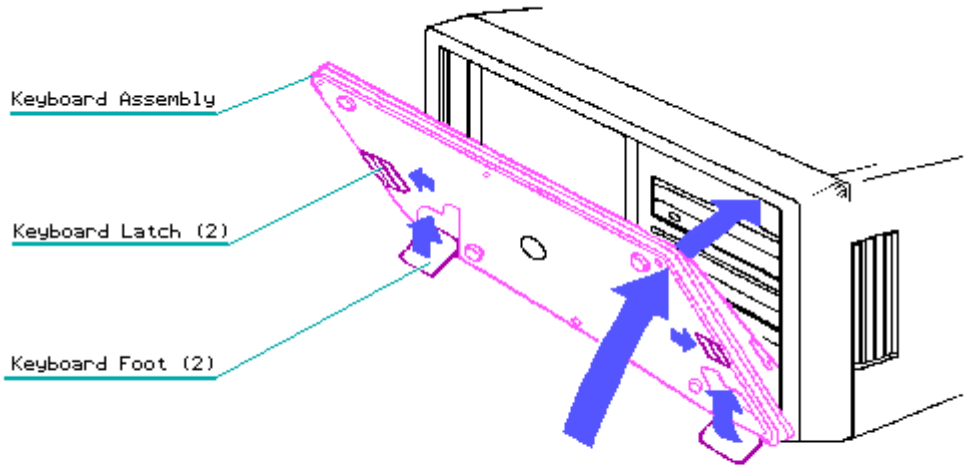


Figure 7-1. Closing and Locking the Keyboard

3. Disconnect any peripheral devices (printers, monitors, and so on) from the computer.

4. Disconnect the AC power cord from the AC outlet and from the computer (Figure 7-2).

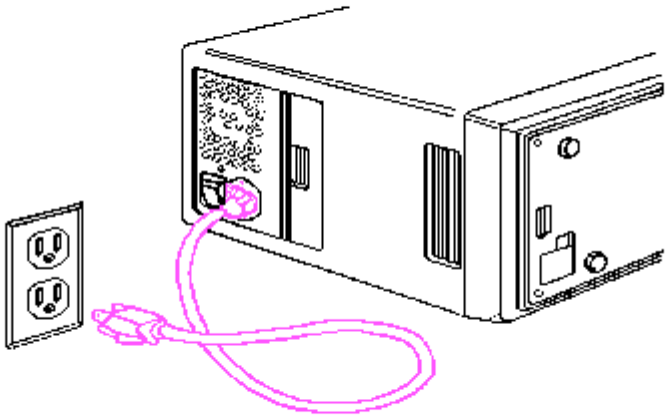


Figure 7-2. Disconnecting the AC Power Cord

Chapter 7.2 AC Power Fuse

To remove the AC power fuse, complete the following steps:

1. Complete the preparation procedures (see Section 7.1).

2. Slide open the AC power compartment to expose the line filter ON/OFF switch (Figure 7-3).

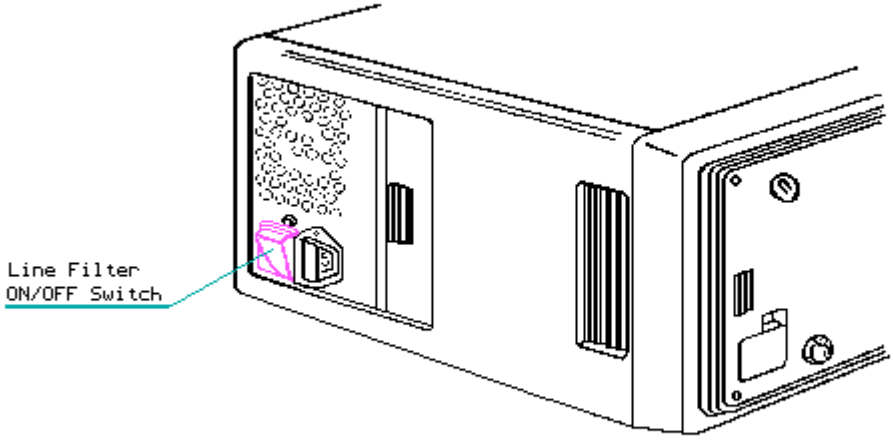


Figure 7-3. Locating the Line Filter ON/OFF Switch Assembly

3. Insert a flat bladed screwdriver along the bottom edge of the fuse box. Lift outward with enough pressure to force the fuse box out of the casing (Figure 7-4).

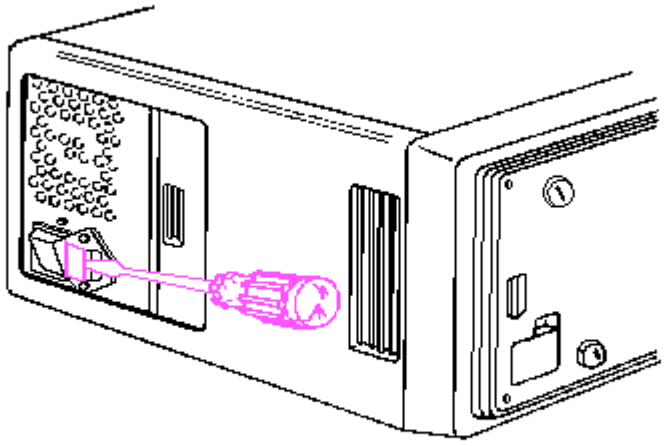


Figure 7-4. Removing the AC Power Fuse

4. When the fuse box is removed, locate the spare fuse provided in the holding area in the fuse box (Figure 7-5). Lift the fuse out and replace it with the spare fuse.

5. Slide the fuse box back into place. Press the cover in until you feel it snap into place.

   NOTE: Use only a fuse with a 3 Ampere current rating.

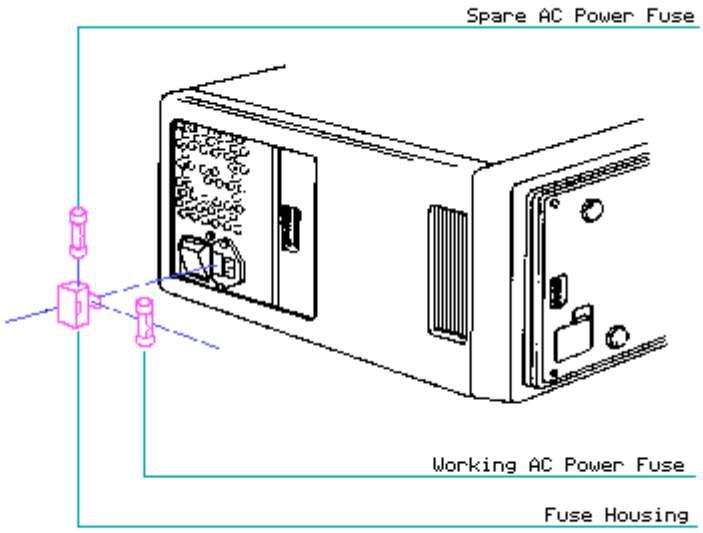


Figure 7-5. Replacing the AC Power Fuse

Chapter 7.3 Outer Access Casing

To remove the outer access casing from the computer, complete the following steps:

1. Complete the preparation procedure (see Section 7.1).

2. Place the computer keyboard side down on a level work surface. Slide open the AC power access panel and the peripheral connector access panels. Using a Torx screwdriver, remove the mounting screws from the AC power and peripheral connector access areas (Figures 7-6 and 7-7).

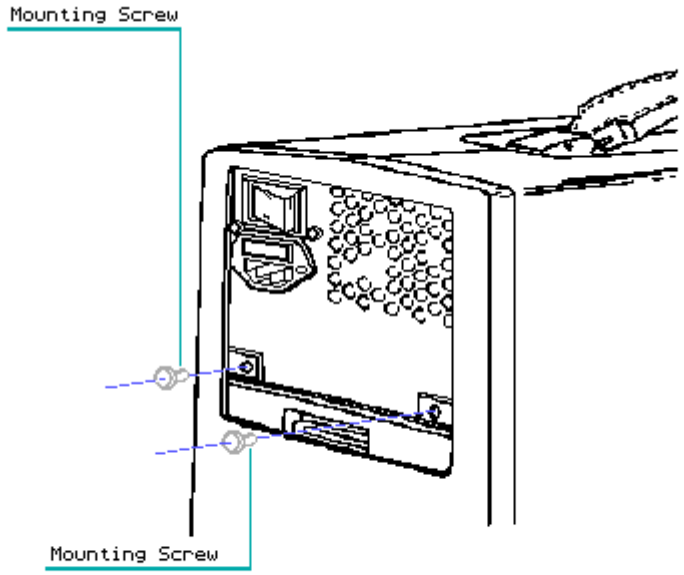


Figure 7-6. Locating and Removing the Outer Access Casing Mounting Screws (Switch Side)

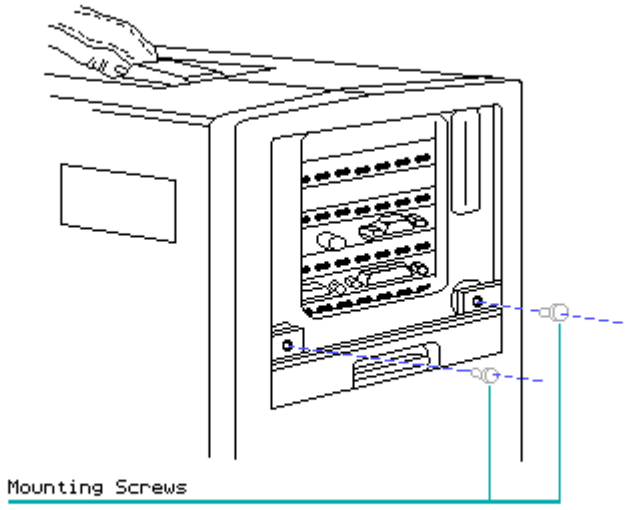


Figure 7-7. Locating and Removing the Outer Access Casing Mounting Screws (Connector Side)

3. Remove the outer access casing by grasping it close to the AC power and peripheral connector access panels and sliding the cover towards you. Slide it completely off the system chassis (Figure 7-8) and set it aside.

   NOTE: The outer access casing is flexible and will bend as needed.

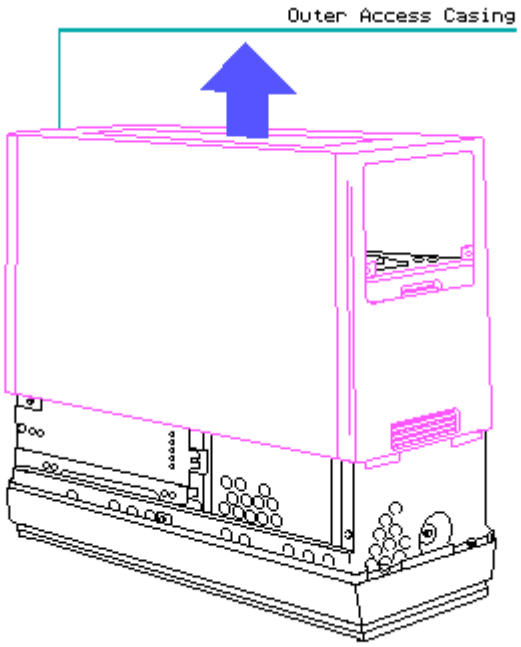


Figure 7-8. Removing the Outer Access Casing

To replace the outer access casing, reverse steps 2 and 3.

Chapter 7.4 Keyboard Assembly

To remove the keyboard assembly, complete the following steps:

1. Remove the cable tray access plate (see Section 7.13).

2. Disconnect the keyboard cable connector from the J116 socket connector on the system board (Figure 7-9).

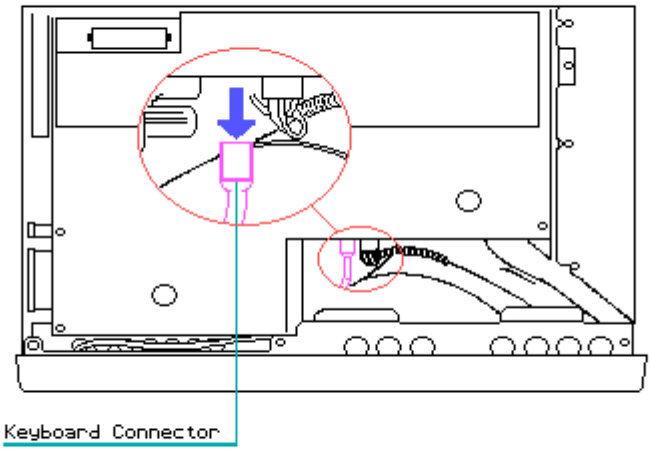


Figure 7-9. Disconnecting the Keyboard Connector from the System Board

3. Lower the rear portion of the computer so that it rests flatly against the work surface (Figure 7-10).

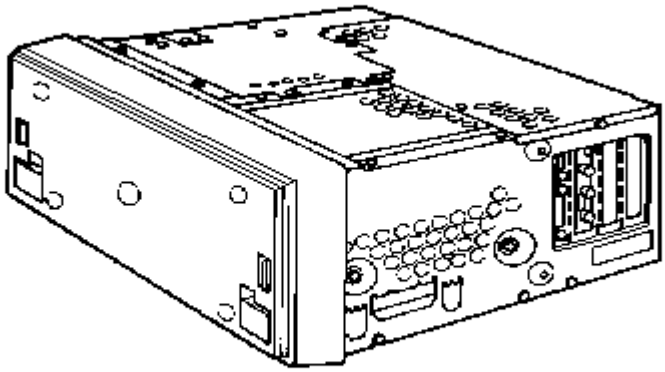


Figure 7-10. Returning the Computer to a Flat Position on the Work Surface

4. Unlock the keyboard latches and gently slide the keyboard cable from inside the computer system chassis (Figure 7-11). Remove the keyboard from the computer system chassis.

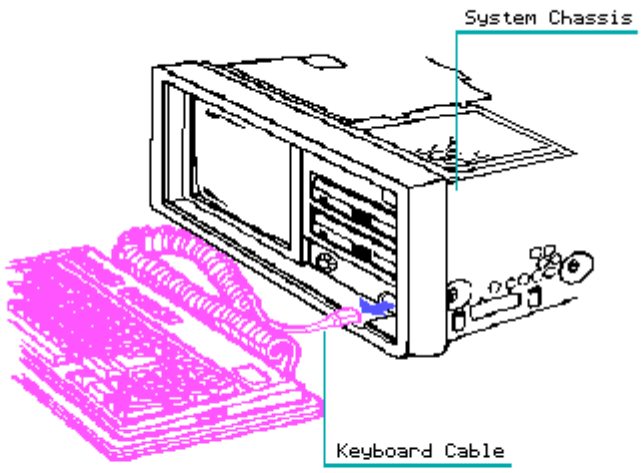


Figure 7-11. Removing the Keyboard

To replace the keyboard assembly, reverse steps 1 through 4.

Chapter 7.5 Front Bezel

To remove the front bezel, complete the following steps:

1. Remove the keyboard assembly (see Section 7.4).
2. Remove the brightness control knob by gently pulling it from the front bezel (Figure 7-12).

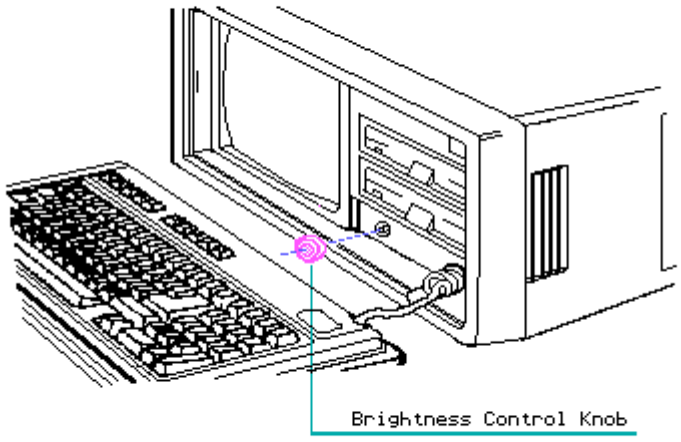


Figure 7-12. Removing the Brightness Control Knob

3. Lift the rear portion of the computer so that it rests firmly on the front bezel (Figure 7-13).

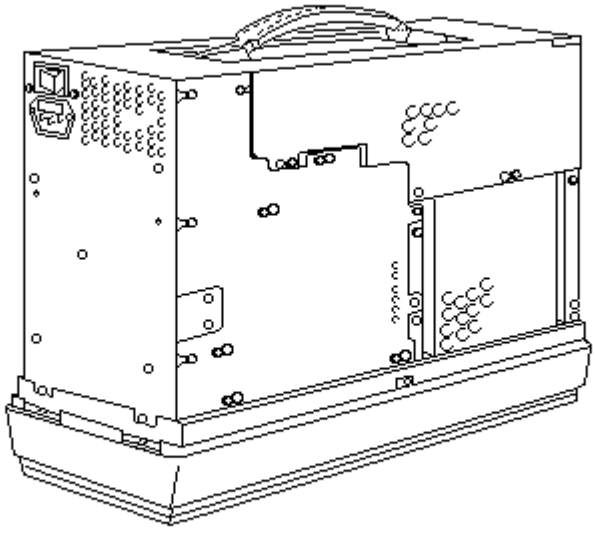


Figure 7-13. Placing the Computer in an Upward Position

4. Using a Torx screwdriver, remove the six screws that secure the front bezel to the system chassis (Figure 7-14).

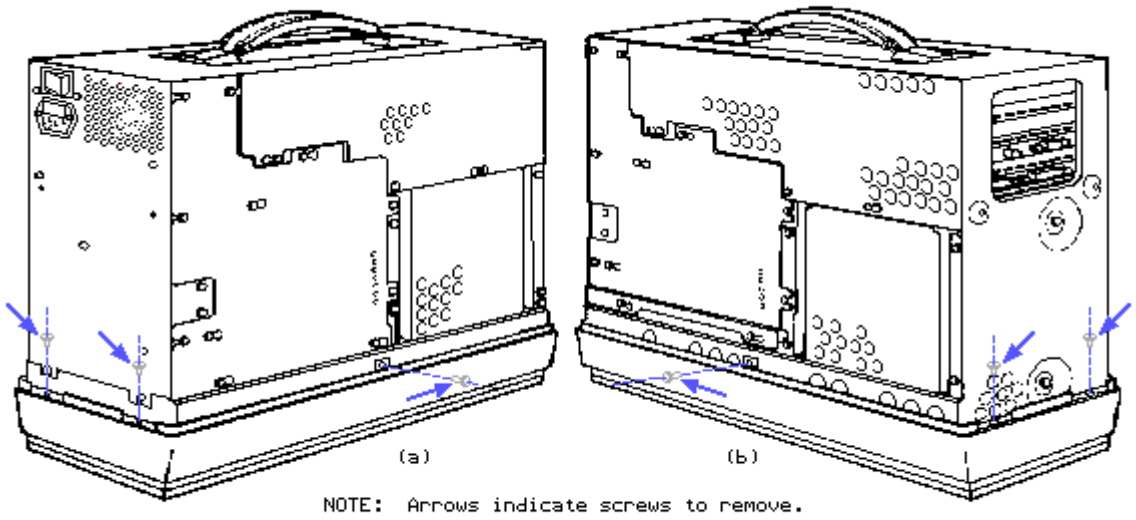


Figure 7-14. Removing the Six Front Bezel Mounting Screws

5. Lower the system chassis and gently pull the front bezel from the system chassis (Figure 7-15).

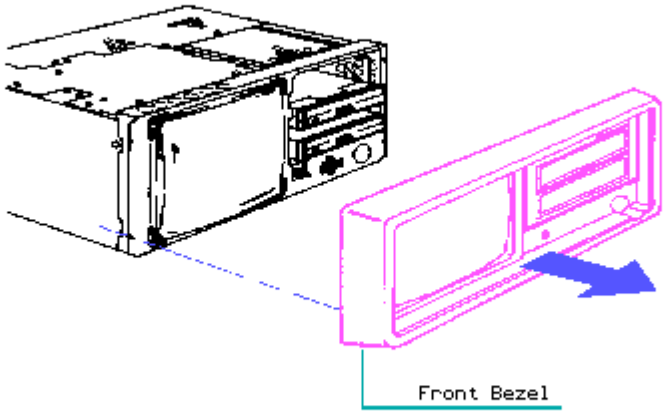


Figure 7-15. Removing the Front Bezel

To replace the front bezel, reverse steps 1 through 5.

Chapter 7.6 Power Supply Subassembly Chassis

The power supply subassembly chassis consists of the following assemblies:

- Line filter ON/OFF switch assembly
- Power supply assembly board
- Power supply insulating paper

To remove the power supply subassembly chassis, complete the following steps:

1. Remove the outer access casing (see Section 7.3).

2. Place the computer on its side (see Figure 7-16).

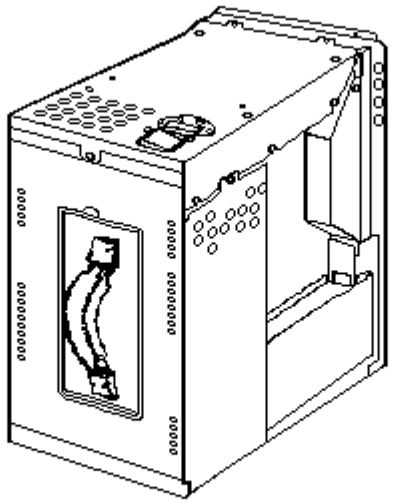


Figure 7-16. Positioning the Computer

3. Using a Torx screwdriver, loosen (DO NOT REMOVE) the nine mounting screws that secure the power supply chassis to the system chassis (Figure 7-17).

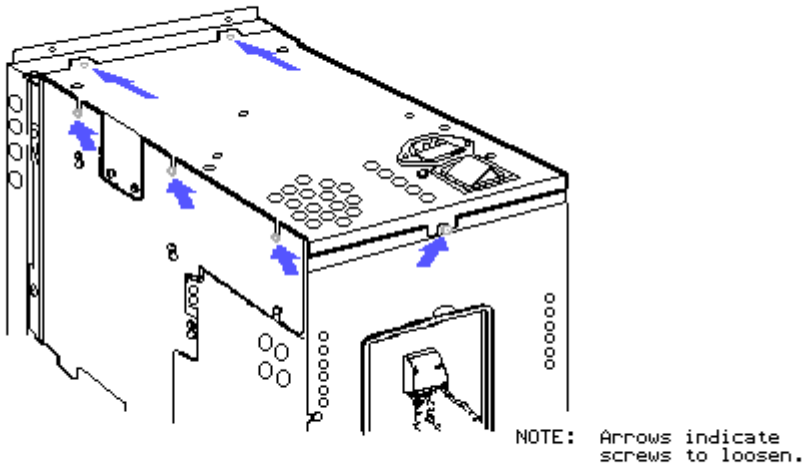


Figure 7-17a. Loosening the Nine Power Supply Mounting Screws (Part 1 of 2)

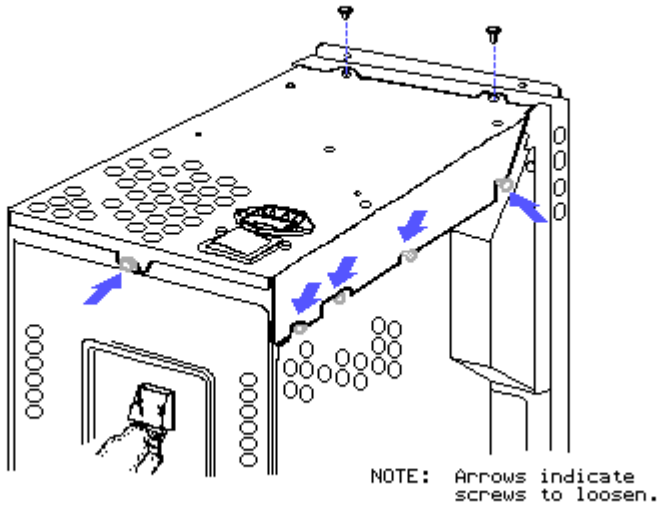


Figure 7-17b. Loosening the Nine Power Supply Mounting Screws (Part 2 of 2)

4. Remove only the two screws indicated in Figure 7-18.

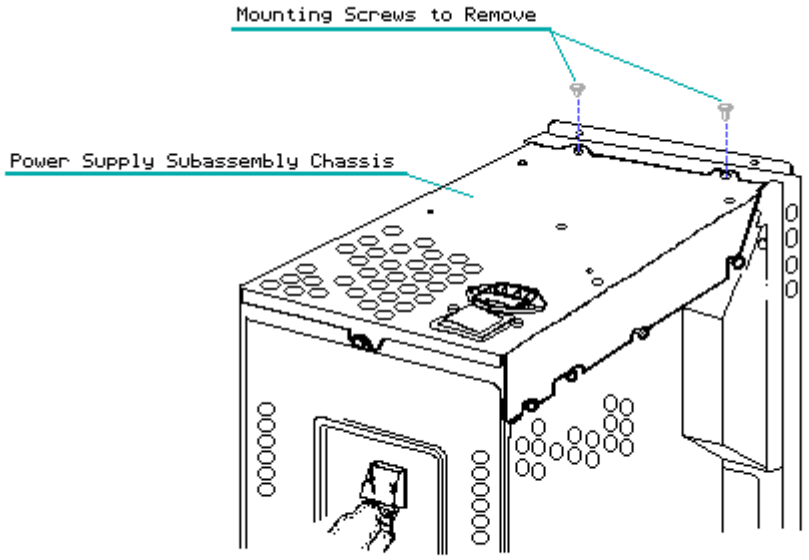


Figure 7-18. Removing the Power Supply Mounting Screws

5. Remove the power supply subassembly chassis by lifting the rear portion of the chassis upward. As this end is raised, lift the front portion of the chassis until the entire power supply chassis is free of the system chassis (Figure 7-19).

   NOTE: Because the power supply connects directly to the system board, be careful when removing the power supply subassembly chassis.

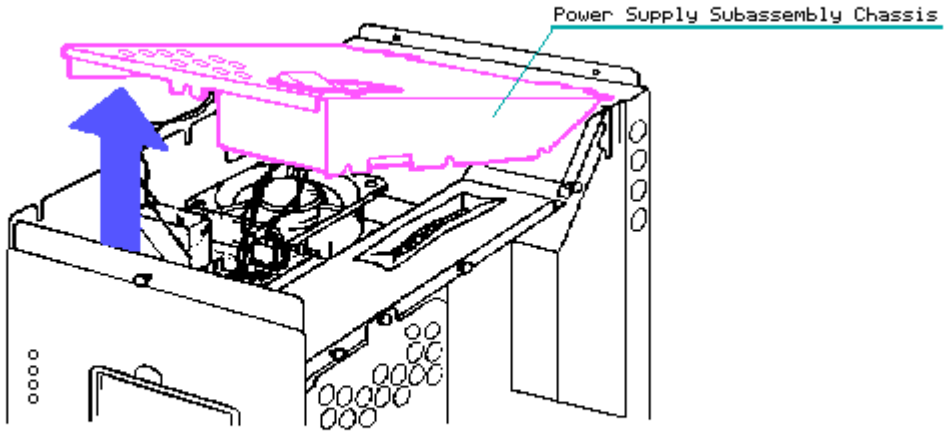


Figure 7-19. Removing the Power Supply

6. To completely remove the power supply subassembly chassis from the

computer system chassis, disconnect the fan assembly and the automatic power switching board (Figure 7-20).

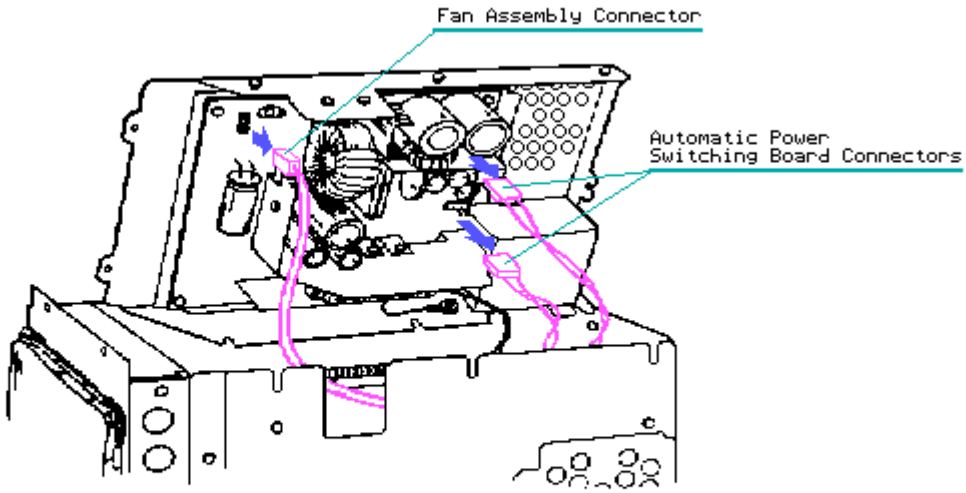


Figure 7-20. Disconnecting the Fan Assembly and Automatic Power Switching Connectors

7. Gently lift and remove the power supply subassembly from the system chassis.

To replace the power supply subassembly chassis, reverse steps 1 through 6.

Chapter 7.7 Fan Assembly

To remove the fan assembly, complete the following steps:

1. Remove the power supply subassembly chassis (see Section 7.6).

2. Cut the two cable ties that secure the fan assembly power cable to the system chassis.

3. Using a Phillips screwdriver, remove the two mounting screws and lift the fan assembly from the system chassis (Figure 7-21).

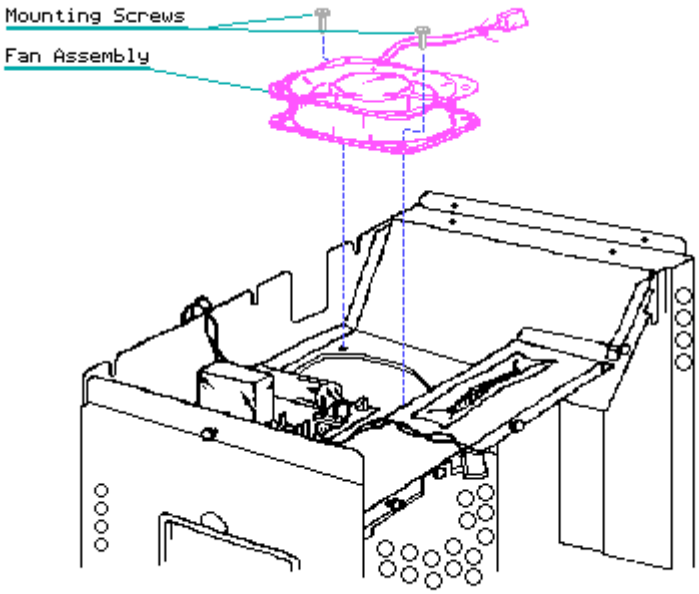


Figure 7-21. Removing the Fan Assembly

To replace the fan assembly, reverse steps 1 through 3.

Chapter 7.8 Automatic Power Switching Board (APS)

1. Remove the power supply subassembly chassis (see Section 7.6).

   NOTE: Carefully remove all three connections to the APS board when removing the power supply chassis. Refer to Figure 7-22 for an illustration of the APS board.

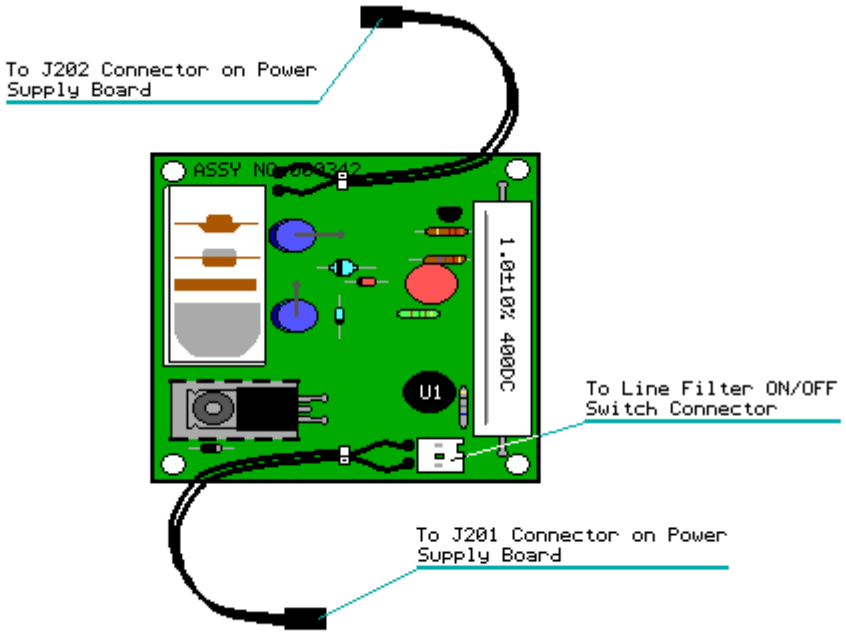


Figure 7-22. Automatic Power Switching (APS) Board (Assy No. 000342)

2. To remove the APS board, which is secured to the system chassis by four PCB standoffs, press the top of each PCB standoff together, one at a time. At the same time, carefully lift that portion of the APS board from the system chassis.

   Repeat this procedure until all board standoffs have been disconnected from the APS board, then lift and remove the APS board from the system chassis (Figure 7-23).

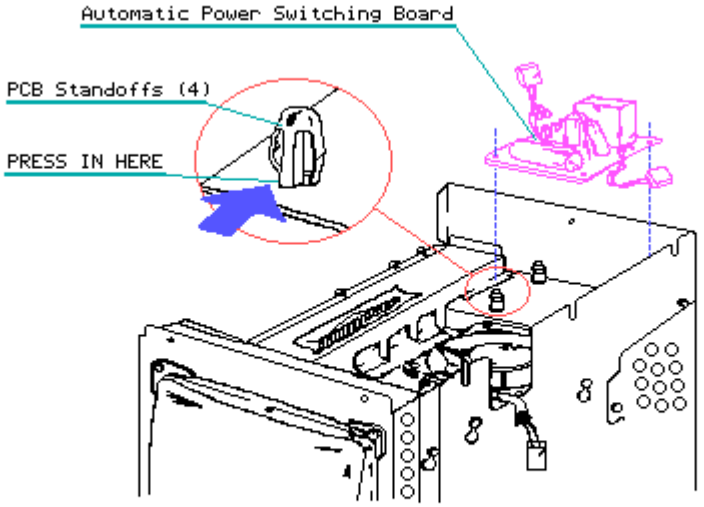


Figure 7-23. Disconnecting and Removing the Automatic Power Switching Board (Assy No. 000342)

To replace the APS Board, complete the following steps:

1. Slide the APS Board into place on the system

2. Press the four PCB standoffs into the holes in the board.

3. Hold the power supply subassembly at a 30 to 45 degree angle as shown in Figure 7-24, and tie the three wire harnesses together with a ty-wrap.

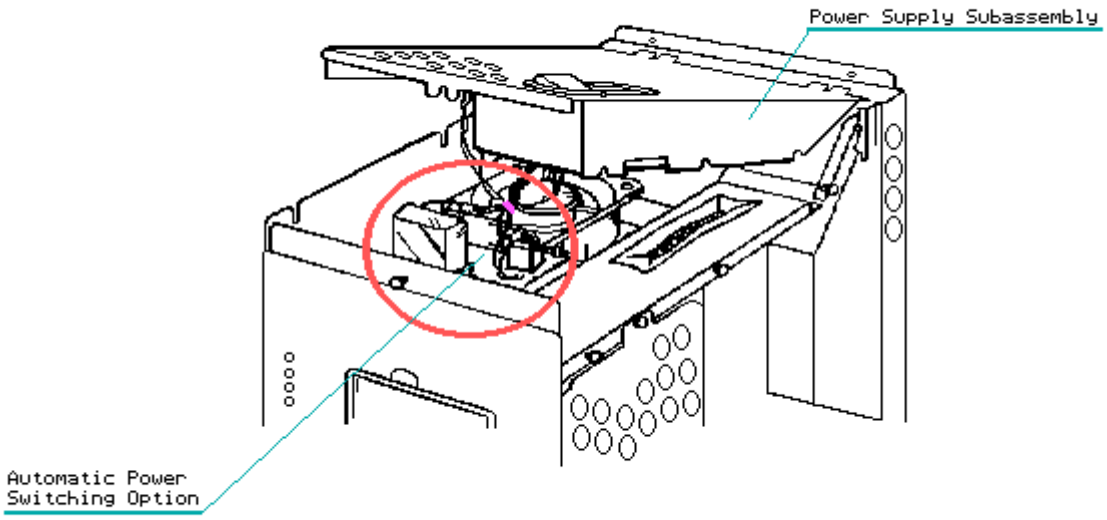


Figure 7-24. Tying Together the Harness Wires

>>>

2. Using a Torx screwdriver, loosen (DO NOT REMOVE) the 11 mounting screws that secure the high voltage monitor access plate to the system chassis. Remove only the three mounting screws indicated in Figure 7-25.

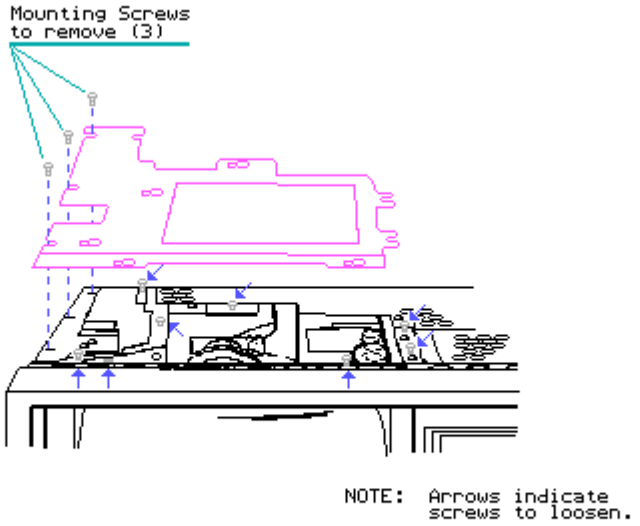


Figure 7-25. Removing High Voltage Access Plate

3. Gently slide the high voltage access plate towards the power supply chassis assembly and lift it up and away from the system chassis.

To replace the high voltage access plate, reverse steps 1 through 3.

Chapter 7.10 Board Slots Access Plate

The board slots access plate is located at the rear of the computer and covers all installed controller and option boards.

To remove the board slots access plate, complete the following steps:

1. Remove the outer access casing (see Section 7.3).

2. Using a Torx screwdriver, loosen (DO NOT REMOVE) the seven mounting screws that secure the board slots access plate to the system chassis. Remove only the one screw indicated in Figure 7-26.

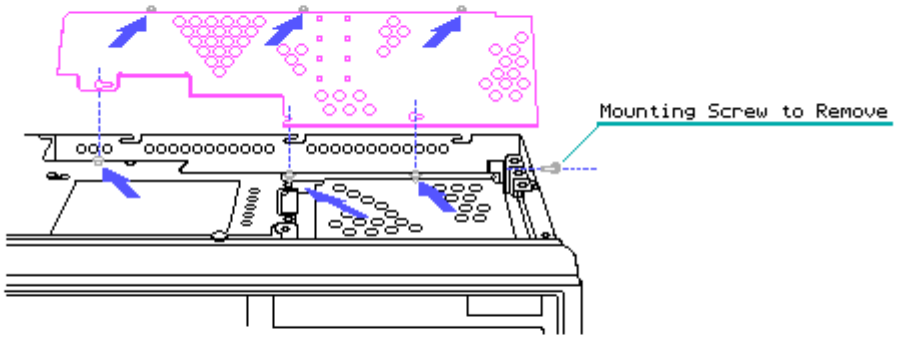


Figure 7-26. Removing the Board Slots Access Plate

3. Gently slide the board slots access plate to the right of the computer and lift it up and away from the system chassis.

To replace the board slots access plate, reverse steps 1 through 3.

Chapter 7.11 Mass Storage Device Access Plate

The mass storage device access plate is located at the front of the system chassis and covers all installed mass storage devices.

To remove the mass storage device access plate, complete the following steps:

1. Remove the outer access cover (See Section 7.3).

   NOTE: It is not necessary to remove the mass storage device access plate before removing the mass storage device enclosure.

2. Using a Torx screwdriver, remove the eight mounting screws that secure the mass storage device access plate to the system chassis and the mass storage device enclosure (Figure 7-27).

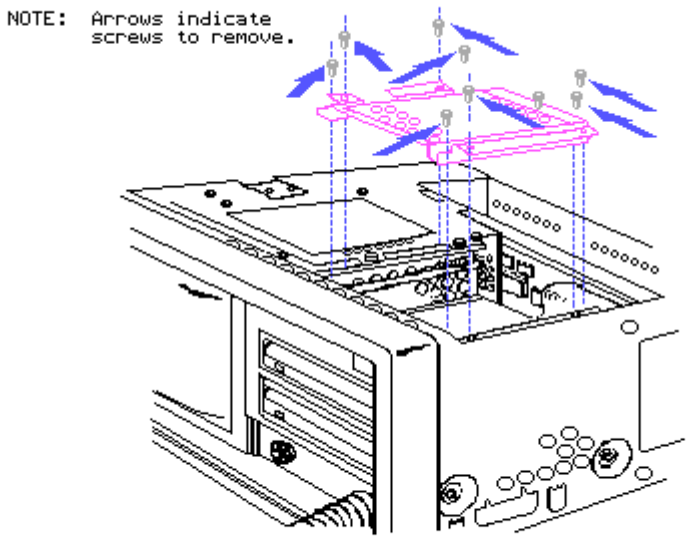


Figure 7-27. Removing the Mass Storage Device Access Plate

3. Carefully lift the mass storage device access plate away from the system chassis.

To replace the mass storage device access plate, reverse steps 1 through 3.

Chapter 7.12 System Memory Board Access Plate

To remove the system memory board access plate, complete the following steps:

1. Remove the outer access casing (see Section 7.3).

2. Turn over the system chassis as illustrated in Figure 7-28.

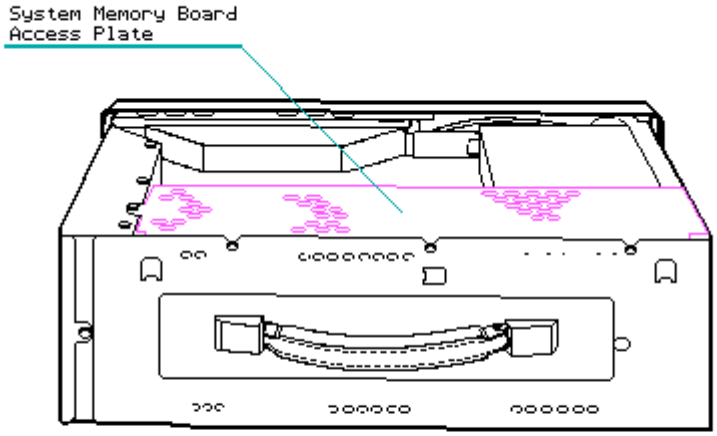
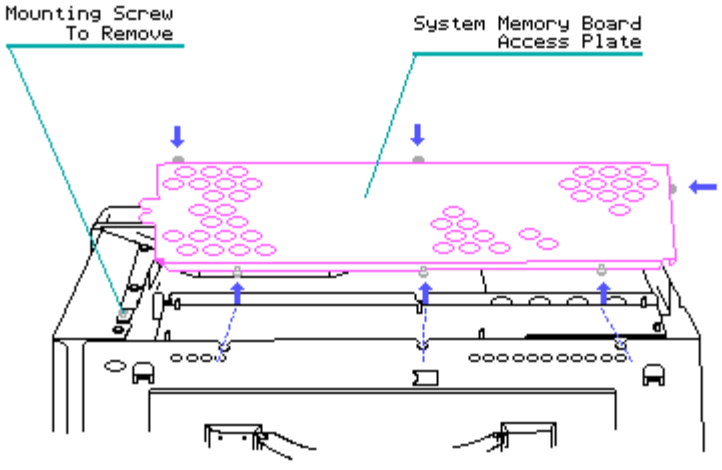


Figure 7-28. Exposing the System Memory Board Access Plate

3. Using a Torx screwdriver, loosen (DO NOT REMOVE) the seven mounting screws that secure the system memory board access plate to the system chassis. Remove only the one mounting screw indicated in Figure 7-29.

4. Carefully lift and remove the system memory board access plate from the system chassis (Figure 7-29).



NOTE: Arrows indicate screws to loosen.

Figure 7-29. Removing the System Memory Board Access Plate

To replace the system memory board access plate, reverse steps 1 through 4.

Chapter 7.13 Cable Tray Access Plate

To remove the cable tray access plate, complete the following steps:

1. Remove the outer access casing (see Section 7.3).

2. Remove the system memory board access plate (see Section 7.12).

3. Turn over the system chassis as illustrated in Figure 7-30.

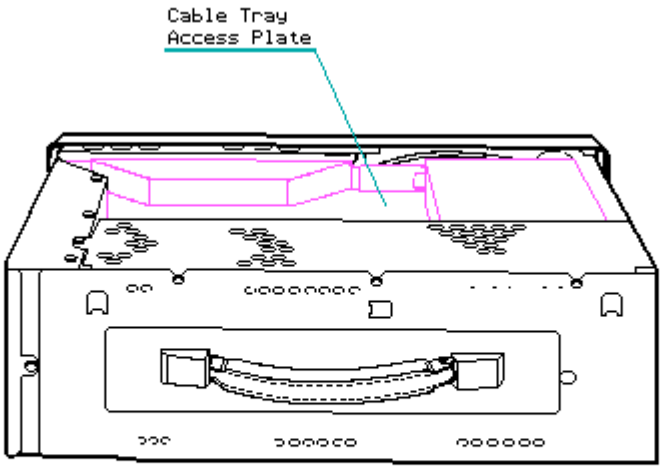
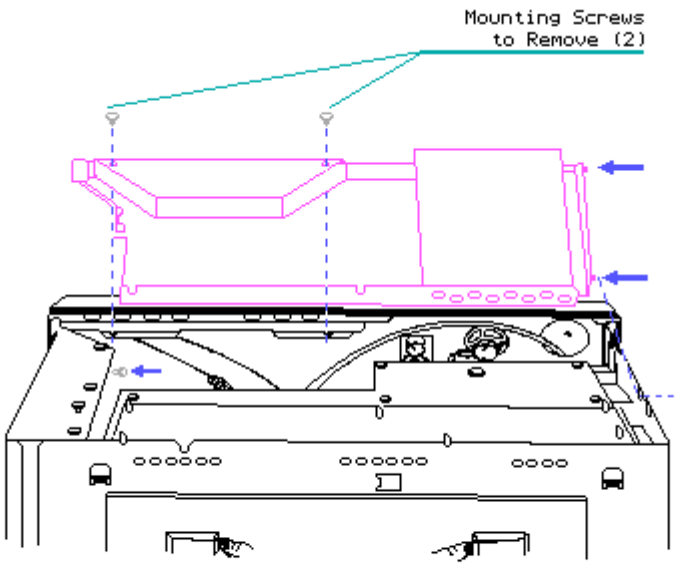


Figure 7-30. Exposing the Cable Tray Access Plate

4. Using a Torx screwdriver, loosen (DO NOT REMOVE) the seven Torx screws that secure the cable tray access plate to the system chassis. Remove only the two mounting screws noted in Figure 7-31.



NOTE: Arrows indicate screws to loosen.

Figure 7-31. Removing the Cable Tray Access Plate

5. Carefully lift and remove the cable tray access plate from the system chassis.

To replace the cable tray access plate, reverse steps 1 through 5.

Chapter 7.14 Multipurpose Controller Board

The Multipurpose Controller Board is located in slot J104 on the system board (Figure 7-32). To remove the multipurpose controller board, complete the following steps.

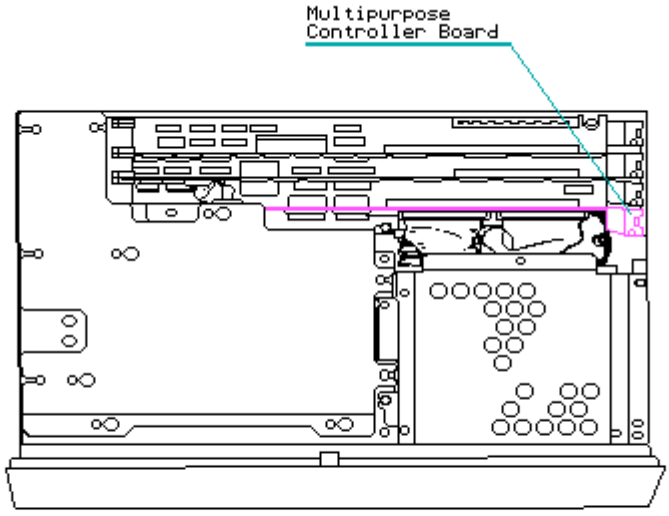


Figure 7-32. Locating the Multipurpose Controller Board

1. Remove the board slots access plate (see Section 7.10).

2. Disconnect the mass storage device signal cable(s) as shown in Figure 7-33.

   NOTE: If a fixed disk drive is installed in the system chassis, you must disconnect two signal cables.

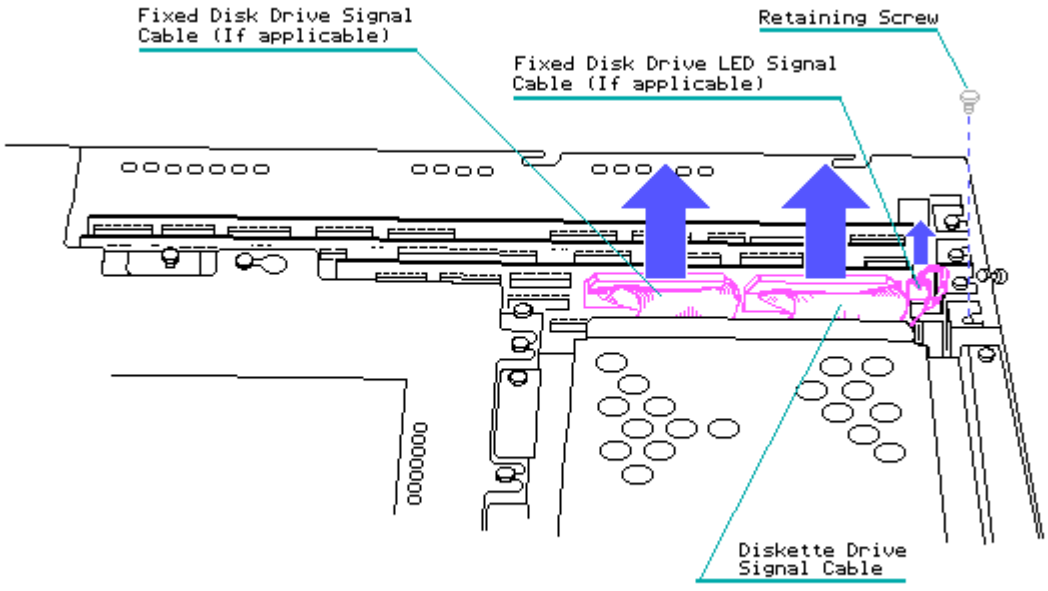


Figure 7-33. Disconnecting the Mass Storage Device Signal Cables from the Multipurpose Controller Board

3. Using a Torx screwdriver, remove the board retaining screw. Carefully

lift the controller board up and out of the system board (Figure 7-34).

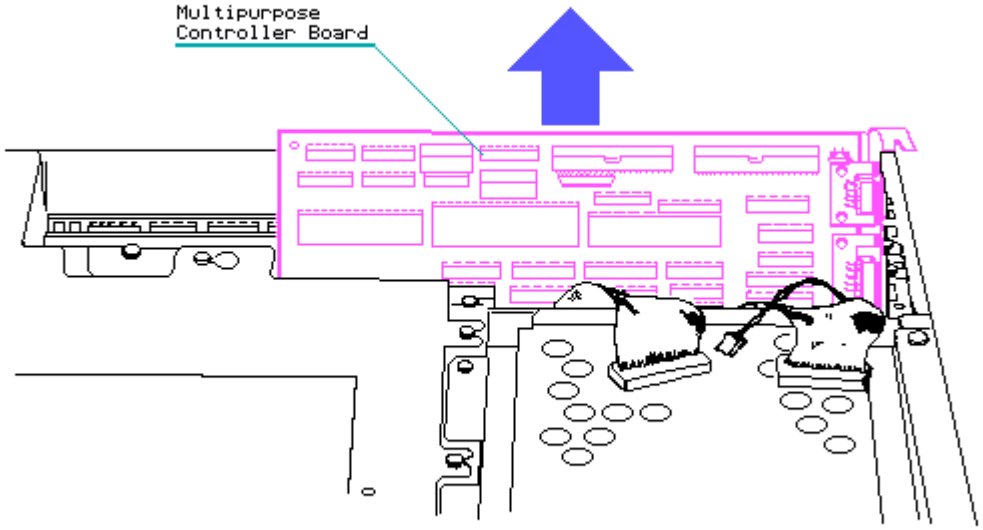


Figure 7-34. Removing the Multipurpose Controller Board

To replace the Multipurpose Controller Board, reverse steps 1 through 3.

Chapter 7.15 Video Display Controller Board

The Video Display Controller Board is located in slot J103 on the system board (Figure 7-35).

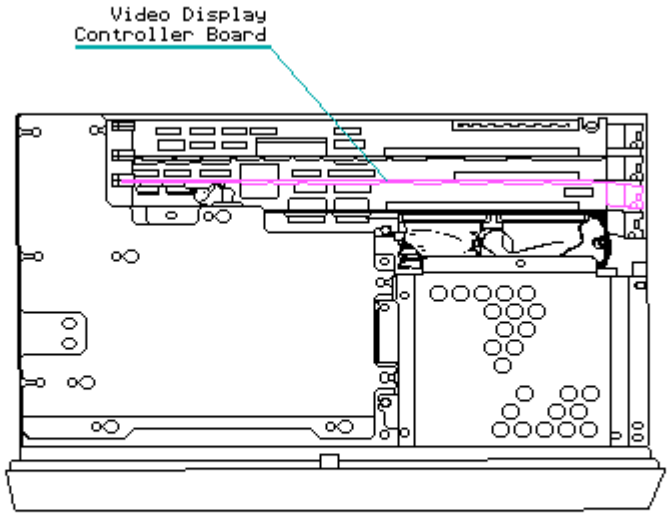


Figure 7-35. Locating the Video Display Controller Board

To remove the Video Display Controller Board, complete the following steps:

1. Remove the board slots access plate (see Section 7.10).

2. Remove the retaining screw that secures the Video Display Controller Board. Then disconnect the video display signal cable (Figure 7-36).

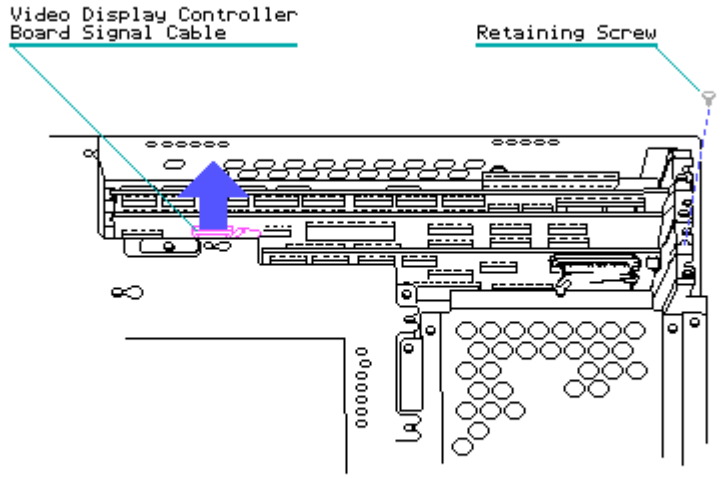


Figure 7-36. Removing the Video Display Controller Board Retaining Screw and Signal Cable

3. Carefully lift the board up and out of its J103 connector location on the system board (Figure 7-37).

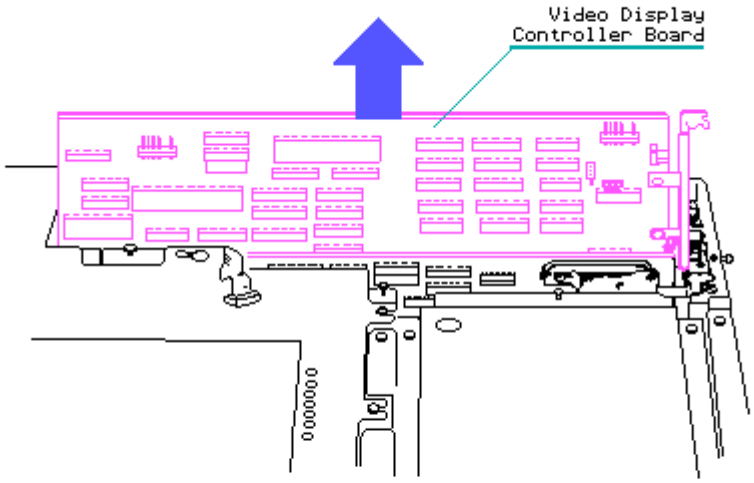


Figure 7-37. Removing the Video Display Controller Board

To replace the Video Display Controller Board, reverse steps 1 through 3.

Chapter 7.16 512/2048 KByte Memory Expansion Board

The COMPAQ PORTABLE II Personal Computer may contain up to two option boards. Option boards are always installed in the back two slots of the system chassis. To remove a COMPAQ 512/2048 Kbyte Memory Expansion Board or any other option board installed in the computer, complete the following steps:

1. Remove the outer access casing (See Section 7.3).

2. Disconnect any signal cables connected to the board.

3. Remove the retaining screw that secures the board to the system chassis.

4. Carefully lift the board up and out of its connector on the system board (Figure 7-38).

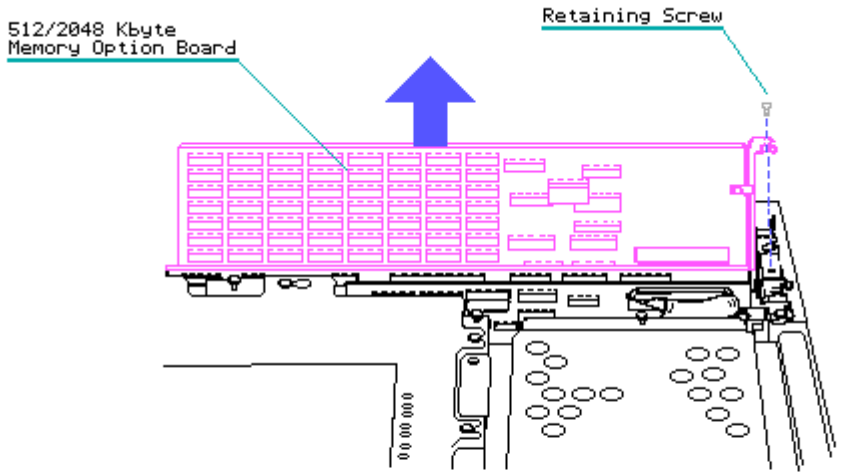


Figure 7-38. Removing the 512/2048 Kbyte Memory Expansion Board or Other Option Boards

To replace the 512/2048 Kbyte Memory Expansion Board or other option boards, reverse steps 1 through 4.

Chapter 7.17 Handle

To remove the handle from the computer, complete the following steps:

1. Remove the outer access casing (see Section 7.3).
2. Remove the board slots access plate (see Section 7.10).
3. Remove all controller and option boards installed (see Sections 7.14 through 7.16).
4. Using a Torx screwdriver, remove the two mounting screws that secure the handle to the system chassis, and remove the handle system chassis (Figure 7-39).

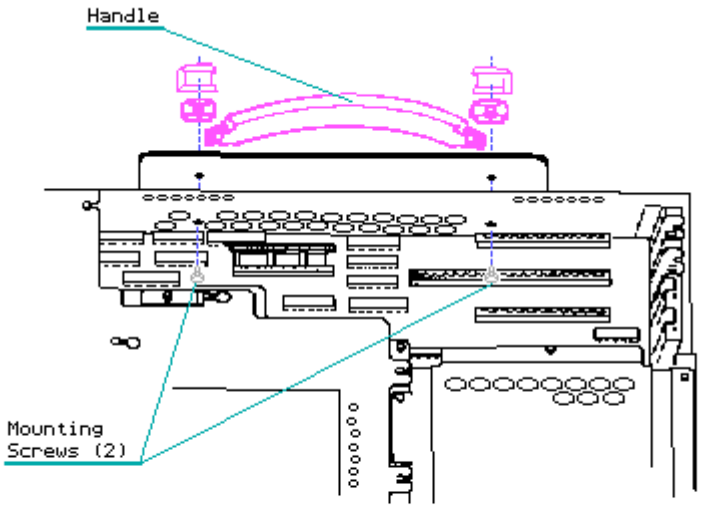


Figure 7-39. Removing the Handle

To replace the handle, reverse steps 1 through 4.

Chapter 7.18 System Board

There are two versions of the system board for the COMPAQ PORTABLE II: version 1 (Assy No. 000318) and version 2 (Assy No. 000324). The two system boards differ in shape, memory chip locations, and jumper locations (Figures 7-40 and 7-41).

Functionally both versions are identical; however, the version 2 system board contains three gate array devices that consolidate the functions of several devices from the earlier version making a smaller system board possible.

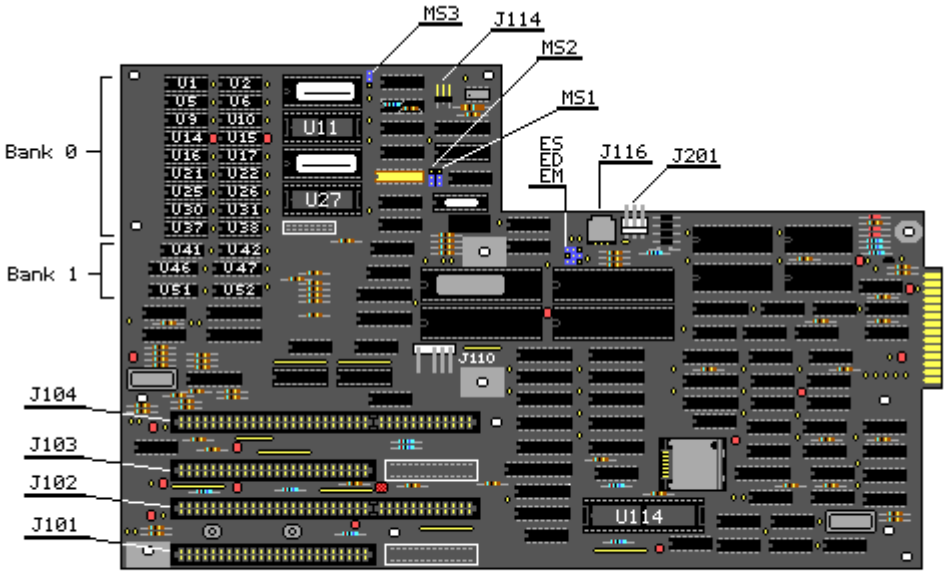


Figure 7-40. COMPAQ PORTABLE II Personal Computer System Board Version 1 (Assy No. 000318)

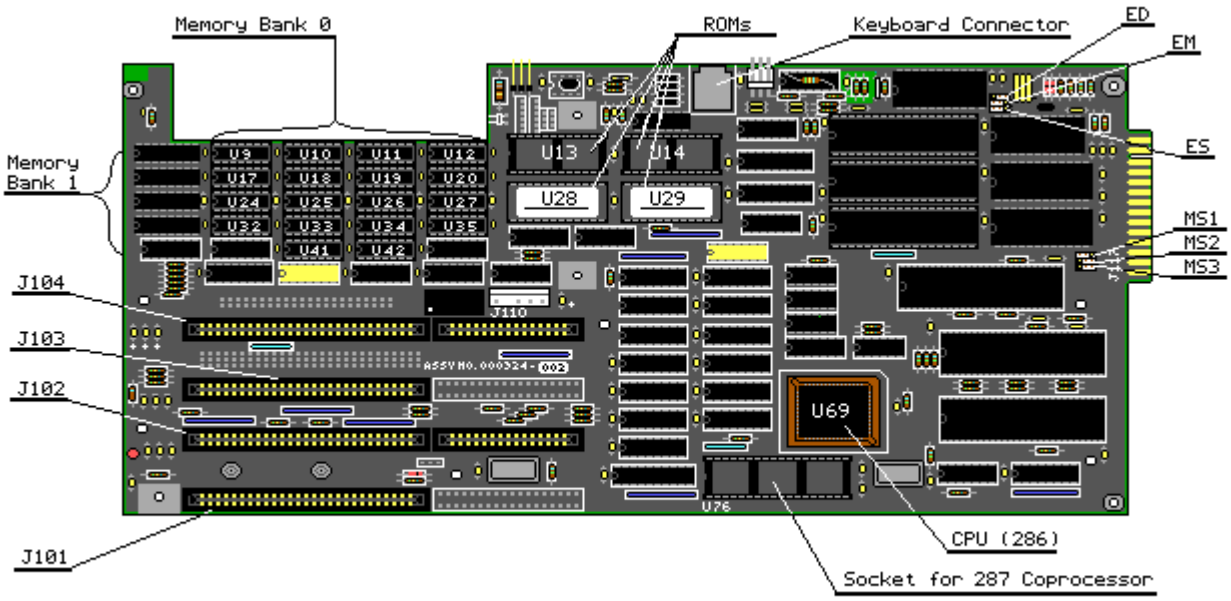


Figure 7-41. COMPAQ PORTABLE II Personal Computer System Board Version 2 (Assy No. 000324)

To remove either system board, complete the following steps:

1. Remove the power supply subassembly chassis (see Section 7.6).

2. Remove the board slots access plate (see Section 7.10).

3. Remove all controller and option boards installed (see Sections 7.14 through 7.16).

4. Using Figure 7-42 as a reference, disconnect the mass storage device power cable.

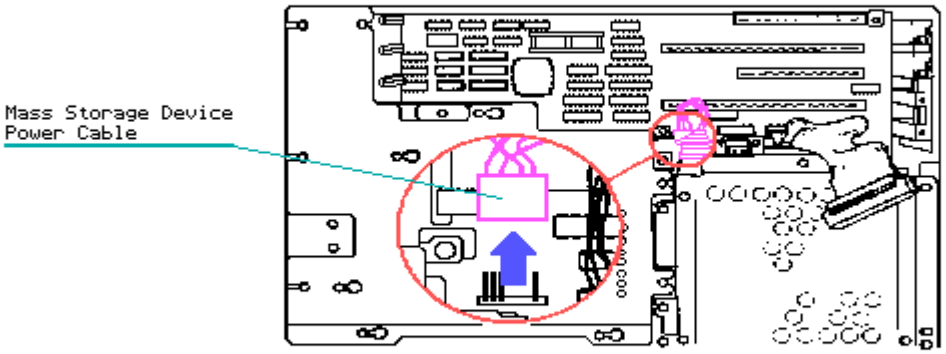


Figure 7-42. Disconnecting the Mass Storage Device Power Cable

5. Remove the system memory board access plate (see Section 7.12).

6. Remove the cable tray access plate (see Section 7.13).

7. Using Figure 7-43 as a reference, disconnect the keyboard power cable, the video display unit power cable, and the speaker/brightness control cable.

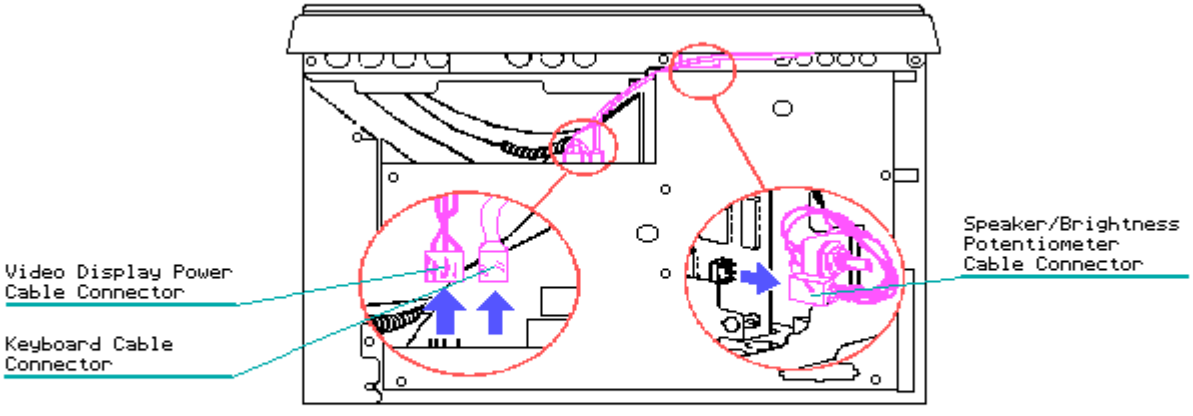
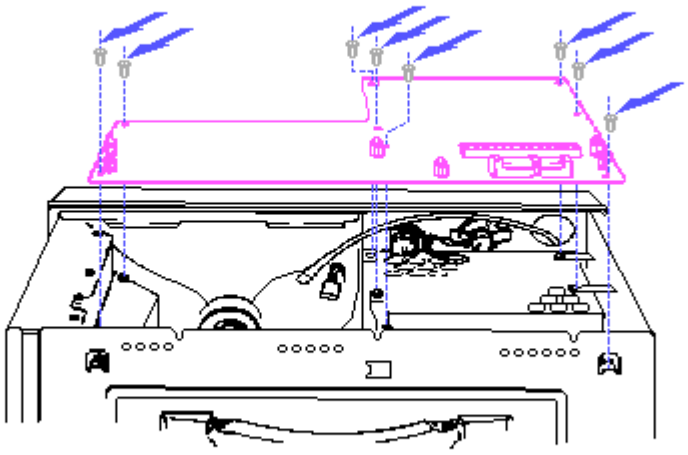


Figure 7-43. Disconnecting the Keyboard Power, VDU Power, and Speaker/Brightness Control Cables

8. Using a Torx screwdriver, remove the eight mounting screws that secure the system board to the system chassis.

9. Carefully lift the system board from the System chassis (Figure 7-44).



NOTE: Arrows indicate screws to remove.

Figure 7-44. Removing the System Board

To replace the system board, reverse steps 1 through 9.

If replacing a Version 1 board with a Version 2 board, the speaker/brightness potentiometer assembly (PN 104185-001 no longer available) should be replaced with an assembly containing a longer wiring harness. Spare part kits containing version 2 boards include a speaker/brightness potentiometer assembly.

NOTE: Be sure to pay special attention to the system board guide/clamp at the rear of the chassis.

Chapter 7.19 Battery

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1. Remove the 512/1536 Kbyte System Memory Board access plate (see Section 7.12).

2. The battery is secured to the bottom of the system board by a snap in connector and cable tie. Remove the battery by first cutting the cable tie then lifting the battery from its snap in connector (Figure 7-45).

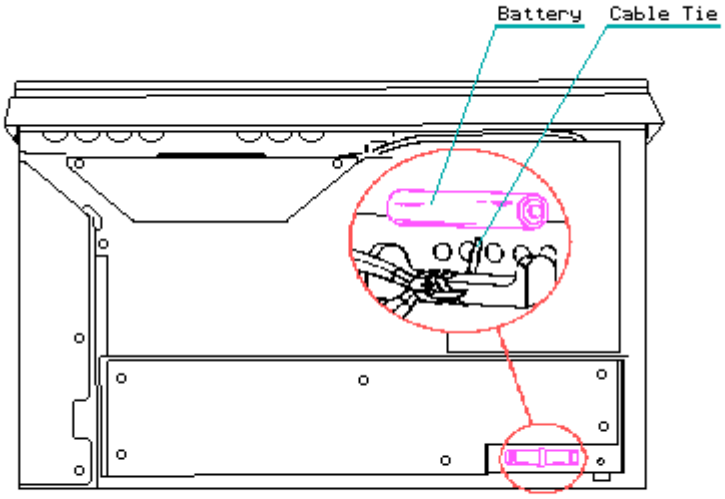


Figure 7-45. Removing the Battery

>>>

steps:

1. Remove the outer access casing (see Section 7.3).

2. Remove the Memory Expansion Board access plate (see Section 7.12).

3. The memory board is secured to the system board by six PCB standoffs. To remove the memory board from the system board, press the top of each PCB stand off together, one at a time. At the same time, gently lift that portion of the memory board from the stand off. Repeat this procedure until all PCB standoffs have been disconnected from the system memory board.

4. Lift the disconnected memory board from the system board (Figure 7-46).

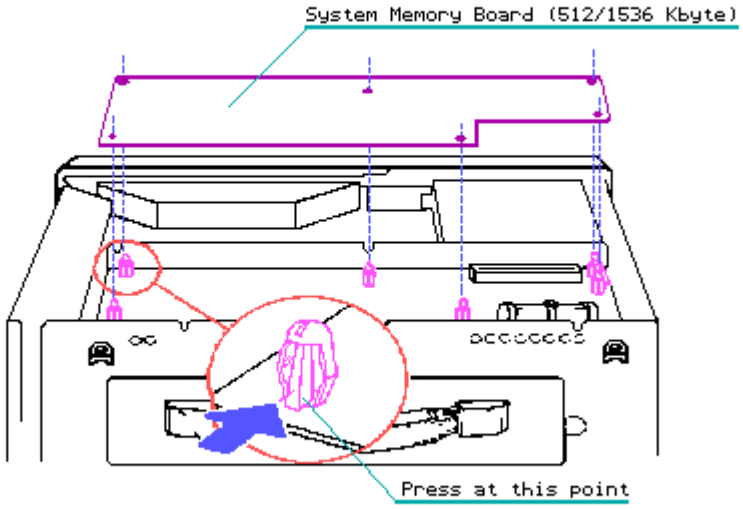


Figure 7-46. Disconnecting and Removing the 512/1536 Kbyte System Memory Board

To replace the memory board, reverse steps 1 through 4.

Chapter 7.21 Memory Option

COMPAQ PORTABLE II Personal Computer memory chips can be installed on the system board, the 512/2048 Kbyte Memory Expansion Board, or the 512/1536 Kbyte System Memory Board.

To remove a memory chip, complete the following steps:

1. Remove the board that contains the memory chip to be replaced. For board removal procedures, refer to one of the following sections.

   - For the 512/2048 Kbyte Memory Expansion Board removal and replacement procedure refer to Section 7.16.

- For the system board removal and replacement procedure refer to Section 7.18.

- For the 512/1536 Kbyte System Memory Board removal and replacement procedure, refer to Section 7.20.

NOTE: The RAM and ROM chip locations on the system boards differ from those on the memory boards (See Figures 7-47 through 7-50).

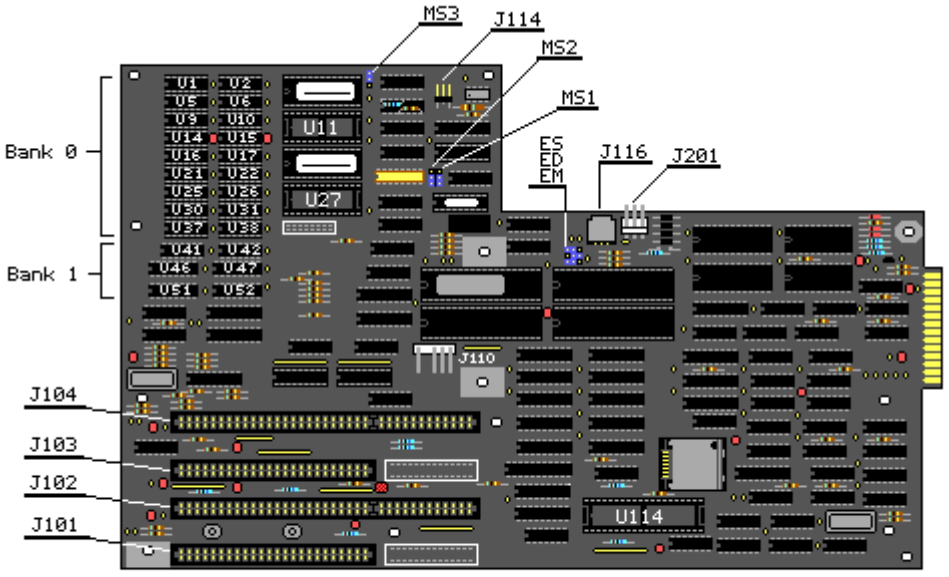


Figure 7-47. COMPAQ PORTABLE II Personal Computer System Board Version 1 (Assy No. 000318)

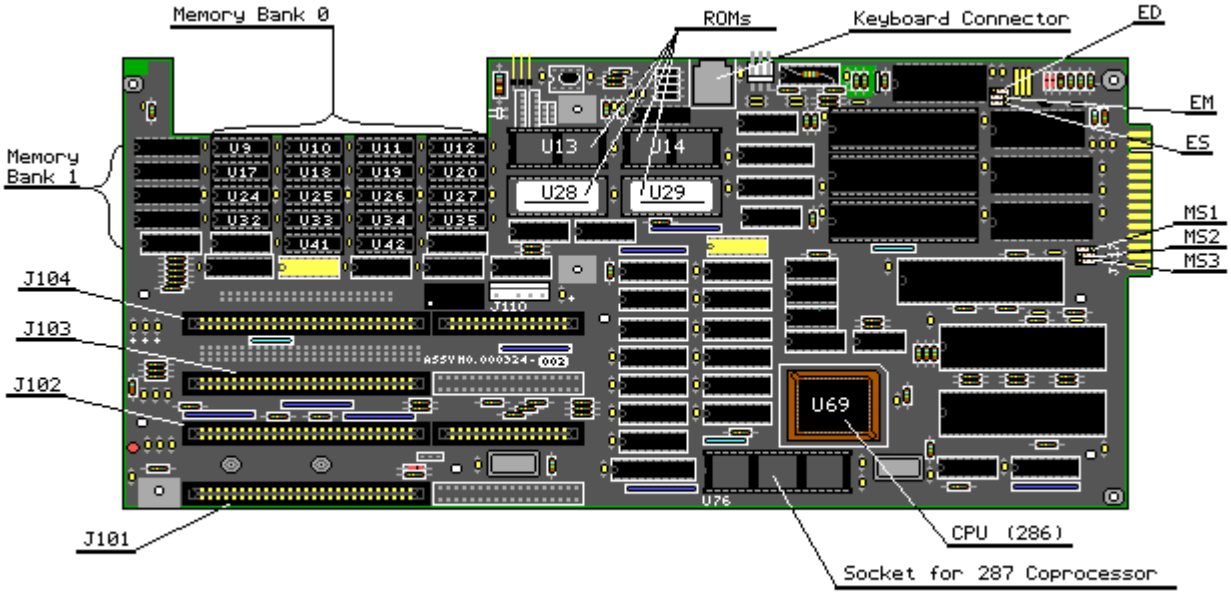


Figure 7-48. COMPAQ PORTABLE II Personal Computer System Board Version 2 (Assy No. 000324)

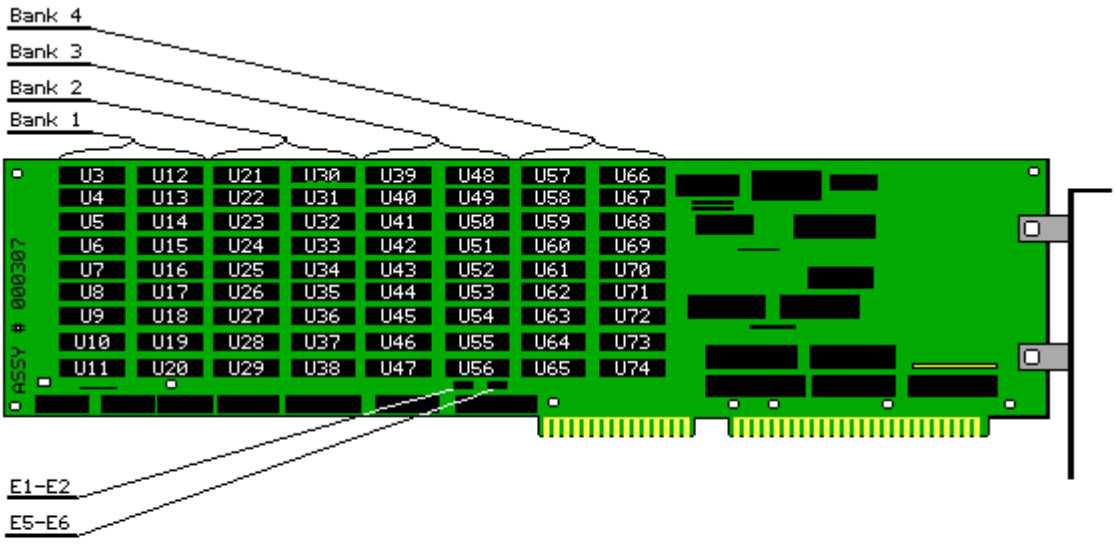


Figure 7-49a. 512/2048 Kbyte Memory Expansion Board (Assy No. 000307) (Part 1 of 2)

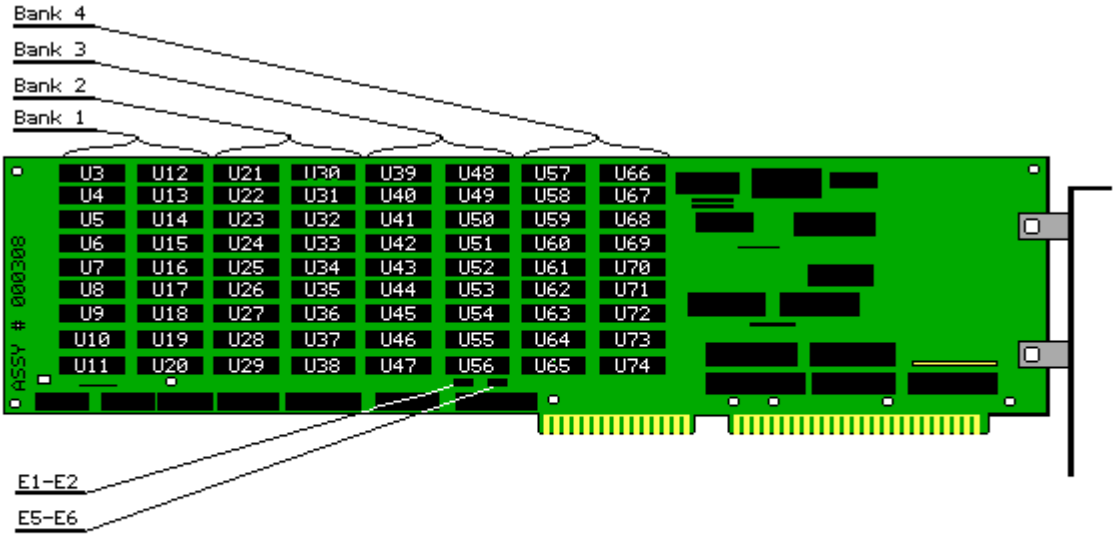


Figure 7-49b. 512/2048 Kbyte Memory Expansion Board (Assy No. 000308) (Part 2 of 2)

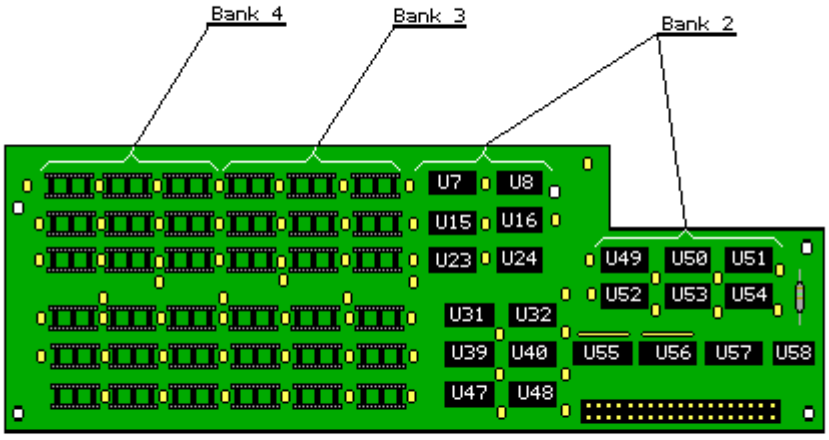


Figure 7-50. 512/1536 Kbyte System Memory Board (Assy No. 000330)

2. Using an IC removal tool, remove the defective RAM or ROM chip(s) (Figure 7-51).

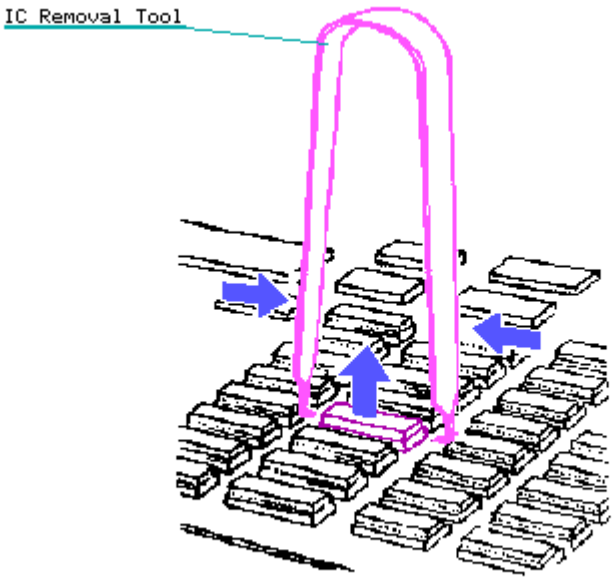


Figure 7-51. Removing a Memory Chip

3. Using a special IC insertion tool, replace the defective RAM or ROM chip(s) (Figure 7-52).

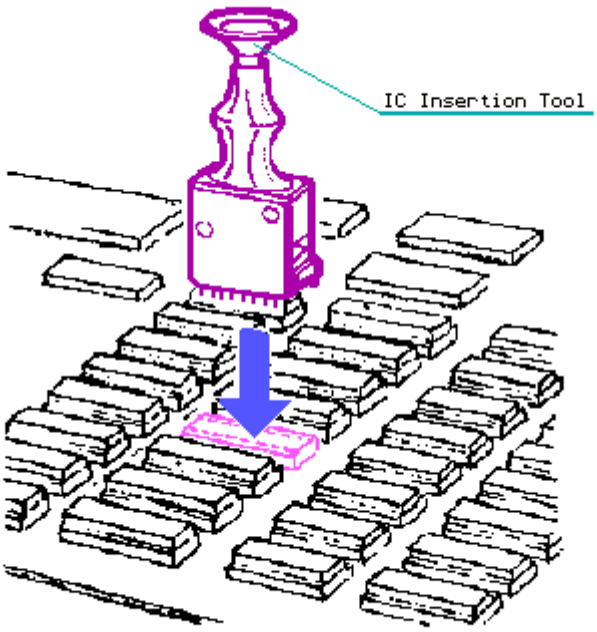


Figure 7-52. Replacing a Memory Chip

4. Replace the memory board.

NOTE: After replacing the defective component, run DIAGNOSTICS to verify proper operation of the computer. (Please consult the SUPPORT SOFTWARE MAINTENANCE AND SERVICE GUIDE for current information on DIAGNOSTICS.)

Chapter 7.22 Mass Storage Device Subassembly

Mass storage device subassemblies for the COMPAQ PORTABLE II Personal Computer consist of fixed disk drives and diskette drives. Refer to Figure 7-53 for the location of each type of mass storage device subassembly.

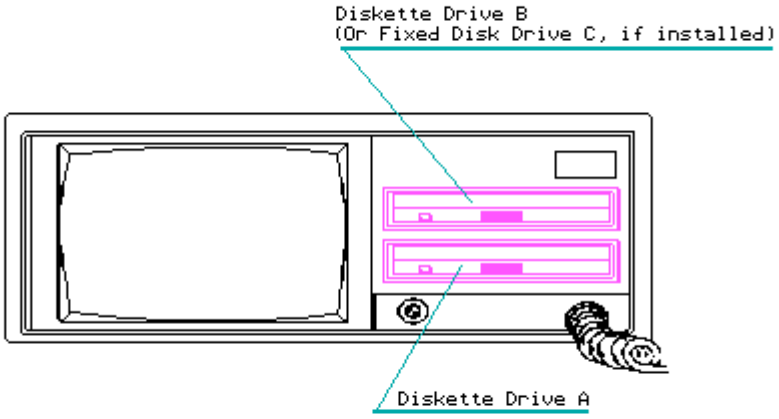


Figure 7-53. Reference Designators for the Mass Storage Device Subassemblies

Mass Storage Device Subassembly Enclosure

To remove the mass storage device subassembly enclosure, complete the following steps:

1. Remove the outer access casing (see Section 7.3).

2. Remove all installed controller and option boards (see Sections 7.14 through 7.16).

3. Using a screwdriver, gently pry the drive bezel from the front bezel (Figure 7-54). This exposes the installed mass storage devices.

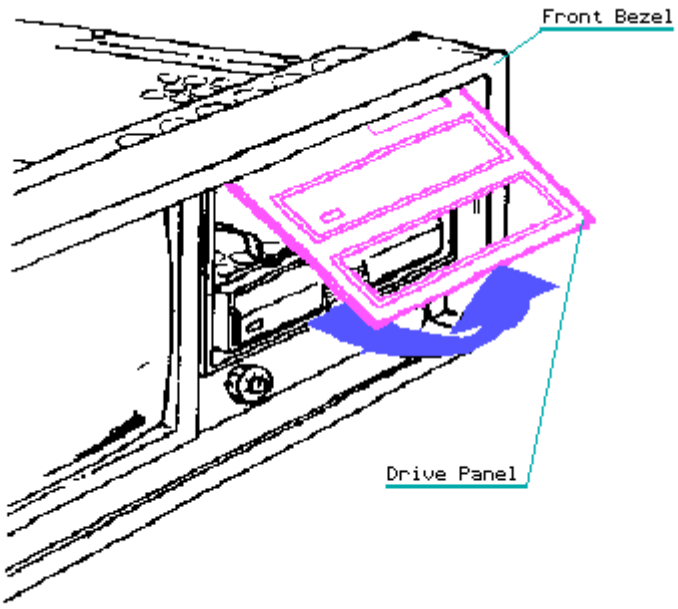


Figure 7-54. Removing the Drive Bezel from the Front Bezel

4. Using a Torx screwdriver, remove the two screws on the right side of the system chassis and the four screws on the top of the mass storage device enclosure. These screws secure the mass storage device subassembly enclosure to the system chassis (Figure 7-55).

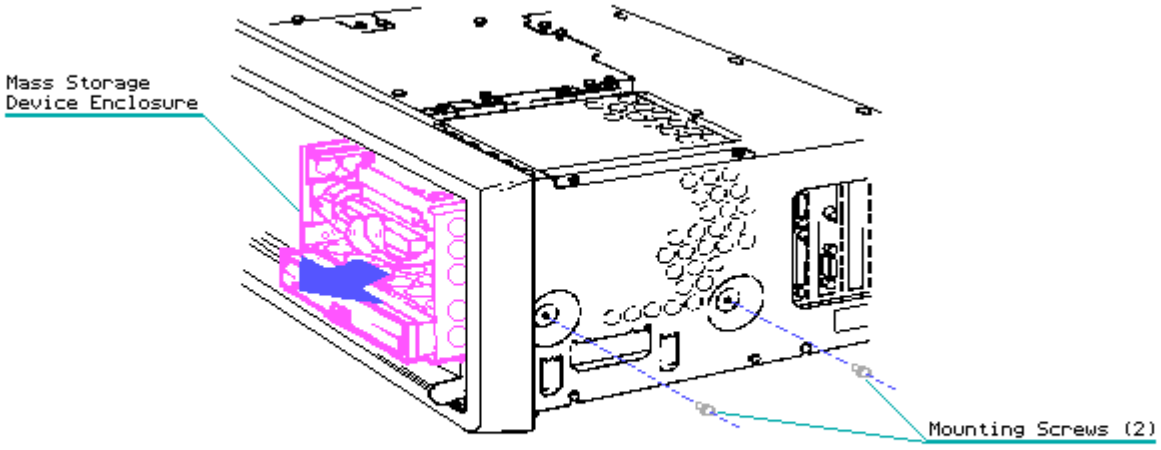


Figure 7-55. Removing the Mass Storage Device Subassembly Enclosure Mounting Screws

5. Slide the mass storage device subassembly enclosure forward a few inches by reaching into the open system chassis area and pushing the enclosure forward. This action will expose the mass storage power cables (Figure 7-56).

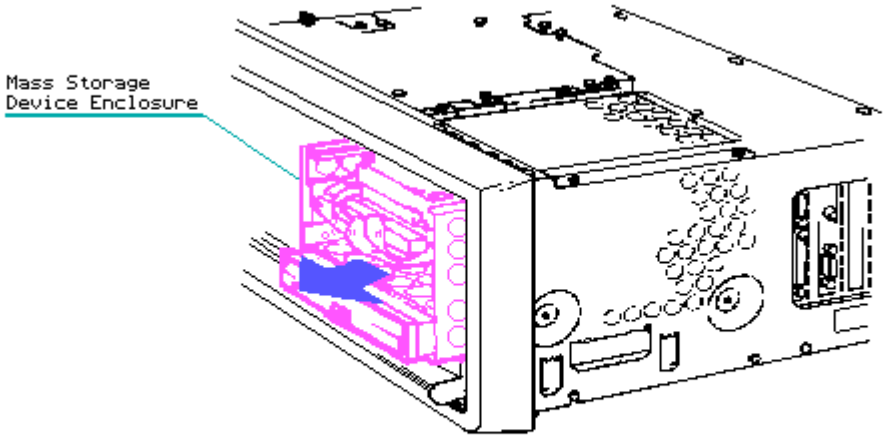


Figure 7-56. Accessing the Diskette Drive and Fixed Disk Drive Power Cable

6. Disconnect the diskette drive and fixed disk drive power cable from the system board (Figure 7-57).

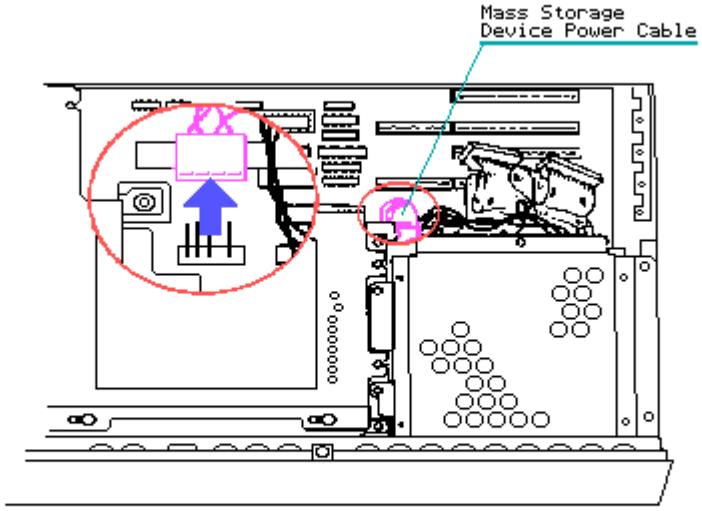


Figure 7-57. Disconnecting the Mass Storage Device Power Cables from the System Board

7. Grasp the sides of the mass storage device subassembly enclosure firmly and slide the enclosure out of the system chassis (Figure 7-58).

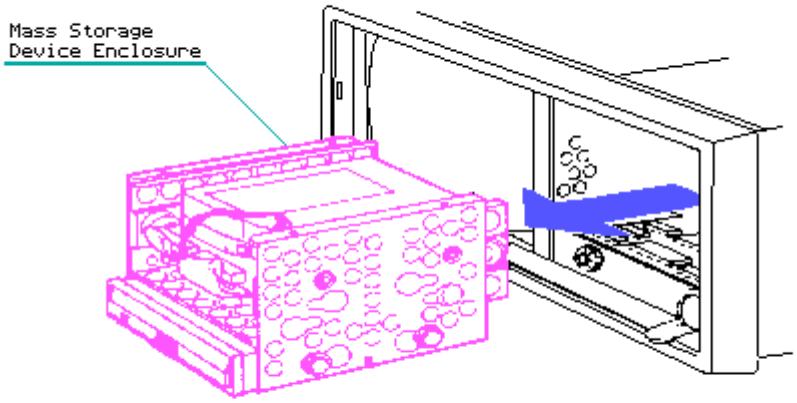


Figure 7-58. Removing the Mass Storage Device Subassembly Enclosure

To replace the mass storage device subassembly enclosure, reverse steps 1 through 7.

Fixed Disk Drive

To remove the fixed disk drive assembly:

1. Remove the mass storage device subassembly enclosure (see Section 7.22).

2. Using a Torx screwdriver, remove the two shoulder bolts and vibration isolators from either side of the mass storage device subassembly enclosure. Slide the fixed disk drive assembly enclosure out of the mass storage device subassembly (Figure 7-59).

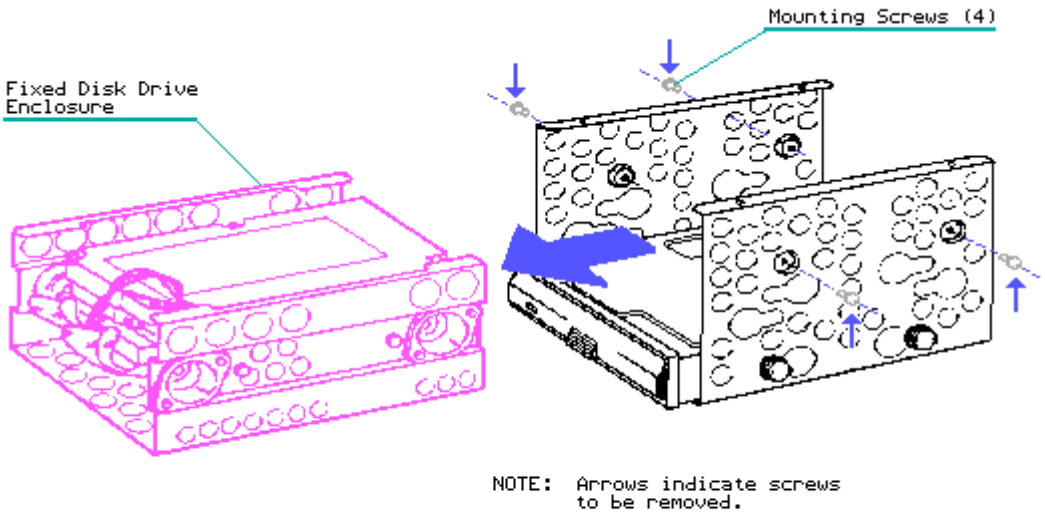


Figure 7-59. Removing the Fixed Disk Drive

NOTE: The fixed disk drive is mounted above the diskette drive.

To replace the fixed disk drive assembly, reverse steps 1 and 2.

Diskette Drive

To remove the diskette drive assembly, complete the following steps:

1. Remove the mass storage device subassembly enclosure (see Section 7.22).

2. Using a Torx screwdriver, remove the two shoulder bolts that secure the diskette disk drive assembly from either side of the mass storage device subassembly enclosure (Figure 7-60).

3. Slide the diskette drive assembly out of the mass storage device subassembly enclosure (Figure 7-60).

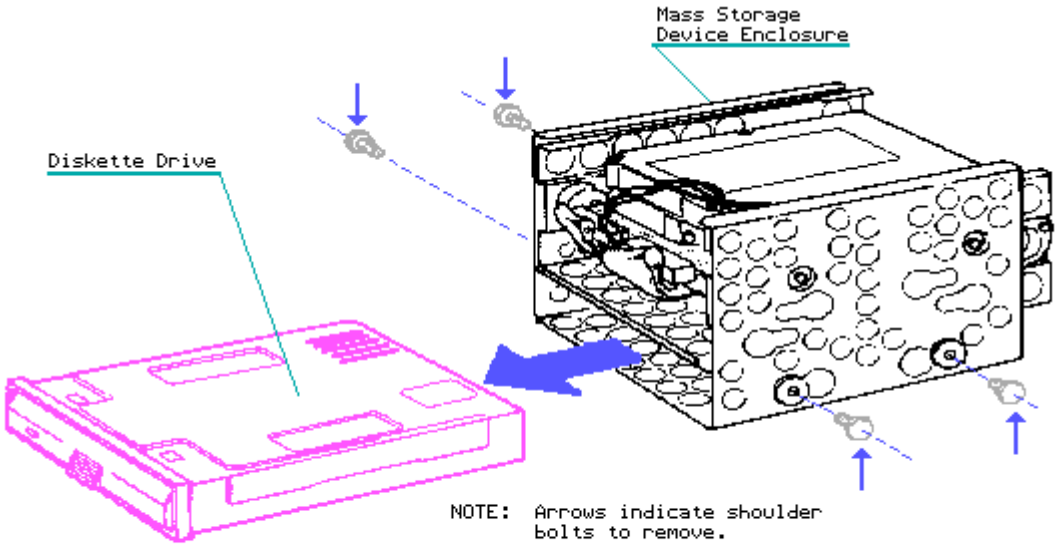


Figure 7-60. Removing the Diskette Drive

NOTE: When removing the diskette drive, make sure the vibration isolators are not lost. You need to reinstall all four isolators when replacing the diskette drive assembly.

To replace the diskette drive assembly, reverse steps 1 through 3.

NOTE: When replacing a flip door style diskette drive with a push button style diskette drive, you should also order a replacement drive bezel (PN 104059-001 (No longer available)) to provide a better fit for the new diskette drive assembly.

Chapter 7.23 Speaker and Brightness Potentiometer

To remove the speaker and brightness potentiometer:

1. Remove the cable tray access plate (see Section 7.13).

2. Remove the mass store device enclosure (see Section 7.22).

3. Remove the front bezel (see Section 7.5).

4. The speaker is secured to the system chassis by a snap in mount and a toe in mount that extend through the system chassis. Remove the speaker by first disconnecting the speaker/brightness control connector from the J114 position on the system board, then pressing the snap in mount inward. At the same time, push the speaker out of the system chassis (Figure 7-61).

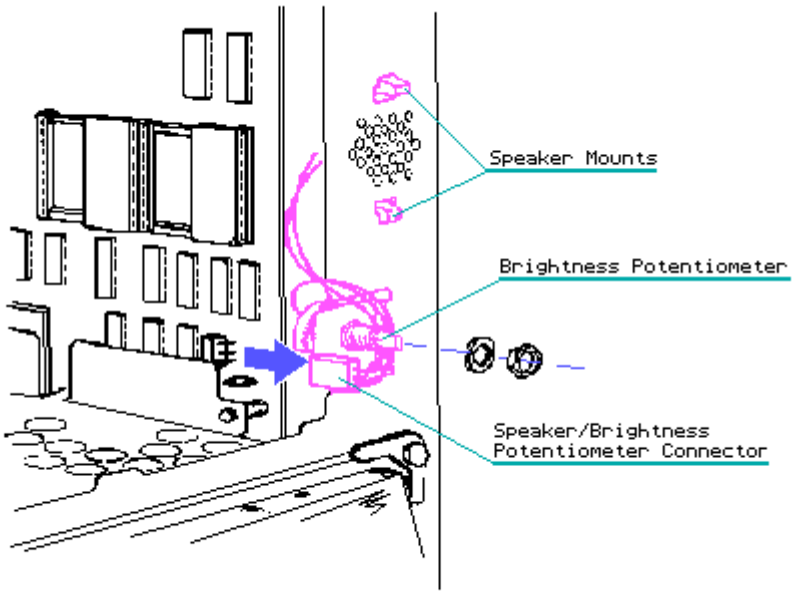


Figure 7-61. Disconnecting the Speaker/Brightness Control Connector and Removing the Speaker

5. Remove the brightness potentiometer by first unscrewing the nut that secures it to the system chassis, then push the brightness potentiometer out of the system chassis (Figure 7-62).

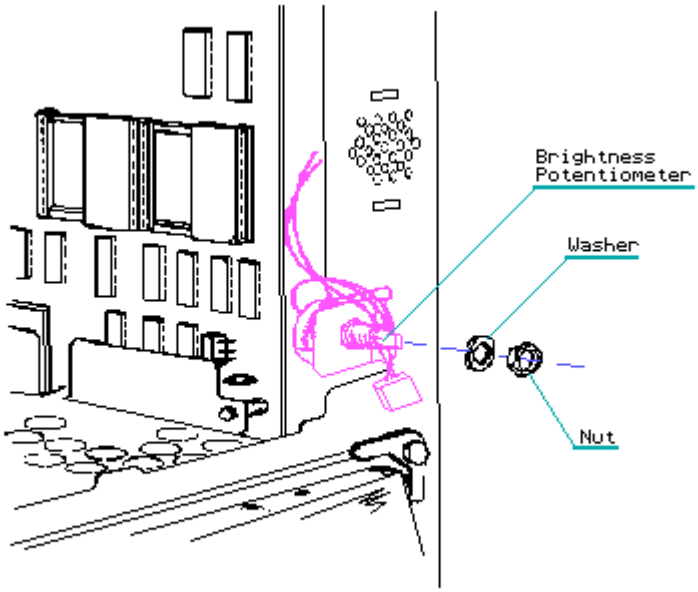


Figure 7-62. Removing the Brightness Potentiometer

To replace the speaker and the brightness potentiometer, reverse steps 1 through 5.

Chapter 7.24 Video Display Unit

To remove the video display unit (VDU), complete the following steps:

1. Remove the keyboard assembly (see Section 7.4).

2. Disconnect the VDU power cable (Figure 7-63).

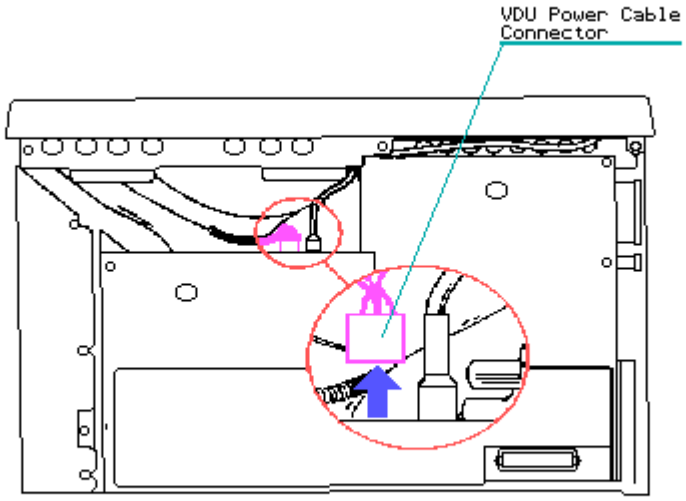


Figure 7-63. Disconnecting the VDU Power Cable

3. Lower the computer so that it appears as illustrated Figure 7-64.

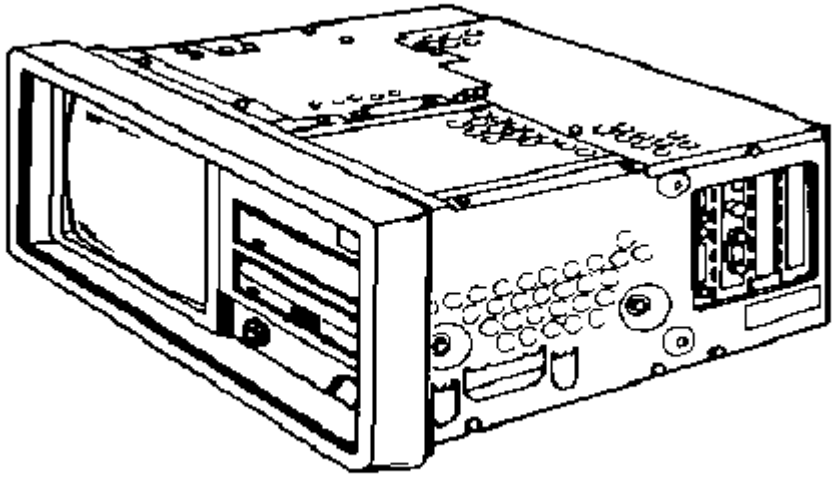


Figure 7-64. Returning the Computer to a Flat Position on the Work Surface

4. Remove the high voltage access plate (see Section 7.9).

5. Remove the front bezel (see Section 7.5)

6. Disconnect the VDU signal cable from the VDU (Figure 7-65).

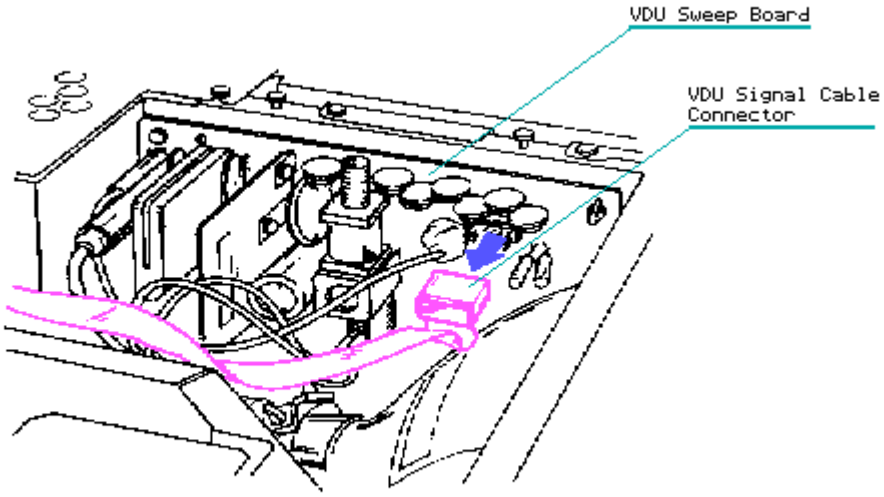


Figure 7-65. Disconnecting the VDU Signal Cable

7. The VDU high voltage cable is secured to the system chassis by a push mount with a snap connector at the end. Disconnect the VDU high voltage cable by first unsnapping the push mount and then removing the cable from its place inside the connector (Figure 7-66).

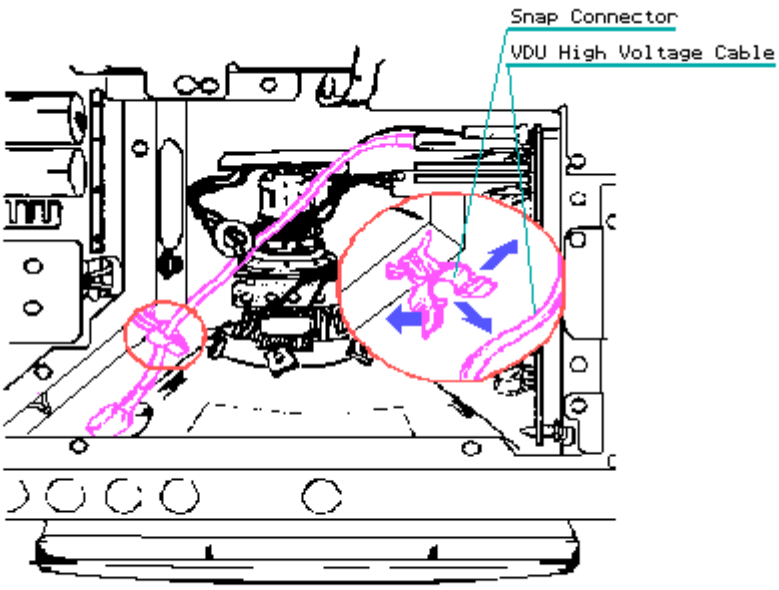


Figure 7-66. Disconnecting and Removing the VDU High Voltage Cable

8. The VDU sweep board is secured to the system chassis by five PCB stand off mounts and one screw. Using a Torx screwdriver, remove the grounding screw in the upper left corner of the VDU sweep board. To remove the VDU sweep, grasp the board and gently slide it upward until the standoffs clear the mounting holes. Once the VDU sweep board clears all standoffs, slide the board out of the mounting holes and let it come to rest on the side of the VDU (Figure 7-67).

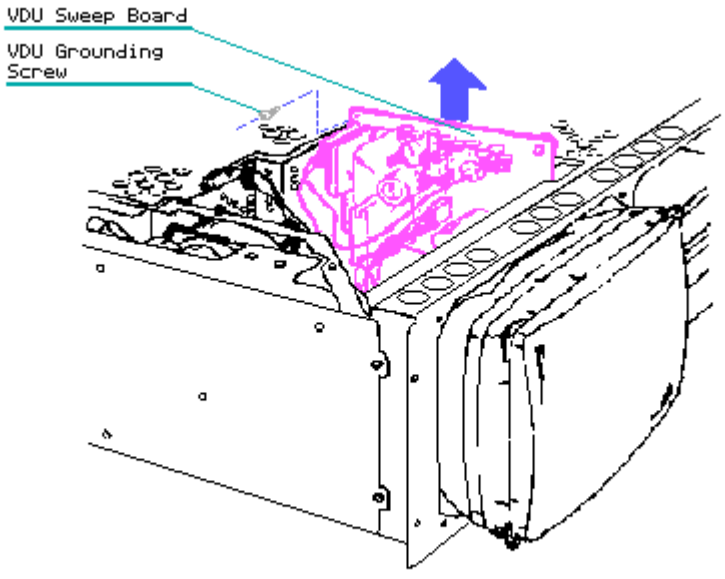


Figure 7-67. Removing the VDU Sweep Board

9. Remove the four screws from the front of the video display unit (Figure 7-68).

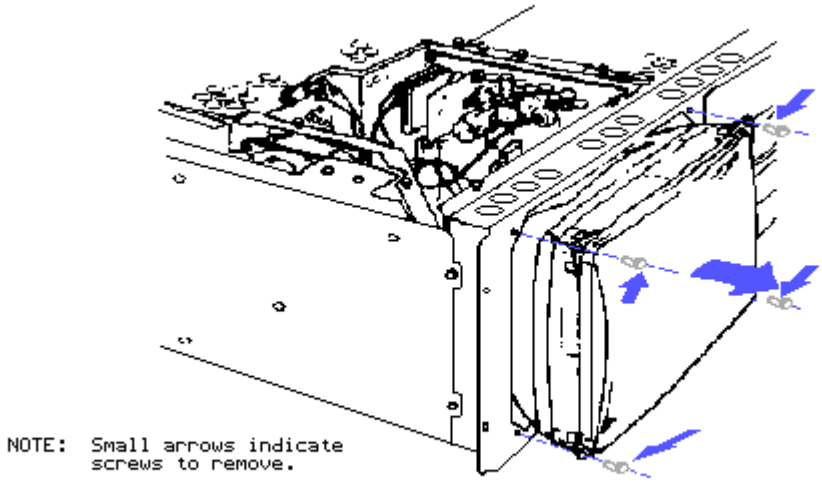


Figure 7-68. Removing the Four VDU Mounting Screws

10. Remove the VDU from the system chassis by grasping the display screen and the VDU sweep board and then sliding the unit until it is out of the system chassis (Figure 7-69).

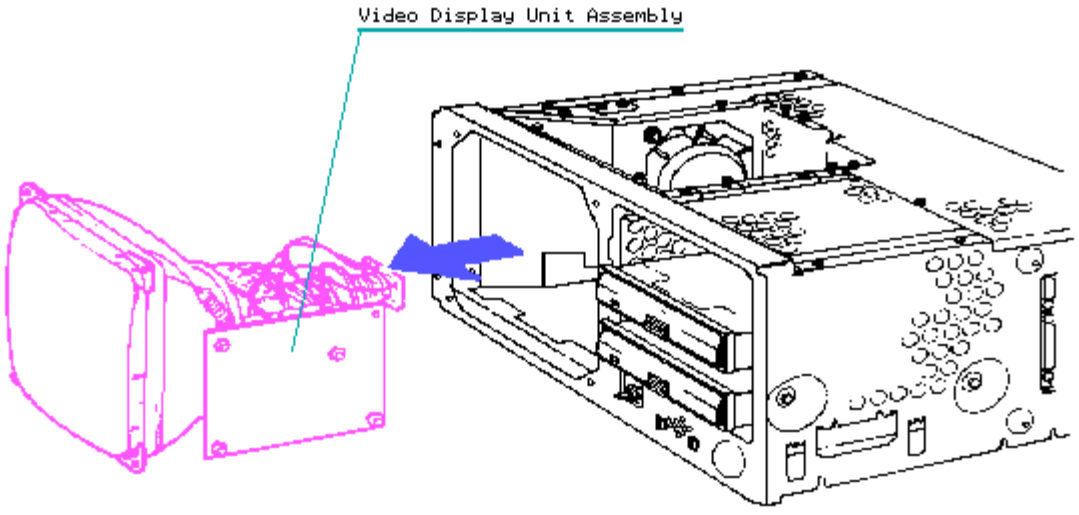


Figure 7-69. Removing the Video Display Unit from the System Chassis

If replacement is necessary, this unit may be shipped back to Compaq Computer Corporation packed only in a Compaq Spare VDU box.

To replace the video display unit, reverse steps 1 through 10.

NOTE: When replacing the VDU assembly, replace the two top front mounting screws first. This helps align the screen correctly.

>>>

Chapter 8. Jumper Settings, Switch Settings, and Adjustments

8.1 System Board Jumper Settings

Figure 8-1 shows the jumper locations on the COMPAQ PORTABLE II Version 1 system board. Refer to Table 8-1 when setting the jumpers on this board.

Figure 8-2 shows the jumper locations on the COMPAQ PORTABLE II Version 2 system board. Refer to Table 8-1 when setting the jumpers on this board.

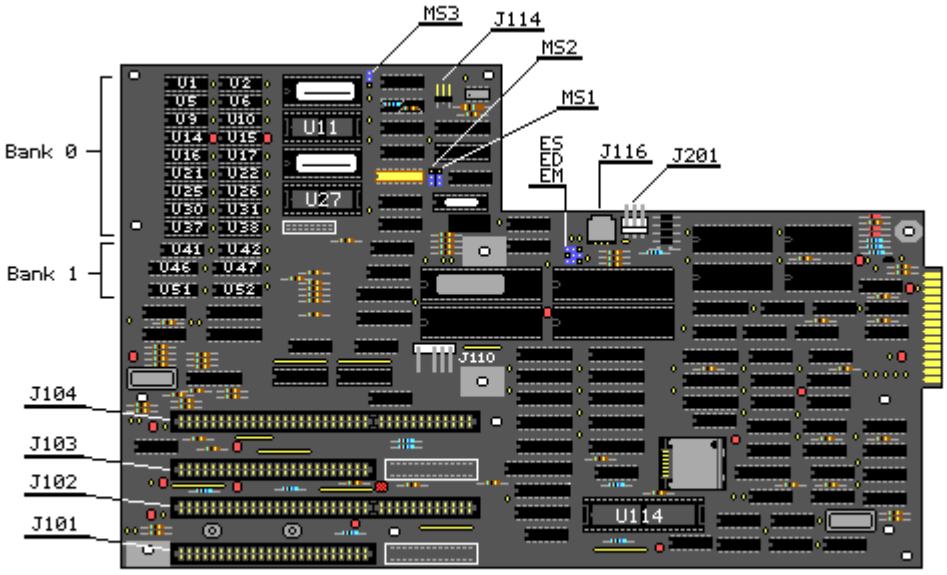


Figure 8-1. COMPAQ PORTABLE II Personal Computer System Board - Version 1 (Assy No. 000318)

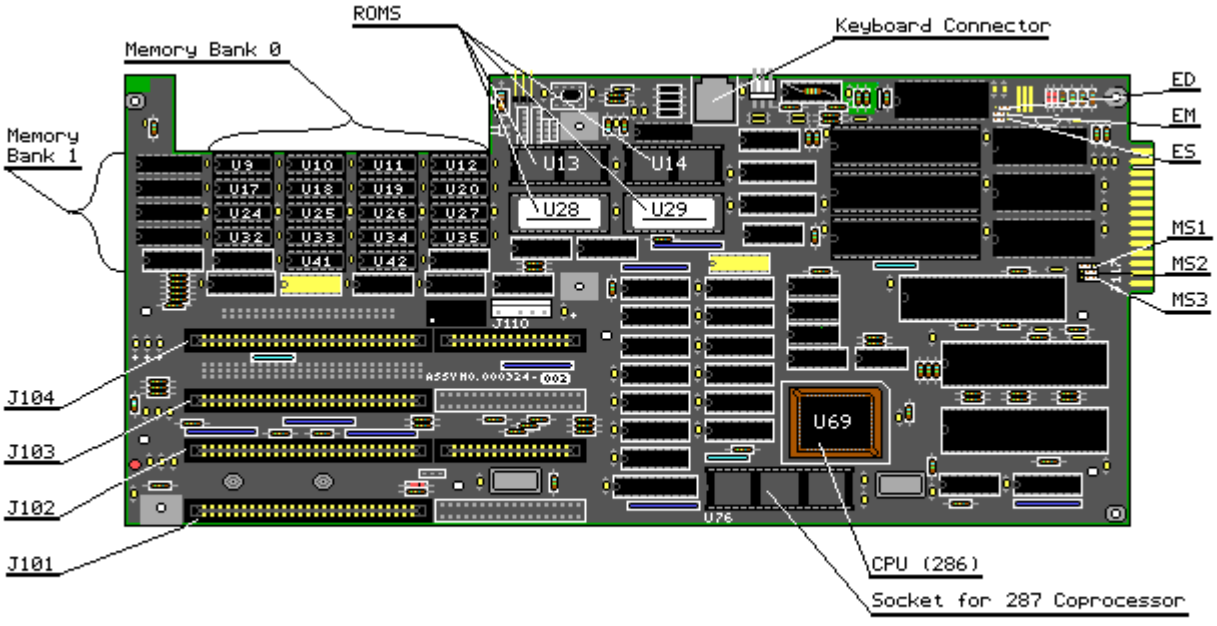


Figure 8-2. COMPAQ PORTABLE II System Board Version 2 (Assy No. 000324)

Table 8-1. System Board Jumper Settings

Jumper Location	Function	Jumper Setting	
ED	Video Display Unit selection	Pins 1-2	Third party monochrome monitor (MDA) installed
		Pins 2-3	COMPAQ VDU, EGA, or RGBI installed (default)
ES	Speed selection	Pins 1-2	Primary or fast speed, 8 MHz (default)
		Pins 2-3	Secondary or common speed, 6 MHz
EM	Reserved	DO NOT CHANGE SETTINGS	
MS1, MS2	System board memory selection	MS1 = G MS2 = G	disable memory and ROM
		MS1 = V MS2 = G	256 Kbyte memory installed
		MS1 = G MS2 = V	512 Kbyte memory installed
		MS1 = V MS2 = V	640 Kbyte memory installed (default)
MS3	Memory Board	MS3 = 1.0 or V	1 Megabyte memory recognized
	512/1536 Kbyte memory selection	MS3 = 1.5 or G	1.5 Megabytes memory recognized (default)

Chapter 8.2 512/2048 KByte Memory Expansion Board Jumper Settings

The jumper settings on the 512/2048 Kbyte Memory Expansion Board correspond to the amount of random access memory installed in the four banks on the option board. There are four possible memory configurations that expand the memory address range from 1.0 megabyte to 3.0 megabytes. Figure 8-3 shows the component layout for the 512/2048 Kbyte Memory Expansion Board.

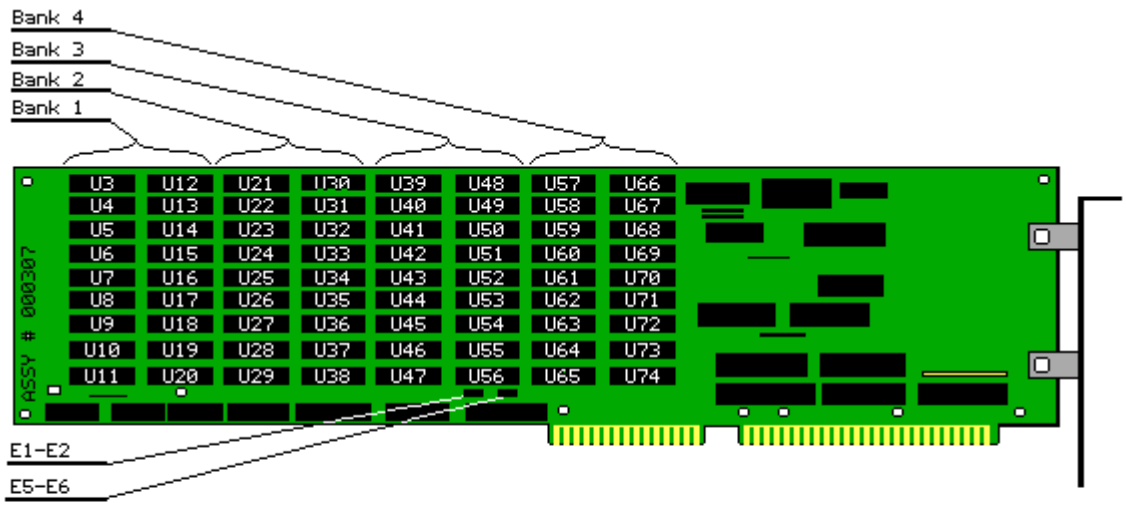


Figure 8-3a. 512/2048 Kbyte Memory Expansion Board (Assy No. 000307) (Part 1 of 2)

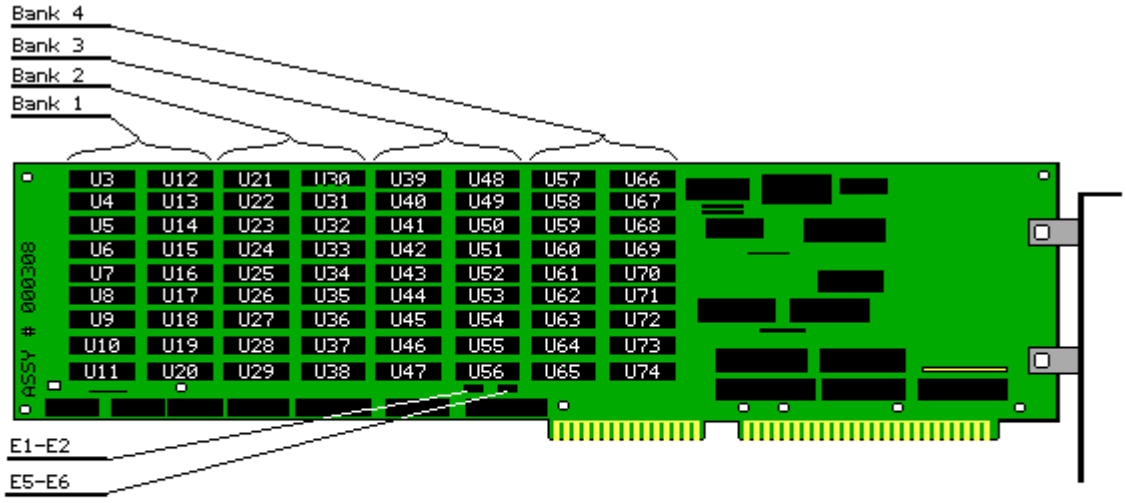


Figure 8-3b. 512/2048 Kbyte Memory Expansion Board (Assy No. 000308) (Part 2 of 2)

Refer to Tables 8-2 and 8-3 when setting the jumpers on the 512/2048 Kbyte Memory Expansion Board.

Table 8-2. 512/2048 Kbyte Memory Expansion Board Jumper Settings

Settings	Address Range *
E1 to E2, E5 to E6	1 to 3 megabyte
E1 to E2, E4 to E5	2.5 to 4.5 megabyte

* Memory address range is dependent on system memory board configuration.

Table 8-3. 512/2048 Kbyte Memory Expansion Board Memory Configuration for Jumper Settings

Banks Installed	Resulting Expansion Memory
Bank 1	512 Kbytes
Banks 1 and 2	1024 Kbytes
Banks 1,2 and 3	1536 Kbytes
Banks 1, 2, 3, and 4	2048 Kbytes

Chapter 8.3 360 KByte Diskette Drive Switch Settings

Figure 8-4 shows the COMPAQ PORTABLE II 360 Kbyte Diskette Drive. This diskette drive contains an 8 position switch. Refer to Table 8-4 when setting switch positions.

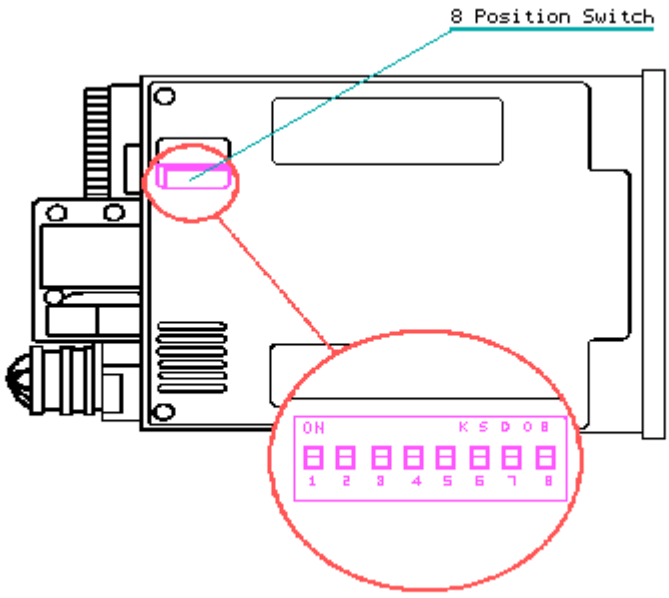


Figure 8-4. 360 Kbyte Diskette Drive

Table 8-4. 360 Kbyte Diskette Drive Switch Settings

Switch Position	Function	Switch Setting
1	Reserved	ON
2	Reserved	OFF
3	Reserved	ON
4	Reserved	OFF
5	Reserved	OFF

6	Reserved	OFF
7	Reserved	OFF
8	Reserved	ON

Chapter 8.4 Multipurpose Controller Board Switch Settings

Figures 8-5 through 8-9 show the switch and jumper locations on the Multipurpose Controller Boards and the Multipurpose Fixed Disk Drive Controller Boards. Refer to Tables 8-5 through 8-7 when setting the switches and jumpers.

Table 8-5. Multipurpose Controller Board Switch and Jumper Settings

Jumper	Pin 1 to Pin 2	Pin 2 to Pin 3
J1	Secondary diskette controller address	Primary diskette controller address (default)
J2 *	Serial interface COM1	Serial interface COM2
J4 *	Serial interface IRQ4	Serial interface IRQ3
J3	Disable parallel interface	Enable parallel interface

* J2 and J4 must be changed together.

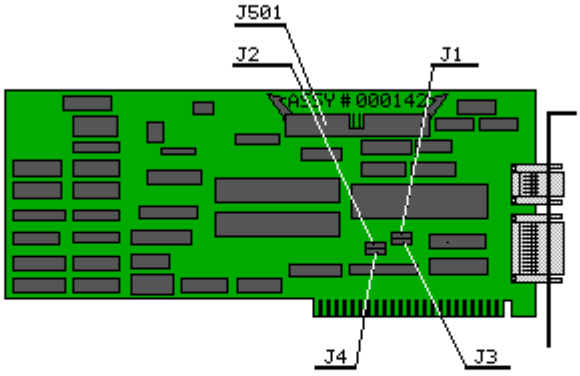


Figure 8-5. Multipurpose Controller Board Switch and Jumper Settings (Assy No. 000142)

Table 8-6. Multipurpose Fixed Disk Drive Controller Board Switch and Jumper

Settings (Assy Nos. 000181-001, 000181-021, 000336-001, and 000336-021)

Jumper/Switch	Setting	Function
J1	Open connectors on left	Serial interface COM1 IRQ4 (default)
	Open connectors on right	Serial interface COM2 IRQ3
J2	Open connectors on left	Primary diskette controller address (default)
	Open connectors on right	Secondary diskette controller address
SW1-1 *	ON	Enable fixed disk drive
	OFF	Disable fixed disk drive
SW1-2	ON	Enable parallel interface (default)
	OFF	Disable parallel interface
SW1-3 **	ON	Enable serial interface (default)
	OFF	Disable serial interface
SW1-4	OFF	Reserved

* SW1-1 reserved on Assy No. 000181 boards.
** SW1-3 reserved on Assy No. 000181-001 (all Revisions) and Assy No. 000336-001 Revision P or earlier boards.

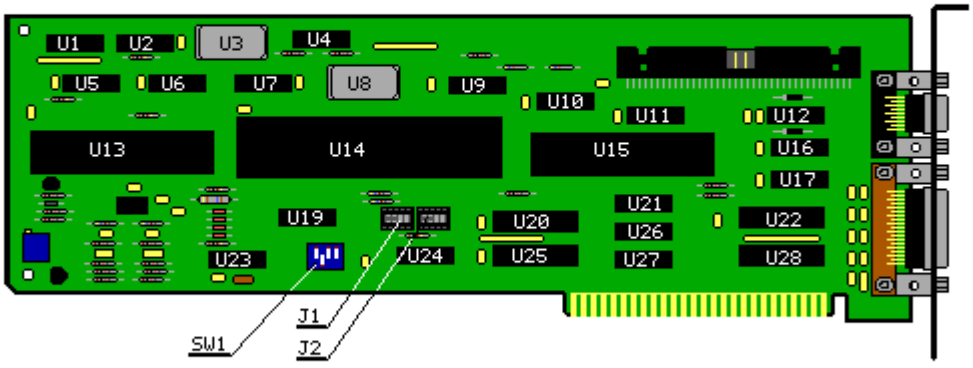


Figure 8-6. Multipurpose Controller Board Switch and Jumper Locations (Assy Nos. 000181-001 and 000181-021)

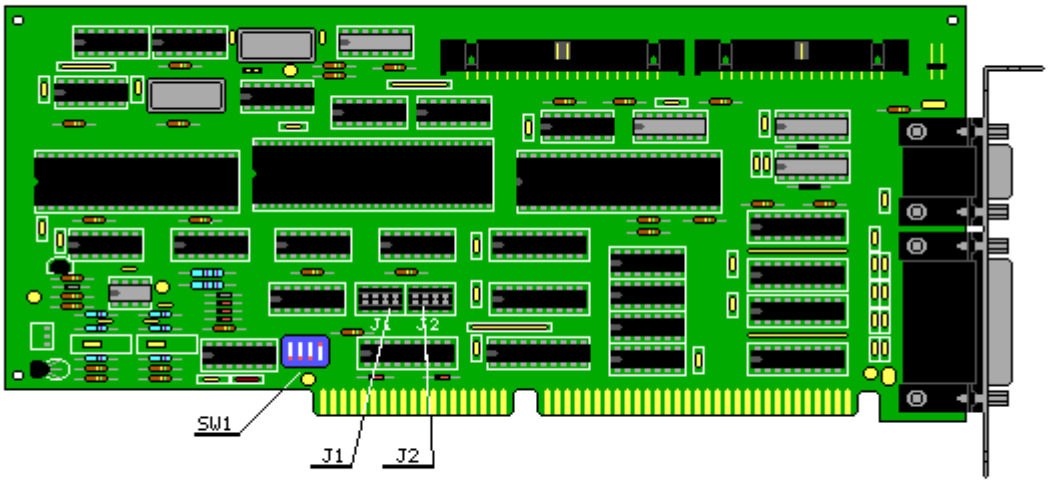


Figure 8-7. Multipurpose Fixed Disk Drive Controller Board Switch and Jumper Locations (Assy No. 000336)

Table 8-7. Multipurpose Fixed Disk Drive Controller Boards Switch and Jumper Settings (Assy Nos. 000519 and 000815)

Switch	Setting	Function
SW500-1	OFF	Primary diskette and fixed disk drive address select (default)
	ON	Secondary diskette and fixed disk drive address select
SW500-2	OFF	Disable high speed transfer rates for systems without 1.2 megabyte diskette drive
	ON	Enable high speed transfer rates for systems with 1.2 megabyte diskette drive (default)
SW500-3	OFF	Enable fixed disk drive (default)
	ON	Disable fixed disk drive
SW500-4	OFF	Serial interface COM1 IRQ4 (default)
	ON	Serial interface COM2 IRQ3
SW500-5	OFF	Enable serial interface (default)
	ON	Disable serial interface
SW500-6	OFF	Enable printer interface (default)
	ON	Disable printer interface

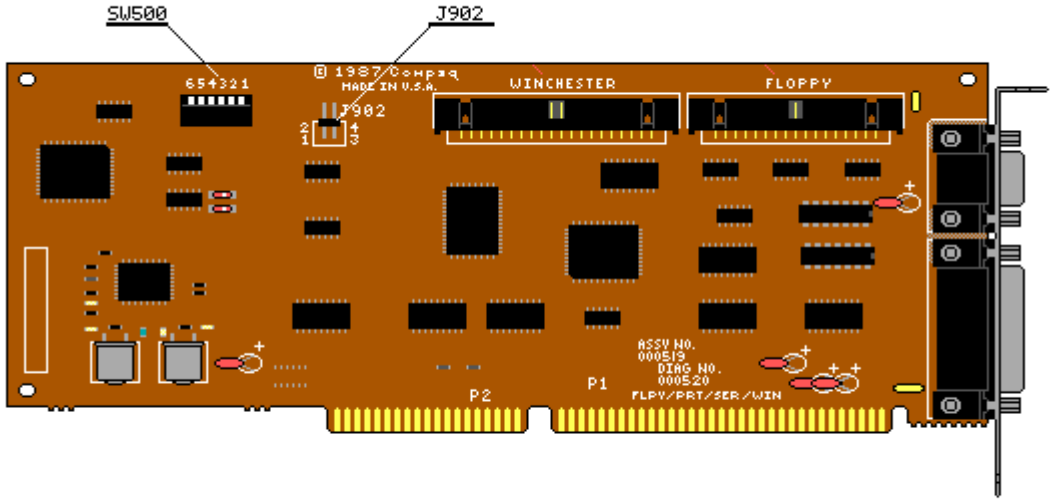


Figure 8-8. Multipurpose Fixed Disk Drive Controller Board Switch and Jumper Locations (Assy No. 000519)

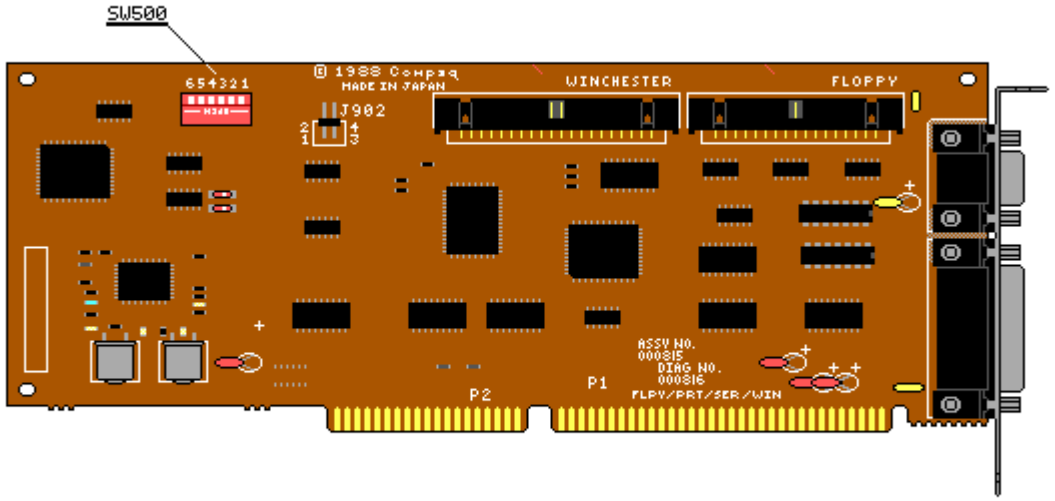


Figure 8-9. Multipurpose Fixed Disk Drive Controller Board Switch and Jumper Locations (Assy No. 000815)

Chapter 8.5 Fixed Disk Drive Controller Board

Figure 8-10 shows the jumper locations for the Fixed Disk Drive Controller Board (Assy No. WD1003IWH). Refer to Table 8-8 when setting the jumpers.

Table 8-8. Fixed Disk Drive Controller Board Jumper Settings
(Assy No. WD1003IWH)

Jumper	Pin	Function

W1		Reserved
W2		Reserved
W3	1-2	20 Megabyte Fixed Disk Drive
	2-3	10 Megabyte Fixed Disk Drive
W4		Reserved

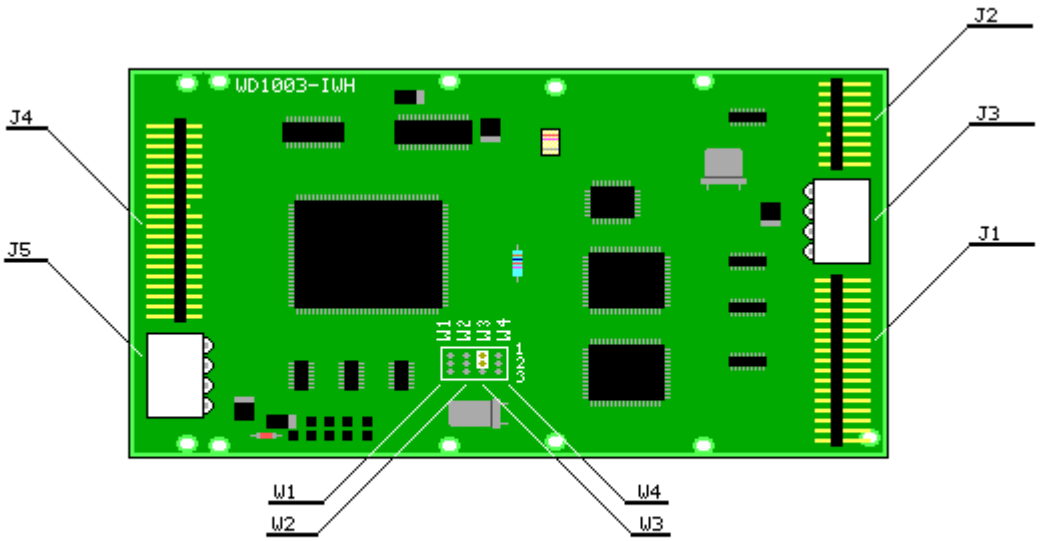


Figure 8-10. Fixed Disk Drive Controller Board Jumper Locations (Assy No. WD1003IWH)

Chapter 8.6 Video Display Controller Board Jumper Settings

Figures 8-11 through 8-14 show the jumper locations on the Video Display Controller Board. Refer to Tables 8-9 and 8-10 when setting the jumpers.

Table 8-9. Video Display Controller Board Switch and Jumper Settings (Assy Nos. 000031, 000160, and 000345)

J3	J5	Function
2-3	2-3	Enable RGBI and composite video - high scan
1-2	1-2	Disable RGBI and composite video - high scan (default)

Table 8-10. Video Display Controller Board Jumper Settings (Assy No. 000525)

Jumper Location	Function	Jumper Setting

J3/J5	RGB enable/disable (high scan)	Pin 1 to Pin 2 - Disables RGB and composite video Pin 2 to Pin 3 - RGB and composite video enable

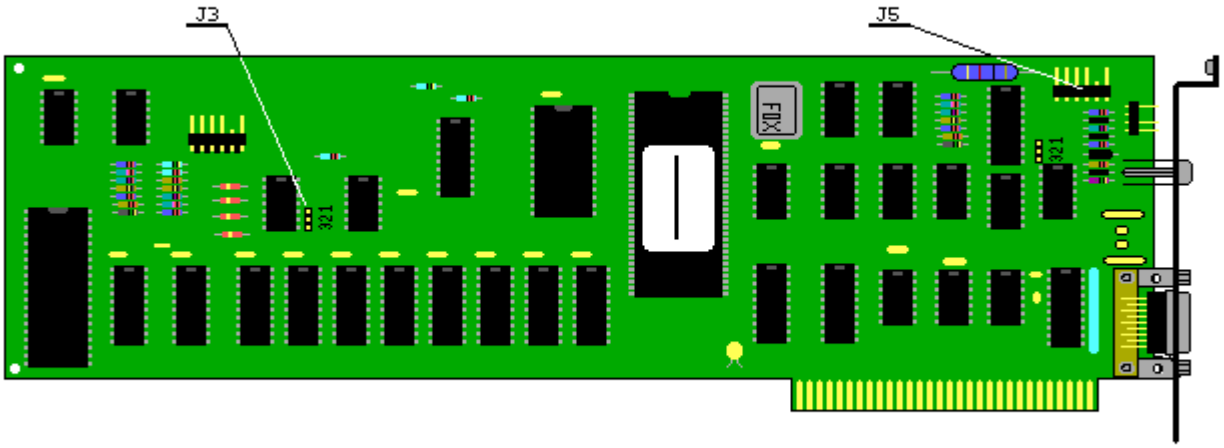


Figure 8-11. Video Display Controller Board Jumper Locations (Assy No. 000031)

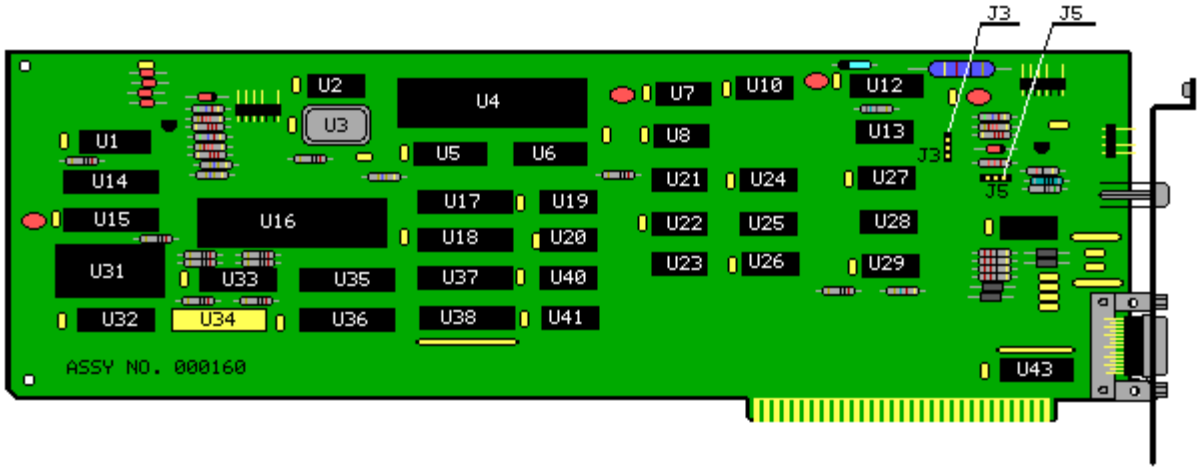


Figure 8-12. Video Display Controller Board Jumper Locations (Assy No. 000160)

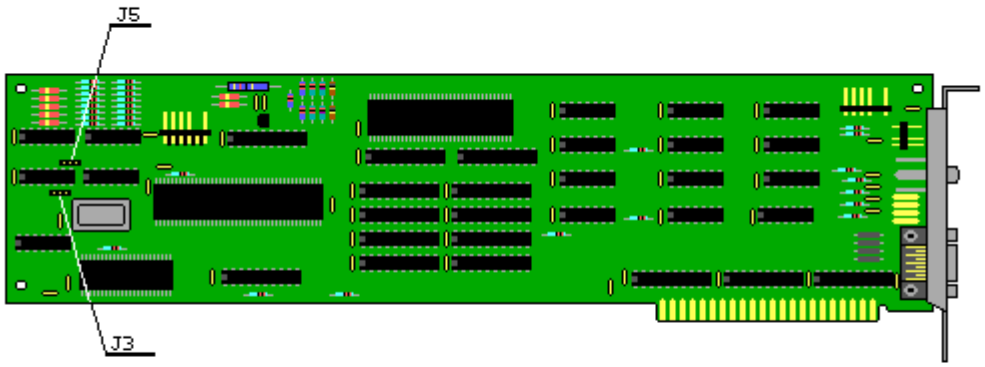


Figure 8-13. Video Display Controller Board Jumper Locations (Assy No. 000345)

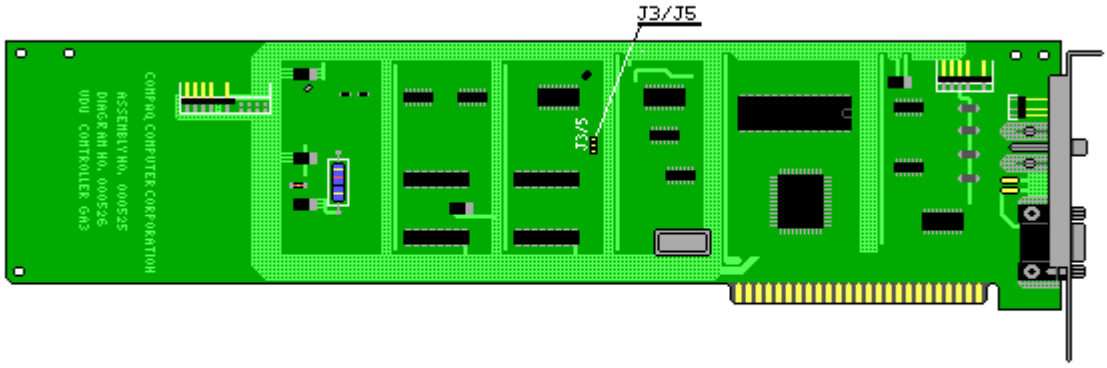


Figure 8-14. Video Display Controller Board Jumper Locations (Assy No. 000525)

Chapter 8.7 Enhanced Color Graphics Board Switch and Jumper Settings

Figures 8-15 and 8-16 show the switch and jumper locations on the Enhanced Color Graphics Boards. Refer to Table 8-11 when setting the switches and jumpers.

Table 8-11. Enhanced Color Graphics Board Switch Settings
(Assy Nos. 000410 and 000471)
==========

SW1 Settings (As the Only or Primary Video Display Controller Board)

1	2	3	4	Monitor Type	Power On Mode	Mode
OFF	ON	ON	OFF	COMPAQ Color, Compatible Enhanced Color or COMPAQ Dual Mode	80 x 25 (640 x 350)	Color
ON	ON	ON	OFF	COMPAQ Color, Compatible Enhanced Color or COMPAQ Dual Mode	80 x 25 (640 x 200)	Color
OFF	OFF	OFF	ON	RGBI Color or COMPAQ Dual Mode	80 x 25 (640 x 200)	Color
ON	OFF	OFF	ON	RGBI Color or COMPAQ Dual Mode	40 x 25 (320 x 200)	Color
OFF	OFF	ON	OFF	COMPAQ Dual Mode	80 x 25 (720 x 350)	Monochrome
OFF	OFF	ON	ON	COMPAQ Color, Compatible Enhanced Color or COMPAQ Dual Mode	80 x 25 (640 x 350)	Color
ON	OFF	ON	ON	COMPAQ Color, Compatible Enhanced Color or COMPAQ Dual Mode	80 x 25 (640 x 200)	Color
OFF	ON	ON	ON	RGBI Color or COMPAQ Dual Mode	80 x 25 (640 x 200)	Color
ON	ON	ON	ON	RGBI Color or COMPAQ Dual Mode	40 x 25 (320 x 200)	Color
OFF	ON	OFF	ON	COMPAQ Dual Mode	80 x 25 (720 x 350)	Monochrome

SW2 Settings

1	2	3	4	5	6	Monitor Type
OFF	OFF					COMPAQ Color or Compatible Enhanced Color (external)
ON	OFF					RGBI Color (external)
OFF	ON					COMPAQ Dual Mode (external)
ON	ON					No external (Portables)
		OFF				COMPAQ Dual Mode internal
		ON				No internal (Desktops)
			ON	OFF	ON	Reserved

NOTE: SW3 Settings (Assy No. 000410 only): If installed, all OFF (Reserved)

Jumper P1 (External Monitor)

Pins	Monitor Type
1 - 2	COMPAQ Color or Enhanced Color
2 - 3	RGBI Color or COMPAQ Dual Mode

Jumper JP3	
Pins	Address Select
1 - 2	3xxh (default)
2 - 3	2xxh

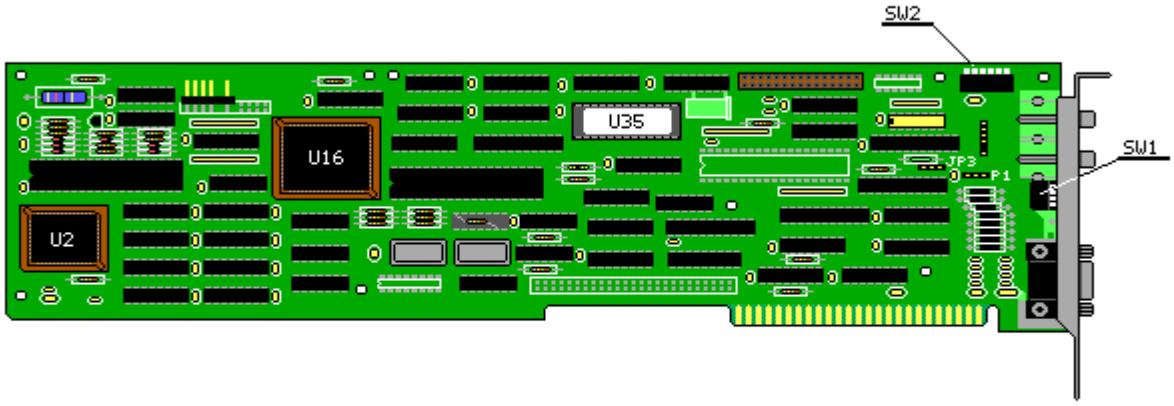


Figure 8-15. Enhanced Color Graphics Board Switch and Jumper Locations (Assy No. 000410)

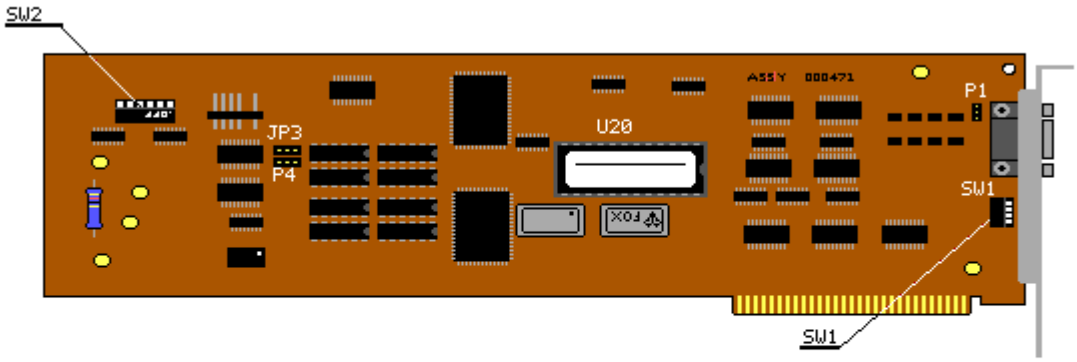


Figure 8-16. Enhanced Color Graphics Board Switch and Jumper Locations (Assy No. 000471)

Chapter 8.8 Serial/Parallel Interface Board Switch Settings

Figures 8-17 and 8-18 show the switch locations on the Serial/Parallel Interface Board. Refer to Tables 8-12 and 8-13 when setting the switches.

Table 8-12. Serial/Parallel Interface Board Switch Settings (Assy No. 000570)

1	2	3	4	5	6	Function
OFF						Reserved
	ON					Enable asynchronous interface
	OFF					Disable asynchronous interface
		ON	ON			Parallel printer interface LPT1
		OFF	ON			Parallel printer interface LPT2
		ON	OFF			Parallel printer interface LPT3
		OFF	OFF			Disable parallel printer interface
				ON	ON	Asynchronous interface COM1
				OFF	ON	Asynchronous interface COM2

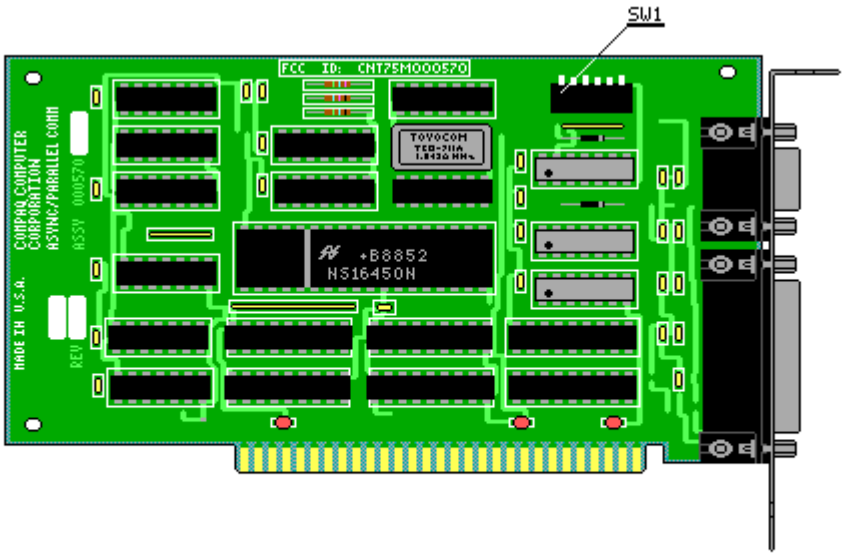


Figure 8-17. Serial/Parallel Interface Board Switch Location (Assy No. 000570)

Table 8-13. Serial/Parallel Interface Board Switch Settings (Assy No. 000990)

1	2	3	4	5	6	Function
OFF						Reserved
	OFF					Reserved
		ON	ON			Parallel printer interface LPT1

OFF	ON		Parallel printer interface LPT2
ON	OFF		Parallel printer interface LPT3
OFF	OFF		Disable parallel printer interface
		ON ON	Asynchronous interface COM1
		OFF ON	Asynchronous interface COM2
		OFF OFF	Disable asynchronous interface

==========

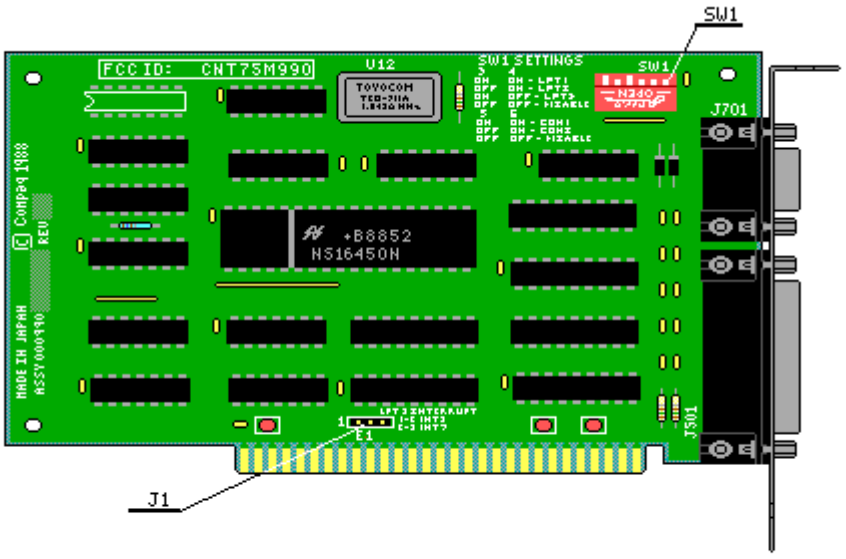


Figure 8-18. Serial/Parallel Interface Board Switch and Jumper Locations (Assy No. 000990)

Chapter 8.9 Video Display Unit Adjustments

The video display unit adjustment (also referred to as CRT adjustment) provides the method for performing any required video display adjustment on your COMPAQ PORTABLE II Personal Computer. To complete the adjustment procedure, the following equipment is required:

- All tools normally required to service a COMPAQ PORTABLE II Personal Computer (see Preface)
- DIAGNOSTICS diskette
- Set of standard television adjustment tools
- COMPAQ PORTABLE II Personal Computer display template (PN 100686-003)

The video display adjustment procedure is arranged in a step by step outline. Read the entire procedure carefully before starting the procedure. Complete each step before continuing to the next step.

CRT Adjustment Procedure:

1. Remove the outer access casing. See Chapter 7 in this guide for the procedure.

2. Place your COMPAQ PORTABLE II Personal Computer in a standard working position with the outer access casing removed (Figure 8-19).

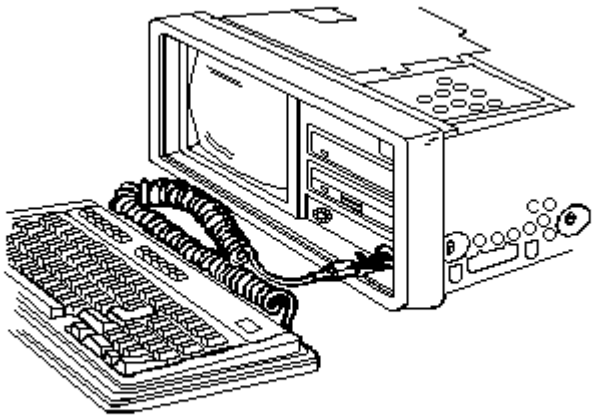


Figure 8-19. Standard Working Position

3. Insert the DIAGNOSTICS diskette in the COMPAQ PORTABLE II Personal Computer and load the DIAGNOSTICS program. (Refer to Chapter 4 in this guide for more information on the COMPAQ PORTABLE II Personal Computer DIAGNOSTICS program.)

4. Select the video display alignment procedure from the video display unit test menu and press ENTER.

After you select the video display adjustment procedure, the screen will appear completely blank. This is normal program operation.

5. Turn the external brightness knob (located on the front bezel) fully clockwise (Figure 8-20.)

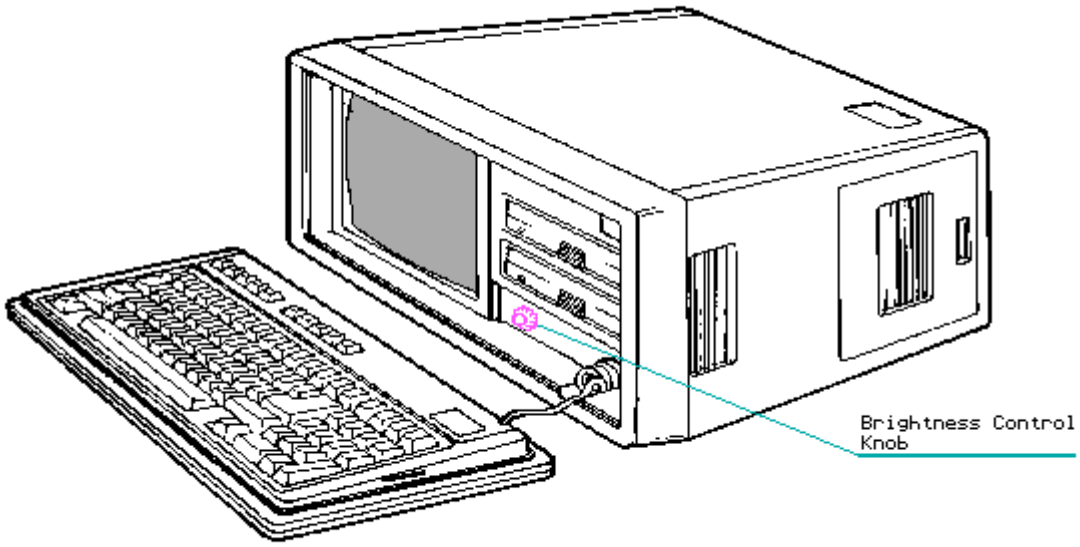


Figure 8-20. Location of Brightness Control Knob

6. Using the label on the high voltage access plate as a guide, locate and adjust the BRITE control until the screen raster appears on the display. Check this display to make sure it is not tilted in any way (Figure 8-21.)

   NOTE: Do not use excessive force when making adjustments on the video display unit.

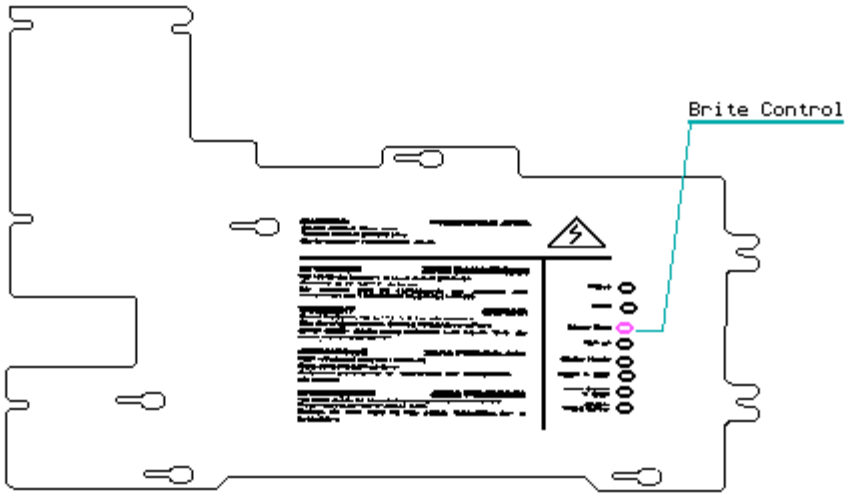


Figure 8-21. Adjusting the CRT BRITE Control

7. Turn the brightness control knob until the raster disappears.

8. Using the VDU adjustment procedure contained in the DIAGNOSTICS program, fill the display screen with a character (refer to Chapter 4 in this guide for more information on the VDU alignment test screen).

9. Using a plastic adjustment tool, turn the WIDTH adjustment potentiometer until the left and right edges of the display screen fall within the range specified on the adjustment template. Width may change between high and low resolution mode and should be adjusted appropriately (Figure 8-22).

   NOTE: When making adjustments on the video display unit do not use excessive force.

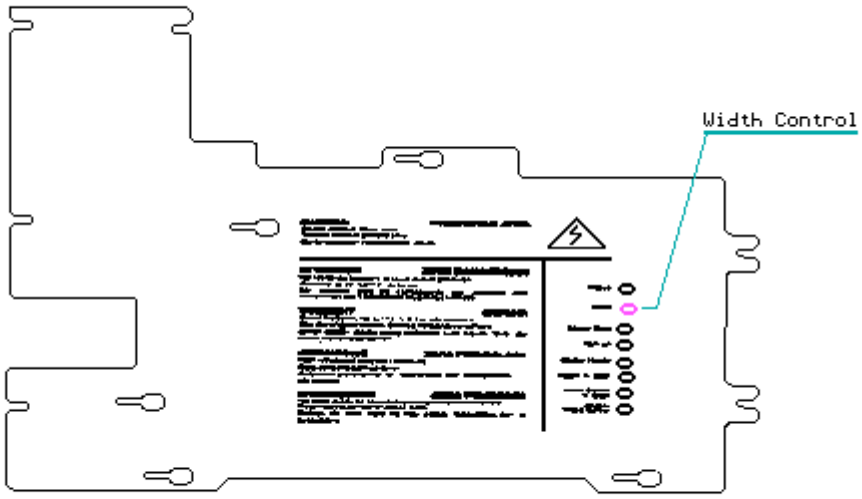


Figure 8-22. Adjusting the CRT Width Control

10. If the characters on your display are not sharp and clear, a focus adjustment is necessary. Using a plastic adjustment tool, adjust the potentiometer labeled FOCUS until all characters are clear and sharp (Figure 8-23).

    NOTE: Do not use excessive force when making adjustments on the video display unit.

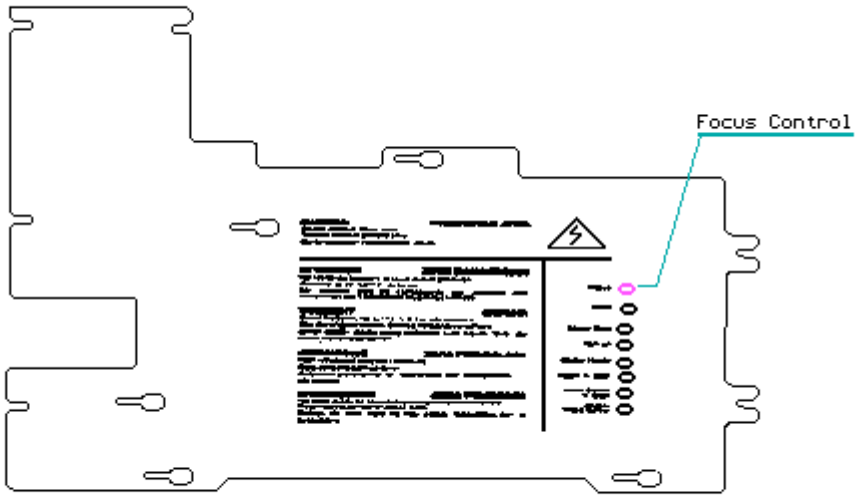


Figure 8-23. Adjusting the Focus Control

11. If the characters on your display appear too short or too tall, adjust the character height. Using the VDU adjustment DIAGNOSTIC program, fill the screen with capital Es (Figure 8-24). See Chapter 4 in this guide for more information on the VDU adjustment program.

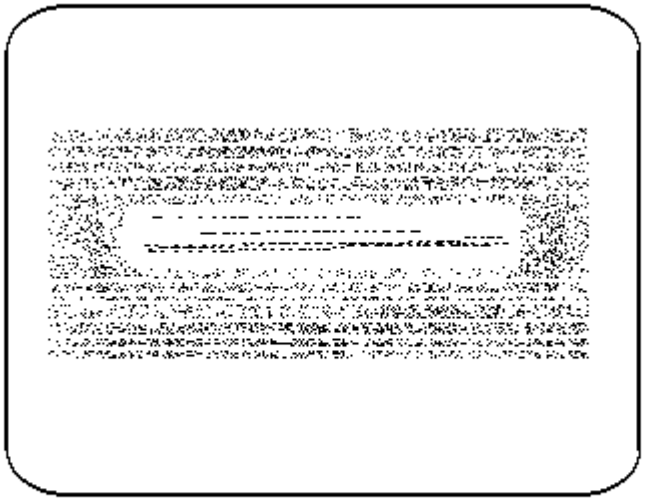


Figure 8-24. Character Height Adjustment

12. Two adjustments control the character height. The first is the master height control potentiometer, which controls the height of characters in both the high and low resolution modes. The second is the high scan height control potentiometer, which controls the height of characters

only while the computer is in a high resolution text mode. (Refer to Figure 8-25 for the location of both control potentiometers.)

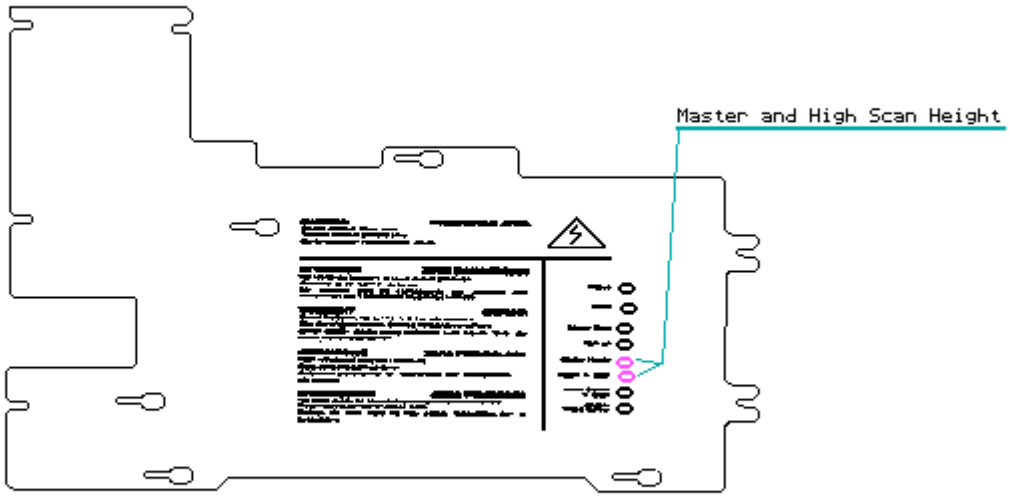


Figure 8-25. Location of Master Height and High Scan Height Control Potentiometers

NOTE: Do not use excessive force when making adjustments on the video display unit.

13. Use the CTRL + ALT + < command to place the display in the low resolution graphics mode. Press and hold the CTRL and ALT keys while pressing the less than (<) key. Then release all three keys. Using the adjustment tool, adjust the master height control potentiometer so that all characters appear at the desired height.

    NOTE: Do not use excessive force when making adjustments on the video display unit.

14. Use the CTRL + ALT + > command to place the display in the high resolution mode. Press and hold the CTRL and ALT keys while pressing the greater than (>) key. Then release all three keys. Using the adjustment tool, adjust the high scan height control potentiometer so that all characters appear at the desired height.

15. If characters near the top of the display appear taller or shorter than those near the bottom of the display, a linearity adjustment is necessary. Using the VDU adjustment DIAGNOSTIC program, fill the screen with capital Es (Figure 8-24 above). See Chapter 4 in this guide for more information on the VDU adjustment program.

16. Using a plastic adjustment tool, turn the vertical linearity control potentiometer until all characters appear equal in height.

    NOTE:

    - This adjustment may affect the character height enough to warrant a readjustment for character height.

- If the vertical linearity control does not exist on your video display unit, it is preset and controlled by the video display unit.

- Step 13 and 14 should be repeated after each adjustment until the character height are equal.

17. If your display is shifted to the right or left, a horizontal display centering adjustment is necessary. Two potentiometers control the centering of the display. One is the master video center potentiometer, which shifts the display in both a high and low resolution mode. The other one is the high scan video center potentiometer, which affects the display in high resolution mode. (Refer to Figure 8-26 for the locations of both the master video center and high scan video center control potentiometers.)

    NOTE: Do not use excessive force when making adjustments on the video display unit.

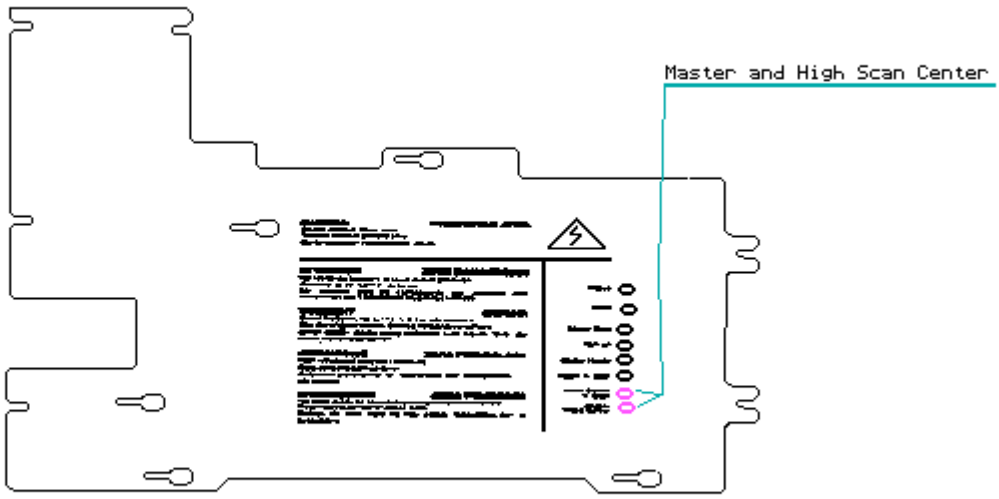


Figure 8-26. Location of the Master Video Center and High Scan Video Center Control Potentiometers

18. Use the CTRL + ALT + < command to place the display in the low resolution graphics mode. Use the adjustment tool, and adjust the master video center control potentiometer so that the display is in the center of the screen.

19. Use the CTRL + ALT + > command to place the display in the high resolution mode. Using the adjustment tool, adjust the high scan video center control potentiometer so that the display is in the center of the screen.

If your display experiences a problem other than those covered in the previous 19 steps, replace the video display unit.

For video display unit removal and replacement procedures, refer to Chapter 7 in this guide.